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How is television time linked to cardiometabolic health in adults? A critical systematic review of evidence for proposed pathways linking TV time with characteristics of sitting, physical inactivity, sleep, diet, and mental health among adults

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How is television time linked to cardiometabolic health in adults? A critical systematic review of evidence for proposed pathways linking TV time with characteristics of sitting, physical inactivity, sleep, diet and mental health among adults

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ABSTRACT

Objective: To assess evidence that TV time is associated with characteristics of sitting, physical inactivity, sleep, diet, and mental health among adults, all of which are potential mediators of the association between TV time and cardiometabolic health, and to assess the evidence that any such associations are likely to be causal.

Design: Systematic review

Data sources: Web of Science and PubMed

Eligibility criteria for selecting studies: Studies that reported on quantitative relationships between TV time (independent variable) and the potential mediators listed above among samples of healthy adults (≥18 years old).

Data extraction and synthesis: Study quality was assessed using the NHLBI Quality Assessment Tools. Due to heterogeneity of study designs and measurements, the findings were synthesised using narrative summary accompanied by custom plots.

Results: We identified 73 studies that met the inclusion criteria. There was some evidence to suggest that TV time might be associated with particularly low energy expenditure and with prolonged sitting, and that it may be causally associated with physical inactivity. High TV time was generally linked to poor diet, but there was sparse evidence of a causal association. The evidence for associations of TV time with sleep and mental health outcomes was inconsistent overall, although there was some evidence for a weak, potentially causal association between TV time and depression.

Conclusion: While there is some evidence of weak causal links between TV time and several of the potential mediators affecting cardiometabolic health examined in this review, none is sufficient to explain the strong association between TV time and cardiometabolic health. The evidence is currently sparse, and further research is needed on how TV viewing in everyday life may influence sedentary time, physical inactivity, diet, sleep and mental health.

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STRENGTHS AND LIMITATIONS OF THE STUDY

- This review synthesises for the first time a diverse body of evidence on posited causal pathways linking TV time with factors (characteristics of sedentary time, lack of physical activity, sleep, diet or mental health) considered to mediate the relationship of TV time with cardiometabolic health
- Custom plots were produced to visualize the findings of individual studies, including the magnitude, direction, and statistical significance of associations
- The outcomes and designs of included studies were too heterogeneous for estimating overall summary estimates using meta-analysis
- There has been little research that directly addresses the causality of proposed links between TV time and these proposed mediators of the effect of TV time on cardiometabolic health

Keywords: TV time, sedentary time, cardiometabolic health, diet, mental health, physical inactivity

INTRODUCTION

Television time has been associated with poor cardiometabolic health outcomes among adults, including cardiovascular mortality,[1, 2] incidence of type 2 diabetes,[1, 2] and increases in adiposity.[3] Television time is generally assumed to be sedentary and sedentary time is now a well-established risk factor for poor cardiometabolic health.[1, 4] However, the association between TV time and poor health outcomes is stronger than the association between total sitting time and poor health outcomes[1, 5] or than time spent sitting in other contexts, such as sitting at work.[6-9] For studies based entirely on self-reported measures, lower measurement error in the assessment of TV time than in the assessment of other

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sedentary behaviors is likely to contribute to such a difference.[10-12] However, the relationship between self-reported TV time and cardiometabolic outcomes is also stronger than the relationship between objectively assessed sedentary time and those same outcomes,[13-15] a finding that cannot be attributed to measurement error. It is possible that TV time has a particularly adverse impact on cardiometabolic health and this suggestion is repeatedly made in the literature.[1, 5, 7, 16] It is therefore important to assess the evidence for potentially causal associations of TV time with mediating factors that are commonly invoked to explain this relationship .

Two of the most common suggestions to explain the association of TV time with poor cardiometabolic health are that sitting while watching TV may be particularly prolonged,[1, 5] which has been experimentally linked to poorer glucose metabolism,[17] or that snacking while watching TV may contribute to its effects.[1, 5, 7, 16] It has also been suggested that TV time may interfere with postprandial glucose and lipid metabolism because it is assumed to take place largely in the evening, after the main meal of the day.[1, 5] Other suggested potential contributors to the pronounced association between TV time and cardiometabolic health include its possible links with inadequate sleep[18] or poor mental health[16, 19] which have both been associated with poor cardiometabolic health.[20-26] However, the evidence lending support to these suggested possible explanations has not, to our knowledge, been either exhaustively or critically evaluated.

The purpose of this review is to improve our understanding of how TV time is linked to cardiometabolic health among adults by systematically and critically evaluating the available literature on associations linking TV time with mediating factors known to affect cardiometabolic health (Figure 1). We evaluate the evidence describing associations between

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TV time and characteristics of sitting, physical inactivity, sleep, dietary factors, and mental health.

METHODS

Eligibility criteria

Studies were eligible for inclusion in the review if they quantitatively described at least one of the following: associations between TV time and characteristics of sitting, physical inactivity (i.e. lack of physical activity rather than sedentary time), sleep, dietary factors, and mental health. Studies could be of any design (e.g., prospective, cross-sectional, experimental, intervention) as long as TV time was the independent/exposure variable. Participants in the studies had to be non-institutionalized adults (≥18 years old) with measurements in adulthood reported in the paper. All studies had to be published in English in 2000 or later as TV viewing practices change over time.

We excluded studies that met any of the following criteria: 1) the study was based on a population characterized by a particular health status (e.g., cancer survivors, only overweight or obese adults), 2) TV time was assessed only as part of a composite measure (e.g., leisure screen use), 3) only the frequency (not the duration) of TV viewing was measured, or 4) TV was the outcome/dependent variable.

Information sources and search strategy

The literature was searched in December 2019 using Web of Science Core Collections and PubMed. The search strategy included terms to identify studies that measured TV time across all disciplines ("television time" OR "television viewing" OR "television watching" OR "sedentary behavio?r" OR "sedentary time" OR "screen time" OR "watching television" OR

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"viewing television"). The reference lists of all included papers were also checked. We also included a relevant study of our own that was published shortly after the search was complete.[27]

Study selection

The titles and abstracts of the retrieved studies were screened for relevance. A subsample (10%) was independently screened by both authors to ensure agreement and consistency; any discrepancies were discussed until an agreed application of inclusion and exclusion criteria was reached, then one author (JW) screened the remainder of titles and abstracts. The full-texts of studies with relevant abstracts were consulted to determine eligibility. Both authors confirmed the eligibility of each full-text, and any cases of disagreement were resolved through discussion. Where more than one paper used the same study or dataset, both authors jointly decided which paper to include in the review, giving preference to the paper that used the most recent data (in the case of longitudinal studies using the same outcome variable at each time point) or had a stronger focus on the relationship between TV viewing duration and the outcome variable of interest to this study (in the case of identical data sets).

Quality assessment

The quality of each eligible full-text was assessed based on the NHLBI Quality Assessment Tools for Observational Cohort and Cross-Sectional Studies (for prospective or cross-sectional studies) and for Controlled Intervention Studies (for experimental and intervention studies) (https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools). In order to draw on the highest standards of evidence, studies deemed to have low internal validity based on the quality appraisal were excluded from the review. The quality assessment of each study

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was confirmed by both authors and any disagreements were discussed until consensus was reached.

Data extraction

The data from each eligible study were extracted into a spreadsheet with pre-defined columns for the general study details (location, date, cohort), study design, methodology, sample characteristics, statistical analyses (including covariates), and main findings describing the association between TV time and the variable(s) of interest.

Synthesis of results

The results of the associations between TV time and the outcome of interest were organized by category of outcome variable (characteristics of sitting, physical inactivity, sleep, diet, mental health). Because of heterogeneity of study outcomes and study designs, meta-analysis was not used. Therefore, the results were synthesized using narrative summary accompanied by visualization of study findings. We did not use forest plots to visualize the results because of the variations in measurements of TV time (i.e., as continuous and categorical), variables adjusted in analyses, and outcome variables. Instead, we constructed custom plots designed to illustrate each study's findings to provide a visual summary of the strength of the evidence for each outcome. Wherever possible, we plotted the effect size of the association between TV time and the outcome variable. If the paper reported a measure of effect size, we extracted this; if studies did not report effect size but reported sufficient information to calculate it (e.g., mean and standard deviation), we calculated the standardized mean difference using the R package *metafor* (Supplementary file 1). If 95% confidence intervals or standard errors were reported, these were converted to standard deviations using the formula suggested in the Cochrane handbook (https://handbook-5-1.cochrane.org/chapter_7).

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When calculating the effect size within experimental studies that used within-subjects designs, we imputed correlations of 0.5 to account for repeated measurements; sensitivity analyses using correlations of 0.6 and 0.8 were also conducted to examine the extent to which the imputed correlation might affect the results (shown in Supplementary file 1). While we acknowledge that the interpretation of the magnitude of effect sizes is arbitrary and context-specific, we classified effect sizes as negligible, small, medium, or large based on standard conventions[28] for the purposes of plotting (standardized mean difference of <0.20, 0.20-0.49, 0.50-0.79, ≥0.80; odds ratio of <1.50 1.50-2.49, 2.50-4.29, ≥4.30, correlation of <0.10, 0.10-0.29, 0.30-0.49, ≥0.50, respectively).

Within each outcome group, we considered first cross-sectional studies of associations between TV time and the outcome, and then studies that can be used to assess the possibility of causality, namely prospective, experimental, and intervention studies.

We separated studies based on whether TV time was constructed as a continuous variable or a categorical variable. For the latter, we plotted the effect size comparing the highest versus lowest TV time for simplicity. Within the plots, the included papers are ordered from most robust to least robust evidence, primarily on the basis of robustness of sampling methods and sample size. Whether the finding of the study was statistically significant is denoted through a filled (black) symbol with the direction of the association denoted by a positive sign (+), negative sign (-), zero (0), or question mark (?) if the direction could not be ascertained; non-significant associations are unfilled. The strength of the effect size is denoted by number of boxes (one=negligible, two=small, three=medium, four=large) based on the reported or calculated effect size. Where the effect size was not reported and could not be calculated in a standardized way (e.g., regression coefficient, graphical presentation of means), we plotted

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the findings with a triangle, filled (or not) to denote statistical significance. Significance was determined based on 1) the calculated or reported confidence intervals not crossing zero, or 2) the study reported a p-value <0.05 where effect size with confidence intervals was not reported and could not be calculated.

RESULTS

Search results

The flow of studies through the review process is shown in Figure 2. From the original 10,649 unique records retrieved by the database searches, 72 were eligible for inclusion. An additional 12 papers were retrieved through the other sources and 11 papers were excluded, leaving 73 studies included in the review (see Supplementary files 2-4 for study details, including information on statistical modelling with covariates listed).

Characteristics of observational studies (cross-sectional and prospective)

Forty-eight studies included in the review were observational, conducted in North America (n=19), Australia/New Zealand (n=12), Europe (n=11), South America (n=3), and Asia (n=3). The majority of studies (n=41) used cross-sectional designs, while the remaining seven used prospective study designs. Sample sizes ranged from 27 to 60,202. The vast majority of studies used samples of adults (men and women) of broad age ranges, but in some cases the samples were limited to women (n=11), or only young (i.e., ≤24 years old, n=5) or older (i.e., mean age >65, n=7) adults.

With the exception of Gemming et al.[29] all observational studies measured TV time by self-report. The most common unit of measurement was hours per day, either as a continuous (n=15) or categorical (n=15) measure. Thirteen studies specifically asked about time spent

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sitting and watching TV, and only one study[30] specified within its questions that, in instances of multitasking, participants should only report TV when it was the main activity. The vast majority of observational studies (n=26) asked about TV time for an unanchored 'usual' or 'typical' day. Six studies specified the same reference period for TV time and the self-reported outcome variable (e.g., past day,[31] past two days,[32] past week,[33, 34] past three months,[35] past year).[36] Five studies used methods to measure TV time and the outcome variable concurrently.[29, 37-40]

Three observational studies reported associations between TV time and characteristics of sitting, eight on physical inactivity, five on sleep, twelve on dietary characteristics, and twenty-one on mental health outcomes. One study[35] reported on outcome variables in two categories so it is counted twice here. Nine studies used objective measures of outcome variables (two for characteristics of sitting; three for physical inactivity, four for mental health); the remainder relied on self-report measures.

Characteristics of experimental studies

Twenty-four studies were experimental, primarily focused on links between TV and food consumption (n=19); the remainder focused on energy expenditure of sitting or movement while watching TV (n=4) or the effects of TV on melatonin and cortisol levels (n=1). Most experimental studies used a within-subjects study design (n=16) and were conducted in Europe (n=11) or North America (n=9), Australia (n=3), and Japan (n=1). Sample sizes ranged from 13 to 120; ten studies included only female participants while one study included only men.

Characteristics of intervention study

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One study was an intervention examining changes in sleep duration and sleep timing when participants (8 university students and 8 older adults in Japan) were asked to watch TV as usual for one week (control) and to limit it to 30 minutes per day the following week (intervention).

Characteristics of sitting

Cross-sectional studies

Prolonged sitting

Two observational studies examined the relationship between self-reported TV time and accelerometer-measured bouts and breaks in sedentary time[27, 41] (Figure 3). One observational study among older adults assessed TV time in the week before sedentary time was measured using accelerometry and found that self-reported TV time was positively associated with time spent in accelerometer-measured prolonged bouts (≥30min) of sedentary time and negatively associated with hourly breaks in sedentary time per day.[41] In our study,[27] pregnant women with high TV time during their second trimester (≥2 hours per day) who wore an inclinometer for one week during that trimester accumulated fewer daily breaks in sedentary time than women with low TV time; there were no differences in prolonged sedentary time between the two groups.

Time of day

In our study,[27] pregnant women with high TV time (≥2 hours per day) had significantly higher activPAL-measured sedentary time in the evenings (6-11pm) compared to women with lower TV time (<2 hours per day), but there were no differences in evening prolonged sedentary time or breaks in sedentary time between the two groups (Figure 3).

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Based on participants' diaries of time spent in various sedentary behaviours including watching TV, Kim and Kang[42] found that TV time was higher in the evening compared to the afternoon or the morning (Figure 3), and significantly more time was spent watching TV in the evenings compared to time spent in other sedentary behaviors, including use of electronic devices, work, and transport (Supplementary file 2).

Experimental studies

Four experimental studies examined associations between watching TV and characteristics of sitting, specifically energy expenditure[43-45] and level of movement.[46]

Energy expenditure and movement

Three experimental studies compared the energy expenditure (EE) of watching TV with the EE of other sedentary behaviors.[43-45] The most robust evidence indicated that the EE of sitting and watching TV was lower than the EEs of typing, playing a handheld video game (both small effect sizes), and playing a screen-based video game while sitting (large effect size; Figure 3).[45] The other two studies found no difference between the EE of watching TV and the EE of typing at a desk,[44] reading at a desk,[44] or working on a laptop computer[43] in their samples (Figure 3; Supplementary file 2), although their relatively small sample sizes and lack of *a priori* power calculations may mean their non-significant findings were due to underpowered samples.

A laboratory-based study compared the level of movement captured using hip-worn, triaxial accelerometry while watching TV (seated on a chair without armrests) compared to other sedentary behaviors among adults.[46] Total movement (which we calculated as vector magnitude of acceleration) was higher while watching TV compared to while reading,

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writing, typing, and sitting both 'naturally' (i.e., with free movement) and motionless, but did not significantly differ from movement while playing cards or preparing food while seated (Figure 3).

Given the laboratory settings of these studies, the external validity of these findings is unclear. In sensitivity analyses using larger imputed correlation coefficients, the effect sizes slightly increased for the difference in the EE of TV compared to typing and for the difference in vector magnitude of acceleration for TV versus reading, preparing food, and sitting naturally (Supplementary file 1).

Summary

There is a small amount of evidence from observational studies to suggest that TV time may be linked with more prolonged bouts of sedentary time and fewer breaks in sedentary time, and that watching TV is an evening pursuit. However, we found no studies that examined whether sedentary time *while watching TV* is particularly prolonged. There is inconsistent evidence on whether TV is a sedentary behavior with particularly low EE in laboratory settings, although the highest-quality evidence suggests its EE is lower than some other sedentary behaviors (e.g., typing, video games). We did not find any studies that reported on levels of muscular activation while watching TV.

Physical (in)activity

Cross-sectional studies

Six cross-sectional studies examined associations between TV time and various measurements of physical (in)activity[34, 35, 47-50] (Figure 4; Supplementary file 2). Bennett et al.[49] reported that TV time was cross-sectionally associated with lower

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likelihood of achieving 10,000 steps per day measured by pedometer. An interaction term for gender was non-significant, suggesting this association was similar for men and women. Tucker et al.[50] found no association between TV time and cardiorespiratory fitness (VO_2 max) among a sample of healthy women. Four studies tested associations between TV time and leisure-time physical activity (LTPA)[34, 47] or LTPA + transport PA[35, 48] with all analyses stratified by gender. Shields and Tremblay[35] reported an association between high TV time and likelihood of being inactive that was stronger among women than men. Sugiyama et al.[34] found an association between high TV time and low PA among women but not in men, while Burton et al.[48] reported high TV time on weekends was associated with lower likelihood of high PA levels among middle-aged men, with no association seen among women. Bertrais et al.[47] reported no associations between TV and LTPA among men or women.

Prospective studies

In a very large study, Kim et al.[51] found that higher TV time at baseline was associated with both lower accelerometer-measured physical activity (measured as average total acceleration) at follow-up 5.7 years later among both women and men and a small but significant decrease in physical activity. Lakerveld et al.[52] reported that high TV time at baseline was associated with higher likelihood of a reduction in PA (LTPA plus walking for transport) five years later among women but not men (Figure 4; Supplementary file 2).

Summary

There is consistent evidence to suggest an association between TV time and physical inactivity (particularly LTPA), including from two large prospective studies, suggesting a

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causal link. However, the effect sizes are negligible to small. We found no studies that tested associations of TV time with objectively measured MVPA.

Sleep

Cross-sectional studies

Five studies examined associations between TV and various measurements of sleep in observational cross-sectional studies[53-57] (Figure 5; Supplementary file 2). Three studies tested associations between TV time and poor sleep quality, with two reporting negligible but significant inverse associations.[54, 56] Buman et al.[54] also reported small, significant associations between TV time and higher likelihood of long sleep onset latency (≥30 minutes) and waking up too early. In a sample of French adults, Andrianasolo et al.[53] reported an inverse association between TV time and likelihood of excessive daytime sleepiness; Buman et al.[54] reported no association with the same outcome. Non-significant associations were reported for TV time and likelihood of short sleep duration,[54, 57] long sleep duration,[54, 57] and waking up in the night.[54] Custers and Van den Bulck[55] reported that TV time was associated with later bedtime on weekend nights but found no associations with the duration of the sleep window or overall tiredness.

Experimental and intervention studies

One small experimental study (n=13)[58] and one small intervention study (n=16)[59] shed some light on possible mechanisms by which TV might affect sleep (Supplementary file 2). Komada et al.[58] tested the effect of TV displays on melatonin and cortisol secretion at night in a laboratory setting, reporting that higher blue light intensity was associated with a suppressive effect on melatonin levels but not on cortisol levels. Melatonin and cortisol are hormones involved in the body's sleep-wake cycles, and the production of melatonin in the

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body is known to be tightly linked to bright light.[60] Asaoka et al.[59] conducted an intervention in which university students and elderly adults limited their TV time to 30 minutes a day for one week following a week-long 'control' in which participants went about normal activities. During the week of restricted TV time, the university students went to bed significantly earlier and slept significantly longer compared to the control week; the intervention had no effect on sleep duration among the elderly.

Summary

In summary, there is inconsistent cross-sectional evidence suggesting that TV time is associated with indicators of poorer sleep, and the effect sizes are very small. One small experimental and one small intervention study suggest possible pathways by which TV could affect sleep, including through suppression of melatonin by blue-light and through possible allocation of time toward TV instead of sleep, at least among young people.

Dietary factors

Cross-sectional studies

Daily/weekly TV time and dietary characteristics in observational studies

Eight cross-sectional observational studies tested associations between TV time and self-reported diet.[32, 35, 36, 61-65] The dietary factors that were most commonly tested (i.e., included in more than one study) were consumption of calorically sweetened beverages,[32, 61, 62] snack foods,[32, 65] fast food,[61, 63] and fruit and vegetables.[35, 61] These associations, plotted in Figure 6, show generally consistent but weak associations with more TV time associated with more consumption of calorically sweetened beverages, snack foods and fast food in all tests reported (apart from with fast food for breakfast or lunch)[63] and with less fruit and vegetables in both tests reported (Supplementary file 3).

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There is also evidence for associations between TV time and other dietary factors from individual studies. Bowman[32] reported associations between TV time and higher total daily energy intake, higher energy consumption at dinner, higher consumption of added sugar and lower intake of dietary fiber. All of these associations were significant but negligible in size (Figure 6). Charreire et al.[36] and Robinson et al.[64] each reported associations between TV time and poorer overall diet scores as determined by principal components analyses specific to each study. Non-significant associations were reported between TV and consumption of alcohol,[61] sweets,[61] and proportion of daily calories from fat, protein, and carbohydrates.[32]

Food and drink consumption while watching TV in observational studies

Four observational studies have examined dietary patterns while watching television in free-living settings among voluntary/convenience samples of adults in New Zealand,[29] African-American women in Chicago,[37] and young adults in the USA[38, 39] (Figure 7; Supplementary file 3).

The most detailed evidence comes from Gemming et al.[29], who linked 24-hour dietary recalls to wearable camera recordings on three separate measurement days, allowing for an in-depth assessment of the contextual details (e.g., location, environment, social context, presence of screens) surrounding each eating episode noted in the recall. In their sample, 17% of eating episodes (22% of meals and 12% of snacks) took place while watching TV. Snacks consumed while watching television were higher in energy than snacks consumed when no TV was viewed. There was no difference in meal energy intake or duration of meals or snacks while watching TV compared to when no screen was viewed.

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The remaining three studies examined the contexts of self-reported eating and drinking using ecological momentary assessment[37] and food diary entries annotated with contextual information.[38, 39] These studies found that 30%[37] and 26%[38] of snacking and eating occasions, respectively, took place while watching TV. Stroebele and de Castro[39] reported that an average of 1.03 meals per day took place with the TV on, but meal size did not differ with or without the TV on. Compared to food and drink consumed while 'not doing anything else,' a larger proportion of snacks[37] and sugar-sweetened beverages[38] were consumed while watching TV. Smaller proportions of water, fruit, vegetables, and cereals/grains consumed were eaten while watching TV compared to while doing nothing else.[38]

Experimental studies

TV versus doing nothing

There is some evidence to suggest that more food is consumed while watching TV compared to while doing nothing else in a laboratory setting (Figure 8; Supplementary file 3). Nine experimental studies compared food consumption while watching TV compared to a control condition spent alone with no distraction.[66-74] Seven studies found that participants had higher consumption in terms of energy intake[66-68, 70, 72] or mass[72-74] during the TV condition compared to the control condition (generally small effect sizes), although this did not reach significance in most cases. In sensitivity analyses, effect sizes increased from small to medium in two studies[66, 68] when a larger correlation coefficient was imputed (Supplementary file 1).

TV versus other activity

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There is little evidence to suggest that more food is consumed while watching TV compared to while doing other activities in a laboratory setting (Figure 8; Supplementary file 3). Seven studies compared food consumption while watching TV with consumption while doing other tasks, including listening to audio stories,[66, 69] listening to classical music,[75] reading,[71] driving,[73] playing video games,[76] eating with a researcher or strangers,[68, 73] and eating with friends.[68] Five of these studies reported no differences in the amount of food consumed between the TV condition and the other experimental condition they tested.[66, 68, 69, 71, 76] The remaining two studies reported higher consumption while watching TV compared to while listening to classical music,[75] driving,[73] or while conversing with a researcher.[73] Sensitivity analyses suggested larger imputed correlations did not materially change the findings (Supplementary file 1).

Effect of TV on later consumption

Three studies tested whether the consumption of a snack while watching TV (compared to while not watching TV) would result in higher food consumption at a meal later on in the same day (consumed without TV)[70, 77, 78] (Supplementary file 3). Two of these studies found that when the earlier snack had been consumed while watching TV, significantly more was eaten in the subsequent TV-free meal;[70, 77] the third did not find subsequent consumption was higher after the eating-with-TV exposure.[78]

The role of TV content

There is little evidence to suggest that the content of TV being watched influences food consumption in experimental settings. Four studies compared food consumption when exposed to food advertisements compared to either neutral (non-food) advertisements[79-81] or no advertisements,[71, 81] all reporting no main effect of commercial condition (although

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Anschutz et al.[79] reported a significant interaction with gender; see Supplementary file 3). Three studies evaluated the role of TV content on concurrent food consumption;[70, 82, 83] two of these reported that participants ate more when watching boring or repeated showings of TV compared to engaging or novel TV,[82, 83] while the third reported no difference in consumption when participants watched funny, boring, or sad TV shows.[70] One study[84] compared snack consumption following exposure to either a clip of a cooking show or a nature show, reporting no difference in overall consumption between the groups (although more chocolate balls were consumed following exposure to the cooking show than the nature show).

Summary

There is a small body of evidence from cross-sectional studies to suggest that those with high TV time tend to consume more sweetened drinks, fast food, and fewer fruit and vegetables than those with lower TV time, although effect sizes are negligible to small. It is important to note that these studies have described links between TV time and typical consumptions of various foods without collecting any information regarding whether consumption took place while watching TV. Other studies have examined eating at the same time as watching and TV and there is some evidence here that snacks consumed while watching TV may be higher in energy than snacks consumed without watching TV. However, there is no evidence of a similar pattern for meals. There is currently no evidence in free-living situations describing what proportion of TV time versus other time is accompanied by eating; this information would help clarify whether TV 'triggers' eating in real-life contexts.

There is some experimental evidence to suggest that more food is consumed while watching TV compared to while doing nothing else, and that eating while watching TV is linked to

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higher consumption later on in the day (without TV), but the laboratory conditions of these findings makes the external validity unclear. There is also little evidence to suggest that more food is consumed while watching TV compared to while doing other things, or that the content of what is viewed on TV (including food advertisements) is linked directly to increased food consumption.

Mental health

Twenty-one studies tested associations between TV time and mental health outcomes[30, 31, 33, 40, 85-101] (Figure 9; Supplementary file 4).

Depression and depressive symptoms

Cross-sectional studies

Seven cross-sectional studies examined links between TV time and depressive symptoms. The two biggest studies[95, 97] found positive and significant associations with symptoms of depression, but others found positive,[91] negative,[88, 89, 92] or zero[33] associations, and these were all non-significant.

Prospective studies

Three out of four prospective studies examining depressive symptoms as an outcome reported positive (very small) associations,[85, 86, 90, 94] which were statistically significant in two cases. Two studies tested links between TV time and incident depression;[87, 94] the largest[87] found a positive (significant) association, while a smaller study[94] found a negative (non-significant) association.

Stress and anxiety

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Cross-sectional studies

Five cross-sectional studies tested associations between TV time and stress, using cortisol levels[96, 98, 101] or self-reported stress[33, 89, 98] as the outcome variable. Two studies found no association between TV time and cortisol levels;[96, 101] the third, which did not control for key confounding variables, found an inverse association between TV and cortisol levels among women and no association among men.[98] No studies found associations between TV and self-reported stress levels.[33, 89, 98]

None of the three cross-sectional studies that tested links between TV time and self-reported anxiety levels reported associations.[33, 88, 93]

Other aspects of mental health

Cross-sectional studies

Two cross-sectional studies, including the largest study,[30] reported significant inverse associations between TV and mental/subjective wellbeing,[30, 99] and another cross-sectional study reported a positive association between TV time and likelihood of feeling sad among men but not women.[95] The remainder of studies, all cross-sectional, reported no associations between TV time and happiness,[100] subjective vitality,[100] life satisfaction,[100] or suicidal thoughts.[95]

In a study using experience sampling methods over the course of a week, Goodwin et al.[40] found that mean positive affect scores were lower while watching TV compared to while engaging in other leisure activities (e.g., hobbies) or while doing productive activities (e.g., work or volunteering); negative affect score did not differ while watching TV compared to while doing any other activities. In sensitivity analysis, only the effect size of the difference

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in positive affect during TV versus productive activities increased (from small to medium) with larger imputed correlations (Supplementary file 1).

Prospective studies

Andrade-Gomez et al.[85] reported no association between TV time at baseline and psychological distress at follow-up 3.3 years later. Using daily diary data over 8 days and lagged models, Bayraktaroglu et al.[31] found that TV time on a given day was not associated with change in positive affect from that day to the next.

Summary

In summary, there is evidence for a positive but weak association between TV time and depressive symptoms and clinical depression, including from prospective studies, suggesting the possibility of a causal association. However, there is no evidence for a relationship between TV time and self-reported stress or anxiety and very little in relation to other aspects of mental health or affect.

DISCUSSION

This review aimed to explore potential mechanisms linking TV time with cardiometabolic health by critically evaluating all available evidence describing links between time spent watching TV and potential mediators associated with cardiometabolic health.

There is some evidence to suggest that TV might be a particularly 'sedentary' behavior. Observational studies in free-living contexts suggest that higher TV time is associated with higher prolonged sedentary time and fewer breaks in sedentary time, although effect sizes are small. Experimental studies testing energy expenditure while watching TV compared to other

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conditions produced inconsistent findings, although the highest quality evidence from one study suggested TV may be a sedentary behavior with particularly low energy expenditure. Differences in energy expenditure within seated postures are likely to reflect differences in levels of muscular activation, and muscular inactivity has been linked to suppression of lipoprotein lipase,[102] an important protein in regulating cardiometabolic risk factors.[103] There is also evidence that more TV viewing takes place in the evening than in the morning or afternoon and that viewing is associated with higher sedentary time in the evening. To our knowledge, there is no evidence directly linking evening sedentary time to cardiometabolic health outcomes. However, there is experimental evidence to suggest that prolonged sitting impedes postprandial glucose metabolism,[17] suggesting that watching TV in the evening could impact health if involves prolonged sitting after a meal. More research is needed to help us understand patterns of sitting and characteristics of sitting *while actually watching TV* within free-living contexts. There is also a small amount of consistent evidence suggesting that TV time is associated with low physical activity, with most evidence relating to leisure time physical activity, although effect sizes are negligible. Overall, there is evidence to support a relationship between TV viewing and overall level of physical activity. It is possible that both particularly low energy expenditure while watching TV and displacement of physical activity by TV during leisure could contribute to links between TV time and poor cardiometabolic health. However, the evidence suggests that such effects are likely to be very small.

There is a small body of consistent evidence from observational studies that TV time is associated with several aspects of diet, including higher consumption of sugar-sweetened beverages and fast food and lower consumption of fruits and vegetables. There is evidence linking high sugar-sweetened beverage consumption, high fast food consumption, and low

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fruit and vegetable consumption with poor cardiometabolic health outcomes.[104-106] When food is placed in front of participants within laboratory settings, there is some evidence to suggest that more food is consumed while watching TV compared to while doing nothing else; however, it is not clear whether these findings are relevant to real-life situations. Further research is needed to examine how or whether TV and eating co-occur in free-living contexts. At present the evidence for a causal link between TV time and a less healthy diet is very weak.

The evidence for associations between TV time and sleep was inconsistent and, where associations were found, the effect sizes were negligible. There is some evidence from one small study that blue light from a TV display suppresses melatonin, which could lead to a disruption in circadian rhythms, and from another small study that a reduction in TV time can lead to an earlier bedtime, at least in young people. Much more research is needed before sleep disruption can be considered a likely mechanism linking TV time and cardiometabolic health.

There was some evidence to suggest that TV time might be weakly linked to mental health outcomes, particularly depressive symptoms and depression, where prospective studies suggest a possible causal association. However, there is also prospective evidence to suggest that depressive symptoms are associated with subsequent increased TV time.[107] Similarly, Bayraktaroglu et al.[31] found, using lagged models over eight consecutive days, that TV time was not linked to change in positive affect, but positive affect was linked to decreases in TV time. These pieces of evidence suggest that while increased TV time may be causally linked to depressive symptoms, this relationship is likely to be bidirectional; further work is needed to explore these associations.

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We have found evidence for weak effects of TV time on sitting characteristics, physical inactivity, and depressive symptoms, with suggestive evidence that it may be causally linked to unhealthy snacking (in experimental contexts) and sleep disturbance. However, overall, there is little evidence that any of the commonly proposed mediating pathways is likely to be able to explain the strong association between TV time and cardiometabolic health. There is, nevertheless, enough suggestive evidence to justify further research, particularly to examine whether TV time is a particularly harmful form of sedentary time and whether it displaces physical activity.

It has been suggested that the association between TV time and cardiometabolic health may be confounded by socioeconomic position (SEP).[19] While the association between SEP and TV time is likely to vary globally across cultural and economic contexts, low SEP is consistently linked to high TV time in 'Western' countries such as the UK,[108] the US,[109] Australia,[110] and Canada.[111] Within these contexts, low SEP itself is associated with poor cardiometabolic health outcomes.[112-115] Thus relationships between TV time and cardiometabolic health could be artefacts of the effects of SEP on TV time and on cardiometabolic health, implying that associations between TV time and the mediators considered here could also be confounded by SEP. To take account of this issue, the vast majority of studies included in this review made some adjustment for socioeconomic position. Twenty-five studies adjusted for one SEP indicator (education in 22 cases) and 9 studies adjusted for two SEP indicators (most commonly education and household income). However, no studies adjusted for all four indicators that have been shown to be independently associated with TV time after mutual adjustment (education, income, occupational/social class, and area-level deprivation),[116] and in any case, removing all variation caused by

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such a powerful latent construct is likely to be impossible. Thus we argue that there is a strong possibility that the links between TV time and variables included in this review may be confounded by SEP. We also note that SEP might influence the observed association between TV and health in other ways besides confounding. For example, SEP has been shown to moderate the impacts that other health-related practices (e.g., smoking, alcohol consumption, physical inactivity) have on health outcomes, with those of lower SEP experiencing disproportionate harm.[117-120] Further research is needed to disentangle relationships between TV time, SEP, and health outcomes.

The findings of this review must be interpreted in light of limitations in the evidence base. The vast majority of included studies reported statistically significant results but often with very small effect sizes, suggesting possible publication bias. The associations seen in this review should therefore be interpreted with caution as statistically significant findings may be over-represented in the literature base. With the exception of one study,[29] TV was self-reported in all observational studies and may therefore be subject to reporting bias. It is also worth noting that only the minority of observational studies (n=13) specifically asked participants to report their posture while they were watching TV, and only one study instructed participants how to report instances of watching TV while doing something else. These measurement issues are especially relevant in an era in which TV can be watched anywhere at any time. It is therefore not possible to assume that TV time is spent sitting, but many researchers do make this assumption (indeed, TV time is often used as a proxy measure of sedentary time in the wider literature.)[121, 122] This must be carefully addressed in future studies. Studies also provide insufficient contextual information about how, where, and when TV fits within everyday lives. Wearable cameras, which have been useful within physical activity research and were used in a study included within this review,[29] may be a

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way forward for addressing some of these gaps by providing an objective assessment of TV time as well as information on the context of TV viewing and how eating practices relate to TV time. Other innovative measurement techniques, such as the use of electromyographic shorts,[123] may also be useful for understanding muscular activation levels while watching TV in real-life situations, thus providing better ecological validity compared to similar experiments within laboratory settings. These methods will also be relevant for examining the contexts of screen-related practices more broadly, beyond just TV time.

CONCLUSIONS

Our findings suggest that while there is some evidence that TV time may be causally associated with some of the factors commonly proposed to mediate its relationship with cardiometabolic health, particularly prolonged sedentary time, lack of physical activity, and depression, the associations are generally weak and inconsistently observed. Before researchers can confidently ascribe the associations of TV time with cardiometabolic health outcomes to TV's effects on the mediators explored here, much more research is needed on how they are affected by viewing TV, and particularly by TV viewing in everyday lives.

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Competing interests

The authors declare that they have no competing interests

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Authors' contributions

JW led the study. Both authors contributed to the design, database searches, data collection, and analysis. JW constructed the plots. Both authors contributed to and approved the final manuscript. f

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Figure 1. Hypothesized mediation pathways between TV and cardiometabolic health. This review aims to evaluate evidence for the links indicated by the dashed lines.

Figure 2. Flow of studies through the review process

Figure 3. Summary plot of effect sizes for associations of TV time with characteristics of sitting

Figure 4. Summary plot of effect sizes for associations between TV time and physical inactivity

Figure 5. Summary plot of effect sizes for associations between TV time and measures of sleep

Figure 6. Summary plot of effect sizes for associations between TV time and dietary characteristics

Figure 7. Summary plot of eating patterns while watching TV

Figure 8. Summary plot of effect sizes for experimental effects of TV on food consumption compared to do nothing else or doing something else

Figure 9. Summary plot of effect sizes for associations between TV time and mental health outcomes

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Supplementary file 1. Effect size calculations, with sensitivity analyses conducted for studies that used within-subjects or repeated-measures designs.

Supplementary file 2. Summary of characteristics, measurements, and findings of included studies examining associations between TV and characteristics of sitting, physical inactivity, and sleep.

Supplementary file 3. Summary of characteristics, measurements, and findings of included studies examining association between TV and dietary characteristics.

Supplementary file 4. Summary of characteristics, measurements, and findings of included studies examining association between TV and aspects of mental health.

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Additional file 4. Summary of characteristics, measurements, and findings of included studies examining association between TV and aspects of mental health.

First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model (SEP listed first where applicable)**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
Andrade-Gomez 2018 [85] Quality = good	Prospective (mean follow-up of 3.3 years) Spain *Seniors-ENRICA cohort*	Stratified cluster sampling	Older adults aged ≥60 in Spain, n=1461 (51%) and n=1766 (52%) included in the depression and distress models, respectively Overall mean age not provided	Average time per week spent sitting or lying while watching TV in the past year *Categorical (≤2, >2-≤3, and >3h/day)*	Education, age, smoking, alcohol consumption, diet score (measuring adherence to Mediterranean diet), BMI, chronic disease, baseline score for each outcome variable, recreational PA, household PA, time spent in other SBs	Depressive symptoms based on Geriatric Depression Scale (score 1 to 10 with higher score = greater symptoms) at baseline and follow-up Psychological distress based on GHQ-12 (score 0 to 12 with higher scores indicating higher short-term psychological distress) at baseline and follow-up	There was a significant sex-interaction for depressive symptoms (p<0.01); among women, those with the highest TV time had higher depressive symptoms than those with the lowest TV time (β=0.37 (95%CI 0.13, 0.62)) but no such association was seen among men (β=0.11 (95%CI -0.70, 0.30)). There was no sex-interaction for psychological distress (p=0.05). With both sexes combined, there was no association between TV and psychological distress (β=0.09 (95%CI -0.15, 0.34)) for highest vs lowest TV time.
Bayraktaroglu 2019 [31] Quality = fair	Prospective USA *National Study of Daily Experiences (subsample of Midlife Development in the United States II project)*	Random subsample of MIDUS II (national survey)	Adults aged 33-83 in the United States, n=1668 (44%) Mean age 56 (12)	Daily time spent watching television (including time spent watching videos) measured via short telephone interview on 8 consecutive days *Continuous (min/day)*	Income level, employment status, age, gender, marital status, health status, personality traits	Positive affect measured via short telephone interview on 8 consecutive days. Participants reported the frequency (0=none of the time, 4=all of the time) with which they experienced positive affective states each day, including 'in good spirits', 'cheerful', 'extremely happy', 'calm and peaceful', 'satisfied', 'full of life', 'close to others', 'like you	TV time was negatively associated with positive affect when measurements taken on the same day were tested (β= -0.00014 (95%CI -0.00026, -0.00001)) TV time on the previous measurement day was not associated with change in positive affect from the previous day to the next in lagged analyses (β= 0.0001 (95%CI -0.0001, 0.0003))

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						belong', 'enthusiastic', 'attentive', 'proud', 'active', and 'confident'	Positive affect on the previous day was associated with decreases in TV time from the previous day to the next (β= -4.176 (95%CI –7.292, -1.060))
Dempsey 2014 [30] Quality = good	Cross-sectional Australia *AusDiab (1999-2000)*	Stratified cluster sampling	Adults in Australia, n=9907 (55%) Mean (95%CI) ages for men and women were 51.5 (51.1, 51.9) and 51 (50.6, 51.4), respectively	Time spent sitting watching TV or DVDs or playing games on the TV in last week, when it was the main activity you were doing (weighted average for weekday/weekend days) *Continuous (h/day)*	Education, employment status, age, gender, diet quality, marital status, diabetes and history of CVD, LTPA, waist circumference	Mental well-being score (sum of vitality, social functioning, mental health, and emotional functioning) as measured by the SF-36 health survey (past 4 weeks)	Significant, negative association (*b*= -0.40 (95%CI -0.69, -0.12))
Goodwin 2005 [40] Quality = fair	Cross-sectional USA	Convenience sampling	Older adults, n=69 (72%) Mean age 72.0 (6.4)	TV was reported on a paper diary when it was the current activity in response to pager prompts which came 6 times a day for 7 days between 8am and 8pm	N/A NB: affect while watching TV was compared to affect while engaged in other leisure activities (e.g., hobbies, reading), maintenance activities (e.g., cleaning, meal preparation), and productive activities (e.g., work or volunteering) which were reported in response to the pager prompts	Positive and negative affect 'in the moment' when prompted by the pager (i.e., while watching TV if that was the current activity). Mood descriptors were rated on a 5-point Likert scale; positive affect descriptors included things like 'enthusiastic', and negative affect descriptors included things like 'upset'	Positive affect was significantly lower while watching TV (mean 26.07 (SD 9.27)) compared to while doing other leisure activities (28.39 (9.35)) and productive activities (29.78 (9.36)), but was not significantly different compared to while doing maintenance activities (26.19 (9.65)) (see Figure 7). Negative affect while watching TV (10.871 (2.93)) did not significantly differ compared to while doing other leisure activities (10.873 (2.86)), during productive activities (11.09 (2.70)), or during maintenance activities (10.86 (2.74)).
Guillen-Royo 2019 [100] Quality = Fair	Cross-sectional Peru	Non-random quota sampling, stratified by age and gender across selected	Adults aged ≥18 in Peru, n=500 (53%)	Average hours of television watched per day *Continuous (h/day)*	Income, education, age, gender, civil status, mother tongue, place of residence, health status, location	General subjective wellbeing, measured as 1) happiness (score on 5-point scale from 'very	TV time was not significantly associated with happiness (*b*= 0.03, p>0.05), life satisfaction (*b*= -0.01, p>0.05), or

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		districts with varying socioeconomic statuses	Mean age 40.3 (17.3), range 18 to 95			happy' (5) to 'very unhappy' (1)); 2) life satisfaction (score on 5-point scale from 'very satisfied' (5) to 'very dissatisfied' (1)); and 3) subjective vitality (6 questions with 5-point scale responses concerning feeling energized and optimistic)	subjective vitality (*b*= 0.02, p>0.05)
Hamer 2014 [86] Quality = good	Prospective (2-year follow-up) England *English Longitudinal Study of Ageing*	Multistage stratified probability sampling	Men and women born on or after 29 February 1952 and living in households in England, n=6359 (55%) 64.9 (9.1) years	Average hours of television watched on ordinary day or evening, weighted for weekday and weekend day *Categorical (<2, 2-<4, 4 to <6, ≥6h/day)*	Last/most recent occupational class, age, sex, smoking, physical activity, alcohol, disability, chronic illness, BMI, internet use, newspaper reading	Depressive symptoms in the past week (possible score ranging 0 to 8), based on CES-D scale	Baseline TV associated with baseline depressive symptoms (*b*=0.49 (95%CI 0.35 to 0.63)) but was not associated with change in depressive symptoms over 2-year follow-up (*b*= -0.09 (95%CI -0.23, 0.04) for <2 vs ≥6h/day)
Jackson 2019 [96] Quality = good	Cross-sectional UK *English Longitudinal Study of Ageing*	Multistage stratified probability sampling	Men and women born on or after 29 February 1952 and living in households in England, n=3555 (67%) Mean age 68.3 (7.9)	Average hours of television watched on ordinary day or evening, weighted for weekday and weekend day *Continuous (h/day)*	Education, wealth, age, sex, ethnicity, limiting long-standing illness, cardiovascular disease, diabetes, smoking status, alcohol intake, physical activity, BMI, depressive symptoms	Hair cortisol concentration (pg/mg)	No association between TV viewing time and hair cortisol concentration (*b*=0.004 (95%CI 0.000, 0.009), p=0.07).
Lucas 2011 [87] Quality = good	Prospective (10-year follow-up) USA *Nurses' Health Study*	Cohort of 121,700 female registered nursed aged 30-55 at enrolment in the US in 1976	Female registered nurses in the US who were free from depressive symptoms at baseline, n=49,821 (100%) Mean age around 63 at baseline	Average weekly time spent sitting at home watching TV *Categorical (0-1, 2-5, 6-10, 11-20, ≥21h/week)*	Age, BMI, marital status, current postmenopausal hormone use, involvement in social/community group, smoking status, total energy intake, coffee intake, diagnosis of chronic disease, PA, physical limitations, mental health score at baseline	Incident clinical depression (new physician's diagnosis or beginning regular use of antidepressant medication)	Likelihood of incident depression was not significantly higher in any categories of TV time compared to the referent (0-1h/day), however the trend was significant (p=0.01)
Nabi 2016 [98] Quality = fair	Cross-sectional USA	Convenience sampling recruited through the university	Undergraduate couples in dating relationships, n=240 (120 couples; 50%)	Hours of TV watched on an average weekday and average weekend day *Continuous (h/day)*	Partners' TV time; cortisol analyses controlled for age and partners' cortisol levels; perceived stress analyses controlled for race	Salivary cortisol concentration measured in the laboratory four times within two hours	No overall association between TV and cortisol levels (β= -0.05, p=0.44). However, a significant interaction by gender was found; stratified

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			Mean age 21 (1.7), range 18-29		and partners' perceived stress	and averaged (log-transformed) Perceived stress score (14-item global measure of perceived stress) over the past month	analyses found that the association was significant among women (β= -0.20, p=0.02) but not men (β= 0.06, p=0.53). No association between TV and perceived stress (β= 0.12, p=0.09); no significant interaction with gender.
Padmapriya 2016 [88] Quality = fair	Cross-sectional Singapore *GUSTO study*	Pregnant women recruited from two major public maternity units when attending first trimester clinic visit	Chinese, Malay, and Indian pregnant women (second trimester) aged ≥18 who were citizens or permanent residents of Singapore, n=1144 (100%) Mean age 30.7 (5.1)	Television viewing per day *Categorical (< or ≥2 h/day)*	Education, household income, age, ethnicity, working during pregnancy, smoking history, parity during pregnancy, pre-pregnancy BMI	Probable depression symptoms in the past week (score ≥15 on Edinburgh Postnatal Depression Scale) Higher state (personality) anxiety symptoms (score ≥75th percentile on Spielberger State-Trait Anxiety Inventory (STAI)) Higher trait anxiety symptoms (score ≥75th percentile on STAI)	No association between TV and depression (OR 0.75 (95%CI 0.44, 1.30)), state anxiety (OR 1.21 (95%CI 0.86, 1.71)), or trait anxiety (OR 1.04 (95%CI 0.75, 1.45))
Rebar 2014 [33] Quality = fair	Cross-sectional Australia *Australian Health and Social Science panel project*	Random selection of households across each Australian state and territory	Australia adults, n=1843 (55%) Median age 58 (interquartile range 48-66)	Time spent sitting watching television in the past week (averaged for work and non-work days) *Continuous (h/day)*	Education, income, sex, age, PA, presence of chronic conditions	Depressive symptoms, anxiety symptoms, and stress symptoms in the past week each with a possible score ranging 0-21, measured by the Depression, Anxiety, and Stress Scale; higher score indicates more severe symptoms	No association between TV time and depression (*b*=0.00 (95%CI -0.03, 0.03)), anxiety (*b*=0.00 (95%CI -0.03, 0.03)), or stress (*b*=-0.02 (95%CI -0.04, 0.00))
Rosenberg 2016 [89] Quality = fair	Cross-sectional USA *Multilevel Intervention for Physical Activity in Retirement Communities*	Participants enrolled in a randomized-controlled trial to increase physical activity (baseline measures reported here)	Older adults living in 11 continuing care retirement communities in San Diego County, n=307 (72%) Mean age 83.6 (6.4)	Time typically spent sitting/lying while watching TV/DVDs (weighted for weekdays/weekend days) *Continuous (h/day)*	Education, age, gender, marital status, MVPA	Depressive symptoms in the past week: continuous score 0-30 based on CES-D scale Perceived stress in the past month: continuous score 0 to 16 based on	No association between TV time and depressive symptoms (β=-0.02 (SE=0.16), p=0.91) or perceived stress (β=-0.05 (SE=0.10), p=0.63)

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						Cohen Perceived Stress Scale	
Russell 2019 [99] Quality = good	Cross-sectional USA *National Health and Aging Trends Study*	Stratified three-stage sampling of Medicare beneficiaries	Medicare beneficiaries living in the continental USA, n=1687 (55%) Mean age 75.2 (0.1)	Time spent sitting or lying down to watch TV in the past seven days *Continuous (h/week)*	Education, age, gender, race/ethnicity	Subjective wellbeing, defined as frequency of feeling cheerful, bored, full of life, or upset in the past month (ranging from everyday=4 to never=0)	TV time was associated with lower subjective wellbeing scores (*b*= -0.01 (SE=0.00), p<0.001)
Sui 2015 [90] Quality = fair	Prospective (9.3-year follow-up) USA *Aerobics Center Longitudinal Study*	Patients who had been examined at the Cooper Clinic in Dallas	Adults who had attended the clinic, n=4802, (21.1%) Mean age at baseline 48.4 (9.8)	Average time viewing TV each week *Categorical (<5, 5-8.9, ≥9h/week) and continuous (h/week)*	Education, employment status, age, gender, marital status, smoking status, BMI, diabetes, MVPA	Depressive symptoms (≥8 on CES-D scale)	Significant association between TV time at baseline and likelihood of depression at follow-up (OR 1.52 (95%CI 1.19, 1.95)); when TV was used as a continuous variable, OR 1.016 (95%CI 1.004, 1.027))
Teychenne 2010 [91] Quality = good	Cross-sectional Australia *2007-08 READI study*	Randomly recruited from 80 Victorian low-SEP neighborhoods (40 rural, 40 urban)	Women living in low SEP Victorian neighborhoods aged 18-45, n=3645 (100%) Mean age 35 (no SD)	Time spent sitting watching TV on a usual day (weighted for weekday and weekend day) *Categorical (<13.7, 13.7-22.7, >22.7h/week)*	Education, income, employment status, age, BMI, marital status, physical health/injury, children living at home, country of birth	Depressive symptoms experienced in the past week (≥10 on CES-D scale)	No association between TV time and depression (OR 1.19 (95%CI 0.99, 1.43))
Teychenne 2016a [92] Quality = fair	Cross-sectional Australia InFANT Extend study (baseline data)	First-time mothers were recruited from first-time parent groups across Melbourne neighborhoods of varying SEP	First-time mothers who were approximately 3 months postpartum, n=406 (100%) Mean age 32 (4.25)	Time spent sitting watching TV/DVD/videos on a usual weekday and weekend day (weighted average) *Continuous (h/week)*	Education, age, country of birth, BMI, marital status, baby sleep time (hours per night), diet quality	Depressive symptoms experienced in the past week (CES-D scale continuous score, ranging 0-30)	No association between TV and depressive symptoms (*b*=-0.059 (95%CI -0.18, 0.06))
Teychenne 2016b [93] Quality = fair	Cross-sectional Australia *Mums, Dads and Kids Study*	Mothers of young children were recruited via preschool/childcare centers in six local government areas of Melbourne with varying levels of disadvantage, supplemented by online recruitment	Mothers of children aged 2 to 5, n=528 (100%) Mean age 37.2 (4.6)	Time spent watching TV/DVD/videos on a typical weekday and weekend day (weighted average) *Continuous (h/day)*	Education, age, BMI, marital status, having a disability or poor health, MVPA	Anxiety symptoms in the past week based on anxiety subscale of the Hospital Anxiety and Depression Scale (continuous variable ranging from 0 to 21)	No association between TV and anxiety symptoms (*b*=0.11 (95%CI -0.05, 0.27))

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Teychenne 2017 [94] Quality = fair	Prospective (3- and 5-year follow-up) Australia *2007-2013 READI study*	Randomly recruited from 80 Victorian low-SEP neighborhoods (40 rural, 40 urban)	Women living in low SEP Victorian neighborhoods aged 18-45 at baseline, n=1033 (100%) Mean age at baseline 36.6 (7.6)	Time spent sitting watching TV (weighted for weekday and weekend day) *Continuous (h/week)*	Education, employment status, age, BMI, marital status, children living at home, self-rated physical health	Depressive symptoms in the past week (CES-D scale continuous score, ranging from 0-30) Being 'at risk' of depression (≥10 on CES-D scale)	No association between TV time and change in depressive symptoms at 5-year follow-up (β=0.01 (95%CI -0.01, 0.02)) or odds of becoming at risk of depression at 3-year (OR 0.999 (95%CI 0.990, 1.008)) or 5-year (OR 0.999 (95%CI 0.991, 1.007)) follow-up
Teychenne 2018 [101] Quality = fair	Cross-sectional Australia *2012-2013 READI Study*	Random sub-sample of women who had taken part in the 2012-3 READI study wave, plus all women who had participated in the READI child sub-study were invited to take part	Women living in low SEP Victorian neighborhoods aged 18-45 when recruited in 2007-8, n=72 (100%) Mean age 43.5 (7.1)	Time spent sitting while watching TV in the past week (weighted for weekday and weekend day) *Continuous (h/week)*	Education, LTPA, and hair dyeing	Hair cortisol levels (ng/g) (indicative of stress levels over the past 1-3 months)	TV viewing was not associated with log-transformed hair cortisol levels (*b*= -0.002 (95%CI -0.008, 0.005))
Werneck 2018 [95] Quality = good	Cross-sectional Brazil *Brazilian National Health Survey 2013*	Multistage stratified cluster design	Adults (aged 18-100) living in Brazil, n=60,202 (57%) Overall mean age not reported; 47% young adults (18-39)	Hours a day usually spent watching TV *Categorical (<2, 2-3.99, ≥4h/day)*	Education, employment status, age, alcohol consumption, smoking, BMI, physical activity	Depression risk (score >9 on Patient Health Questionnaire with possible score range 0 to 27) based on past 2 weeks Feelings of sadness and suicidal thoughts on at least several days over the past two weeks (singular questions extracted from Patient Health Questionnaire), binary	Among men, compared to those who watched <2h/day, those who watched ≥4h/day had higher risk of depression (OR 1.34 (95%CI 1.03, 1.73)) and feelings of sadness (OR 1.41 (95%CI 1.18, 1.67)) but not higher risk of suicidal thoughts (OR 1.39 (95%CI 0.95, 2.02)). Among women, compared to those who watched <2h/day, those who watched ≥4h/day did not have significantly higher likelihood of depression (OR 1.09 (95%CI 0.93, 1.27)), feelings of sadness (OR 1.00 (95%CI 0.89, 1.12)), or suicidal thoughts (OR 1.17 (95%CI 0.95, 1.44)).
Yu 2019 [97]	Cross-sectional China	Adults who had received physical	Adults in Tianjin, n=18,994 (46%)	Hours per day spent watching TV	Education, income, occupation, age, sex, BMI,	Depressive symptoms (score ≥45 on the	Higher TV time was associated with higher

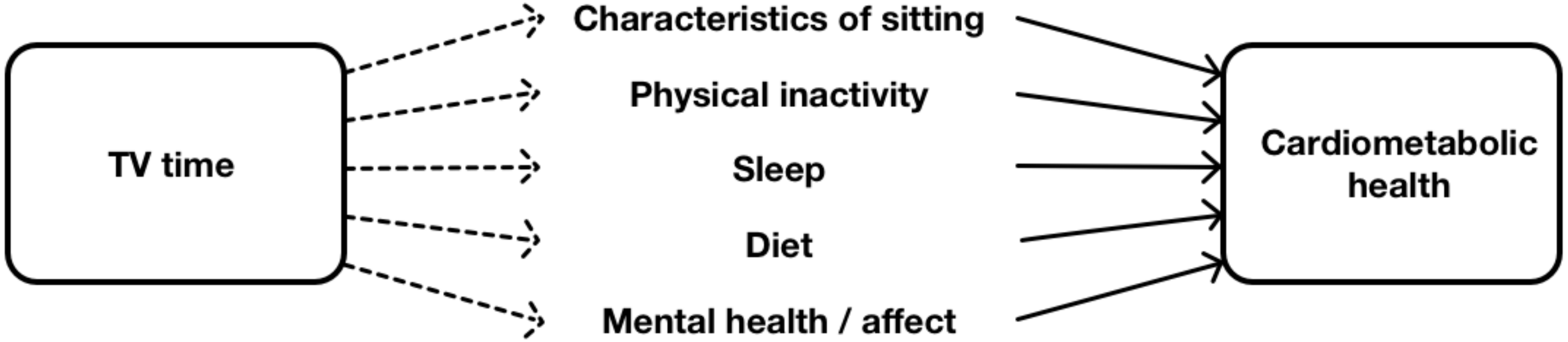
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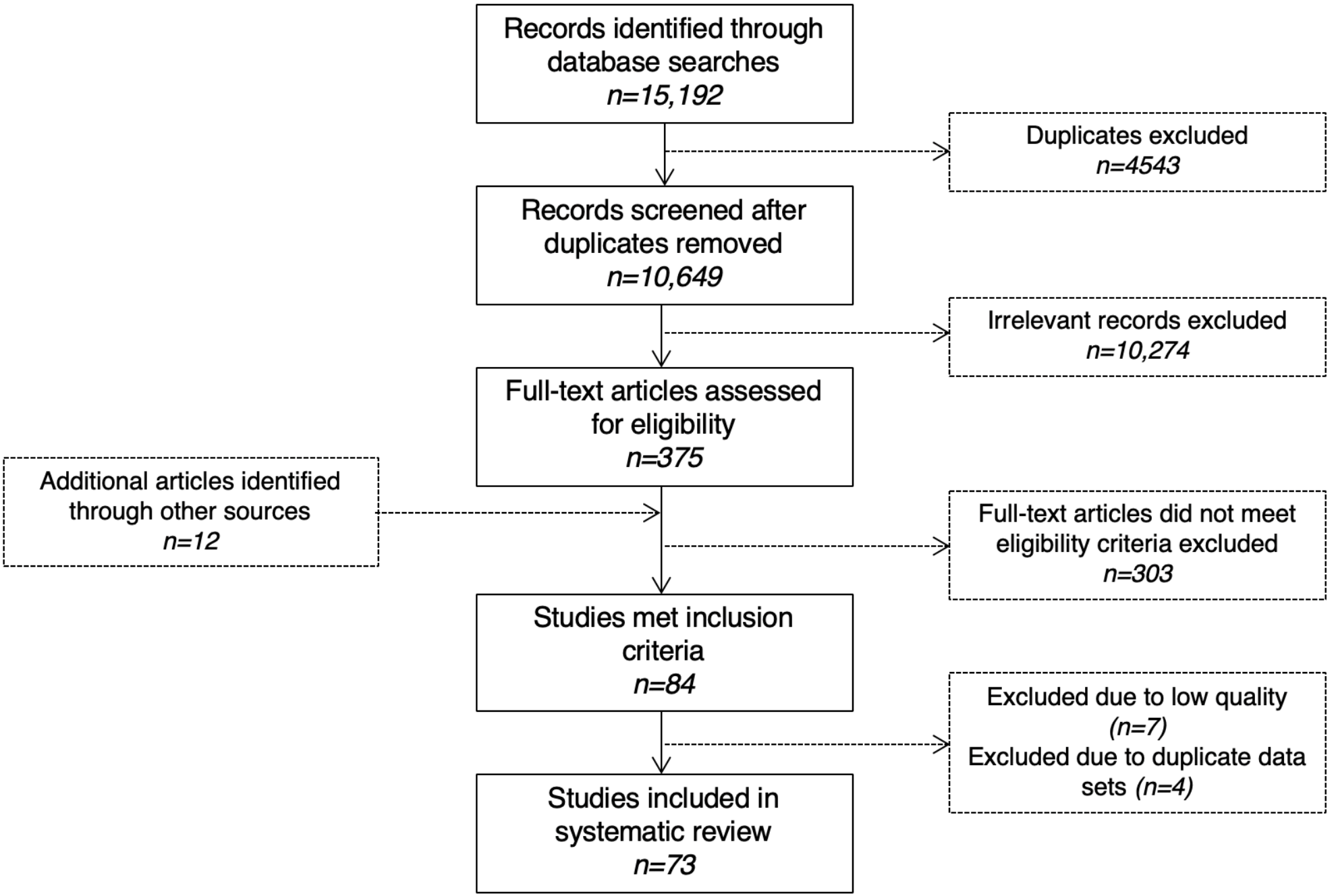
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Quality = good	*Tianjin Chronic Low-grade Systemic Inflammation and Health Cohort*	examinations at Tianjin Medical University General Hospital Health Management Center were recruited	Mean age 42.2 (11.9)	*Categorical (<1, 1-3, 3-5, 5-10, >10h/day)*	waist circumference, marital status, cohabitants, social contact, physical activity, sitting time, sleep duration, total energy intake, metabolic syndrome, smoking and drinking, internet addiction, dietary patterns, computer/mobile device time	Chinese version of the Self-Rating Depression Scale	likelihood of depressive symptoms; OR 1.95 (95%CI 1.10, 3.35) for those who watched >10h/day versus those who watched <1h/day; trend was significant ($p<0.0001$)

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TV time
Characteristics of sitting
Physical inactivity
Sleep
Diet
Mental health / affect
Cardiometabolic health


Records identified through database searches n=15,192
Duplicates excluded n=4543
Records screened after duplicates removed n=10,649
Irrelevant records excluded n=10,274
Full-text articles assessed for eligibility n=375
Additional articles identified through other sources n=12
Full-text articles did not meet eligibility criteria excluded n=303
Studies met inclusion criteria n=84
Excluded due to low quality (n=7)
Excluded due to duplicate data sets (n=4)
Studies included in systematic review n=73

Study	Study design (n)	Outcome measure	Association		
			Overall	Women	Men
Cross-sectional studies					
Prolonged sitting					
Shibata 2019	Cross-sectional (n=287)	ST accumulated in prolonged bouts (≥30min)	▲+		
		Number of hourly breaks in ST	▲-		
Wagnild 2020	Cross-sectional (n=167)	>137.1 minutes of ST accumulated in prolonged bouts (≥30min)		□+	
		Number of daily breaks in ST		▲-	
Time of day					
Wagnild 2020	Cross-sectional (n=167)	Total ST in the evenings (6-11pm)		▲+	
		Prolonged ST in the evenings		□+	
		Breaks in ST in the evenings		△-	
Kim 2019a	Cross-sectional (n=27)	Time spent watching TV in the evening vs the morning	■+ ■+ ■+ ■+		
		Time spent watching TV in the evening vs the afternoon	■+ ■+ ■+ ■+		
Experimental studies					
Energy expenditure (EE)					
Mansoubi 2015	Within subjects (n=51)	EE TV vs typing	■- ■-		
		EE TV vs handheld game (PSP)	■- ■-		
		EE TV vs seated video game (Wii)	■- ■- ■- ■-		
Newton 2013	Within subjects (n=25)	EE TV vs typing	□-		
		EE TV vs reading	□-		
Creasy 2016	Between subjects (n=36)	EE TV vs computer	□- □-		
Level of movement					
van der Berg 2019	Within subjects (n=18)	Acceleration TV vs reading	■+ ■+ ■+		
		Acceleration TV vs writing	■+ ■+ ■+ ■+		
		Acceleration TV vs typing	■+ ■+ ■+ ■+		
		Acceleration TV vs playing cards	□0		
		Acceleration TV vs preparing food while seated	□+		
		Acceleration TV vs 'natural' sitting	■+ ■+ ■+		
		Acceleration TV vs sitting motionless	■+ ■+ ■+ ■+		

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Study	Study design (n)	Outcome measure	Association		
			Overall	Women	Men
Cross-sectional studies					
Continuous TV					
Burton 2012	Cross-sectional (n=9121)	LTPA + walking for transport		-	-
Bennett 2006	Cross-sectional (n=486)	≥10,000 pedometer steps per day	-		
Categorical TV					
Shields 2008	Cross-sectional (n=42612)	LTPA + walking/cycling for transport		- -	-
Sugiyama 2008	Cross-sectional (n=2046)	LTPA		- -	-
Bertrais 2004	Cross-sectional (n=7404)	LTPA		-	+
Tucker 2015	Cross-sectional (n=302)	Cardiorespiratory fitness (VO_2 max)		- -	
Prospective studies					
Continuous TV					
Kim 2019b	Prospective (n=91648)	Change in average acceleration (mg)		▲-	▲-
Categorical TV					
Lakerveld 2011	Prospective (n=4841)	LTPA + walking for transport		-	-

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Study	Study design (n)	Outcome measure	Association
Continuous TV			
Buman 2015	Cross-sectional (n=843)	Short sleep duration (<7h)	+
		Long sleep duration (>8.5h)	+
		Poor sleep quality	+
		Excessive daytime sleepiness	+
		Long sleep onset latency (≥30min)	+
		Waking up in the night	+
		Waking up too early	+
Kakinami 2017	Cross-sectional (n=658)	Short sleep duration (<7h)	+
		Long sleep duration (>9h)	+
		Poor sleep quality	+
Categorical TV			
Adrianasolo 2016	Cross-sectional (n=4179)	Excessive daytime sleepiness	-
de Souza 2017	Cross-sectional (n=959)	Poor sleep quality	+
Custers 2012	Cross-sectional (n=711)	Bedtime (weekends)	+
		Bedtime (weeknights)	+/-
		Rising time	+/-
		Sleep window	+/-
		Overall tiredness	+/-

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Study	Study design (n)	Outcome measure	Association		
			Overall	Women	Men
Continuous TV					
Compernolle 2016	Cross-sectional (n=6001)	Fruit (≥7 portions/week)	-		
		Vegetables (≥7 portions/week)	-		
		Sugar-sweetened beverages (≥2/week)	+		
		Fast food (>0.5 portions/week)	+		
		Alcohol (>2/week)	+		
		Sweets (≥3portions/week)	+		
Robinson 2004	Cross-sectional (n=6125)	Prudent diet score (see table)		-	
Thomson 2008	Cross-sectional (n=613)	Frequency of energy-dense snack consumption	+ +		
		Frequency of healthy snack consumption	-		
Categorical TV					
Shields 2008	Cross-sectional (n=42612)	Low fruit/veg consumption (<3/day)		+ +	+ +
Bowman 2006	Cross-sectional (n=9157)	Total kcal/day	+		
		Dinner kcal/day	+		
		Snacks kcal/day	+		
		Added sugars (%kcal)	+ +		
		Dietary fibre (per 1000kcal)	-		
		Regular soft drinks (g)	+		
		Energy-rich snack-type foods (g)	+		
		Breakfast kcal/day	0		
		Lunch kcal/day	+		
		Total fat (%kcal)	-		
		Saturated fat (%kcal)	+		
		Carbohydrate (%kcal)	+		
		Protein (%kcal)	0		
Scully 2009	Cross-sectional (n=1495)	Fast food for breakfast (≥once/week)	+		
		Fast food for lunch (≥once/week)	+		
		Fast food for dinner (≥once/week)	+		
		Fast food for snacks (≥once/week)	+ +		
Rehm 2008	Cross-sectional (n=9916)	Sugar-sweetened soda (≥1/day)	+ +		
Chareire 2011	Cross-sectional (n=1359)	Sex-specific dietary patterns (see table)		▲	▲

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Study	Method	Variable of interest	While watching TV compared to	Finding
Gemming 2015	**Wearable cameras** worn for three days linked with 24-hour diet recalls on those days (n=40)	Snack energy intake (kJ)	At home with no TV	▲+
		Meal energy intake (kJ)	At home with no TV	△+
Ghosh Roy 2018	**Ecological momentary assessment** (5 prompts per day) to ascertain the context of any snack or beverage consumption that occurred since the previous prompt (n=79)	Snacks consumed (%)	While doing nothing else	■+ ■+
		Sugar-sweetened beverages consumed (%)	While doing nothing else	□-
Laska 2011	**7-day food diary** with contextual information provided for each instance of eating or drinking (n=48)	Sugar-sweetened beverages consumed (%)	While doing nothing else	▲+
		Water consumed (%)	While doing nothing else	▲-
		Fruit and vegetables consumed (%)	While doing nothing else	▲-
		Cereals and grains consumed (%)	While doing nothing else	▲-
		Cookies and sweetened baked goods consumed (%)	While doing nothing else	△-
		Candy/gummy fruit snacks consumed (%)	While doing nothing else	△0
		Frozen desserts consumed (%)	While doing nothing else	△0
		Salty snacks consumed (%)	While doing nothing else	△+
		Fried side dishes consumed (%)	While doing nothing else	△-
Stroebele 2004	**7-day food diary** in which participants specified whether the TV was on in each instance (n=78)	Meal size (kJ)	Without the TV on	△?

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Study	Study design (n)	Outcome measure	Association	
			Overall	Women
TV vs doing nothing				
Hetherington 2006	Within subjects (n=37)	Meal intake (kJ)	+ +	
Martin 2009	Within subjects (n=48)	Meal intake (kcal)	-	
Rosenthal 2017	Within subjects (n=20)	Meal intake (kcal, g)	+	
Moray 2007	Within subjects (n=20)	Meal intake (ounces)	+	
Braude 2014	Within subjects (n=62)	Snack intake (kJ)		+
Bellisle 2004	Within subjects (n=48)	Meal intake (kJ)		+ +
Bellisle 2009	Within subjects (n=40)	Meal intake (kJ, g)		?
Ogden 2013	Between subjects (n=81)	Snack intake (g)		+ +
Mittal 2010	Between subjects (n=32)	Snack intake (kJ)		+ + +
TV vs other condition				
Hetherington 2006	Within subjects (n=37)	With friends (kJ)	-	
Hetherington 2006	Within subjects (n=37)	With strangers (kJ)	+	
Martin 2009	Within subjects (n=48)	While reading (kcal)	+	
Bellisle 2004	Within subjects (n=48)	Listening to audio story (kJ)		-
Bellisle 2009	Within subjects (n=40)	Listening to audio story (kJ, g)		?
Bellisle 2009	Within subjects (n=40)	With strangers (kJ, g)		+
Blass 2006	Within subjects (n=20)	Listening to music (kcal)	+	
Lyons 2012	Between subjects (n=120)	With handheld video games (kcal)	-	
		With motion-controlled video games (kcal)	- -	
Ogden 2013	Between subjects (n=81)	While driving (g)		+ + +
Ogden 2013	Between subjects (n=81)	While talking to researcher (g)		+ + +

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Study	Study design (n)	Outcome measure	Association		
			Overall	Women	Men
Cross-sectional studies					
Depression and depressive symptoms					
Continuous TV					
Rebar 2014	Cross-sectional (n=1843)	Depressive symptoms	△0		
Rosenberg 2016	Cross-sectional (n=307)	Depressive symptoms	△		
Teychenne 2016a	Cross-sectional (n=406)	Depressive symptoms		△-	
Categorical TV					
Werneck 2018	Cross-sectional (n=60,202)	Depressive symptoms		+	+
Yu 2019	Cross-sectional (n=18,994)	Depressive symptoms	+ +		
Teychenne 2010	Cross-sectional (n=3645)	Depressive symptoms		+	
Parmapriya 2016	Cross-sectional (n=1144)	Depressive symptoms		-	
Stress and anxiety					
Continuous TV					
Jackson 2019	Cross-sectional (n=3555)	Hair cortisol	▲		
Rebar 2016	Cross-sectional (n=1843)	Stress symptoms	△		
		Anxiety symptoms	△0		
Rosenberg 2016	Cross-sectional (n=307)	Perceived stress	△		
Teychenne 2016b	Cross-sectional (n=528)	Anxiety symptoms		△+	
Nabi 2016	Cross-sectional (n=240)	Salivary cortisol	△	▲	△+
		Perceived stress	△+		
Teychenne 2018	Cross-sectional (n=72)	Hair cortisol		△-	
Categorical TV					
Parmapriya 2016	Cross-sectional (n=1144)	State anxiety		+	
		Trait anxiety		+	
Other mental health outcomes					
Continuous TV					
Dempsey 2014	Cross-sectional (n=9907)	Mental wellbeing	▲		
Russell 2019	Cross-sectional (n=1687)	Subjective wellbeing	▲		
Guillen-Royo 2019	Cross-sectional (n=500)	Happiness	△+		
		Life satisfaction	△		
		Subjective vitality	△+		
Goodwin 2005	Cross-sectional (n=69)	*Positive affect*			
		TV vs other leisure	- -		
		TV vs productive activities	- -		
		TV vs maintenance activities	-		
		Negative affect			
		TV vs other leisure	0		
		TV vs productive activities	-		
		TV vs maintenance activities	0		
Categorical TV					
Werneck 2018	Cross-sectional (n=60,202)	Feelings of sadness		0	+
		Suicidal thoughts		+	+
Prospective studies					
Depression and depressive symptoms					
Continuous TV					
Teychenne 2017	Prospective (n=1033)	Incident depression		-	
		Depressive symptoms		△-	
Sui 2015	Prospective (n=4802)	Depressive symptoms	+		
Categorical TV					
Lucas 2011	Prospective (n=49,821)	Incident clinical depression		+	
Hamer 2014	Prospective (n=6359)	Depressive symptoms	△		
Andrade-Gomez 2018	Prospective (n=1461)	Depressive symptoms		▲	△+
Other mental health outcomes					
Categorical TV					
Andrade-Gomez 2018	Prospective (n=1766)	Psychological distress	△		
Bekhtaroglu 2019	Prospective (n=1668)	Positive affect score (lagged model)	△		

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Additional file 1. Effect size calculations, with sensitivity analyses conducted for studies that used within-subjects or repeated-measures designs.

Study	Condition comparison or outcome variable	Standardized mean difference (95% CI)	Notes
Characteristics of sitting			
Creasy 2016 [43]	Energy expenditure (kcal/15min) TV vs computer	-0.23 (-0.88, 0.43)	
Kim 2019a [42]	Time spent watching TV in the evening vs the morning	1.53 (0.98, 2.09)	Imputed correlation 0.5
		1.70 (1.11, 2.29)	Imputed correlation 0.6
		2.29 (1.57, 3.01)	Imputed correlation 0.8
	Time spent watching TV in the evening vs the afternoon	1.08 (0.60, 1.55)	Imputed correlation 0.5
		1.20 (0.71, 1.70)	Imputed correlation 0.6
		1.70 (1.11, 2.29)	Imputed correlation 0.8
	Time spent watching TV vs time spent using other electronic devices (non-work related) in the evening	1.10 (0.62, 1.58)	Imputed correlation 0.5
		1.22 (0.72, 1.72)	Imputed correlation 0.6
		1.67 (1.08, 2.25)	Imputed correlation 0.8
	Time spent watching TV vs time spent in other non-work related sedentary behavior in the evening	0.38 (-0.01, 0.77)	Imputed correlation 0.5
		0.42 (0.03, 0.81)	Imputed correlation 0.6
		0.59 (0.18, 1.00)	Imputed correlation 0.8
	Time spent watching TV vs time spent in work-related screen-based sedentary behavior in the evening	1.36 (0.83, 1.88)	Imputed correlation 0.5
		1.51 (0.96, 2.06)	Imputed correlation 0.6
		2.08 (1.41, 2.75)	Imputed correlation 0.8
	Time spent watching TV vs time spent in work-related non-screen-based sedentary behavior in the evening	1.47 (0.92, 2.01)	Imputed correlation 0.5
		1.63 (1.05, 2.21)	Imputed correlation 0.6
		2.23 (1.53, 2.94)	Imputed correlation 0.8
	Time spent watching TV vs time spent in screen-based sedentary behavior during transport in the evening	2.07 (1.40, 2.74)	Imputed correlation 0.5
		2.12 (1.44, 2.80)	Imputed correlation 0.6
		2.22 (1.52, 2.93)	Imputed correlation 0.8
	Time spent watching TV vs time spent in non-screen-based sedentary behavior during transport in the evening	1.88 (1.25, 2.50)	Imputed correlation 0.5
		2.01 (1.35, 2.67)	Imputed correlation 0.6
		2.39 (1.65, 3.13)	Imputed correlation 0.8
Mansoubi 2015 [45]	Energy expenditure (METs) TV vs typing	-0.41 (-0.70, -0.12)	Imputed correlation 0.5, total sample
		-0.45 (-0.74, -0.17)	Imputed correlation 0.6, total sample
		-0.61 (-0.91, -0.31)	Imputed correlation 0.8, total sample
	Energy expenditure (METs) TV vs PSP	-0.30 (-0.58, -0.02)	Imputed correlation 0.5, total sample
		-0.33 (-0.62, -0.05)	Imputed correlation 0.6, total sample
		-0.47 (-0.76, -0.18)	Imputed correlation 0.8, total sample
	Energy expenditure (METs) TV vs seated Wii	-1.66 (-2.08, -1.24)	Imputed correlation 0.5, total sample
		-1.78 (-2.22, -1.33)	Imputed correlation 0.6, total sample
		-2.11 (-2.61, -1.62)	Imputed correlation 0.8, total sample
Newton 2013 [44]	Energy expenditure (METs) TV vs typing	-0.19 (-0.59, 0.20)	Imputed correlation 0.5
		-0.22 (-0.61, 0.18)	Imputed correlation 0.6

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		-0.31 (-0.71, 0.09)	Imputed correlation 0.8
	Energy expenditure (METs) TV vs reading	-0.05 (-0.45, 0.34)	Imputed correlation 0.5
		-0.06 (-0.45, 0.33)	Imputed correlation 0.6
		-0.08 (-0.47, 0.31)	Imputed correlation 0.8
van der Berg 2019 [46]	Vector magnitude (cpm) TV vs reading	0.79 (0.26, 1.32)	Imputed correlation 0.5
		0.84 (0.30, 1.37)	Imputed correlation 0.6
		0.96 (0.40, 1.52)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs writing	0.97 (0.41, 1.53)	Imputed correlation 0.5
		0.99 (0.43, 1.55)	Imputed correlation 0.6
		1.03 (0.46, 1.60)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs typing	0.92 (0.37, 1.47)	Imputed correlation 0.5
		0.95 (0.40, 1.51)	Imputed correlation 0.6
		1.04 (0.47, 1.62)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs playing cards	0.00 (-0.46, 0.46)	Imputed correlation 0.5
		0.00 (-0.46, 0.46)	Imputed correlation 0.6
		0.01 (-0.46, 0.47)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs preparing food while seated	0.14 (-0.32, 0.61)	Imputed correlation 0.5
		0.16 (-0.31, 0.62)	Imputed correlation 0.6
		0.22 (-0.24, 0.69)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs natural sitting	0.71 (0.19, 1.22)	Imputed correlation 0.5
		0.78 (0.25, 1.31)	Imputed correlation 0.6
		1.02 (0.45, 1.59)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs imposed sitting	0.94 (0.39,1.50)	Imputed correlation 0.5
		0.97 (0.41, 1.52)	Imputed correlation 0.6
		1.01 (0.44, 1.58)	Imputed correlation 0.8
Physical inactivity			
Sugiyama 2008 [34]	LTPA (min/day)	-0.28 (-0.42, -0.14)	Women
Sugiyama 2008 [34]	LTPA (min/day)	-0.11 (-0.29, 0.07)	Men
Tucker 2015 [50]	VO_2 max	-0.20 (-0.50, 0.10)	Fully adjusted model
Diet			
Bowman 2006 [32]	Total kcal/day	0.07 (0.01, 0.13)	
	Breakfast kcal/day	0.00 (-0.06, 0.06)	
	Lunch kcal/day	0.05 (-0.01, 0.11)	
	Snacks kcal/day	0.09 (0.03, 0.15)	
	Dinner kcal/day	0.06 (0.00, 0.12)	
	Soft drink consumption (g)	0.12 (0.06, 0.18)	
	Energy-rich snack foods (g)	0.10 (0.04, 0.16)	
	Added sugars (%kcal)	0.20 (0.14, 0.26)	
	Fiber intake	-0.13 (-0.19, -0.07)	
	Total fat (%kcal)	-0.01 (-0.07, 0.05)	

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	Saturated fat (%kcal)	0.01 (-0.05, 0.07)	
	Carbohydrates (%kcal)	0.03 (-0.03, 0.09)	
	Protein (%kcal)	0.00 (-0.06, 0.06)	
Experimental diet			
Bellisle 2004 [66]	Food intake (kJ), TV vs control	0.46 (0.16, 0.76)	Imputed correlation 0.5
		0.51 (0.21, 0.82)	Imputed correlation 0.6
		0.73 (0.41, 1.05)	Imputed correlation 0.8
Hetherington 2006 [68]	Food intake (kJ), TV vs control	0.34 (0.01, 0.67)	Imputed correlation 0.5
		0.38 (0.05, 0.71)	Imputed correlation 0.6
		0.52 (0.18, 0.86	Imputed correlation 0.8
Mittal 2011, experiment 1 [70]	Food intake (kJ), TV vs control	0.59 (-0.13, 1.29)	Snack consumed while watching TV
	Food intake (kJ), TV vs control	0.51 (-0.19, 1.22)	Meal consumed after TV/no-TV condition
Ogden 2013 [73]	Food intake (g), TV vs control	0.45 (-0.17, 1.07)	
Bellisle 2004 [66]	Food intake (kJ), TV vs audio story	-0.04 (-0.32, 0.25)	Imputed correlation 0.5
		-0.04 (-0.32, 0.24)	Imputed correlation 0.6
		-0.06 (-0.34, 0.23)	Imputed correlation 0.8
Hetherington 2006 [68]	Food intake (kJ), TV vs eating with friends	-0.13 (-0.46, 0.19)	Imputed correlation 0.5
		-0.15 (-0.47, 0.18)	Imputed correlation 0.6
		-0.21 (-0.53, 0.12)	Imputed correlation 0.8
Lyons 2012 [76]	Food intake (kcal), TV vs handheld video game	-0.06 (-0.50, 0.37)	Total intake
	Food intake (kcal), TV vs motion-controlled video game	0.36 (-0.09, 0.80)	Total intake
Ogden 2013 [73]	Food intake (g), TV vs driving	0.77 (0.13, 1.40)	
	Food intake (g), TV vs talking to researcher	0.73 (0.08, 1.37)	
Mental health			
Goodwin 2005 [40]	Positive affect, TV vs other leisure	-0.25 (-0.49, -0.01)	Imputed correlation 0.5
		-0.28 (-0.52, -0.04)	Imputed correlation 0.6
		-0.39 (-0.63, -0.14)	Imputed correlation 0.8
	Positive affect, TV vs productive activities	-0.39 (-0.64, -0.15)	Imputed correlation 0.5
		-0.44 (-0.69, -0.19)	Imputed correlation 0.6
		-0.62 (-0.88, -0.36)	Imputed correlation 0.8
	Positive affect, TV vs maintenance activities	-0.01 (-0.25, 0.22)	Imputed correlation 0.5
		-0.01 (-0.25, 0.22)	Imputed correlation 0.6
		-0.02 (-0.25, 0.22)	Imputed correlation 0.8
	Negative affect, TV vs other leisure	0.00 (-0.24, 0.24)	Imputed correlation 0.5
		0.00 (-0.24, 0.24)	Imputed correlation 0.6
		0.00 (-0.24, 0.24)	Imputed correlation 0.8
	Negative affect, TV vs productive activities	-0.08 (-0.31, 0.16)	Imputed correlation 0.5
		-0.09 (-0.32, 0.15)	Imputed correlation 0.6
		-0.12 (-0.36, 0.12)	Imputed correlation 0.8
	Negative affect, TV vs maintenance activities	0.00 (-0.23, 0.24)	Imputed correlation 0.5

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		0.00 (-0.23, 0.24)	Imputed correlation 0.6
		0.01 (-0.23, 0.24)	Imputed correlation 0.8

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Additional file 2. Summary of characteristics, measurements, and findings of included studies examining associations between TV and characteristics of sitting, physical inactivity, and sleep stratified by outcome category

First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model (SEP listed first where applicable)**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
Characteristics of sitting							
Creasy 2016 [43] Quality = good	Experimental (between subjects) USA	Recruited at university (convenience)	Adults aged 18 to 40 with BMI between 20-35, n=74 (58%)* Mean age 24.2 (3.6) *In full study. Only those whose baseline conditions were sitting while watching TV or sitting at the computer (n=18 in each condition) are described here	Lab-based exposure (15 minutes) *Experimental aim: to examine the cumulative energy expenditure of various combinations of sitting, standing, and walking*	N/A	Energy expenditure (kcal per 15min, indirect calorimetry)	TV EE (18.66 (4.01) kcal/15min) not significantly* different from computer EE (19.63 (4.37) kcal/15min) *Based on confidence intervals crossing zero in our calculation of the standardized mean difference between the two groups; the paper itself did not test this (see Additional file 1)
Kim 2019a [42] Quality = good	Cross-sectional USA	Convenience sampling	Adults aged ≥18 without physical or medical disabilities, n=27 (44%) Mean age 33.3 (11.4), range 20-64	All sedentary behaviors (including TV) were self-recorded in 10-minute blocks over 4 consecutive days while wearing a wearable camera (criterion measure)	N/A	Amount of time spent watching TV at different times of day (self-report)	Self-reported TV time was significantly higher in the evening (6pm to midnight; 78.8 (38.3) minutes) compared to the afternoon (12 to 6pm; 37.4 (36.2) minutes) and morning (midnight to midday; 24.3 (28.7) minutes))
Mansoubi 2015 [45] Quality = good	Experimental (within subjects) UK	Recruited at university and in community in 2x2 format to obtain equal numbers of men/women and healthy weight/obese participants (convenience)	Adults aged ≥18, 47% of whom were obese, n=51 (51%) Overall mean age not reported	Lab-based exposure (10 minutes) *Experimental aim: to measure the energy expenditure of sedentary behaviours in normal weight and obese participants*	N/A	Energy expenditure (METs, indirect calorimetry)	TV EE (METs) (1.33 (0.24)) was significantly* lower than typing (1.45 (0.32)), playing a handheld computer game (1.41 (0.28)), and playing a screen-based computer game (2.06 (0.50)) * Based on confidence intervals crossing zero in our calculation of the standardized mean change between the conditions; the paper

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							itself did not test this (see Additional file 1)
Newton 2013 [44] Quality = fair	Experimental (within subjects) USA	Recruited in community (convenience)	African American adults aged ≥18, n=25 (60%) Mean age 38.2 (11.4), range 20-56	Lab-based exposure (30 minutes) *Experimental aim: to examine the energy expenditure of common sedentary behaviours*	N/A	Energy expenditure (METs, whole-room calorimetry)	TV EE (METs) (1.03 (0.15)) was not significantly different from EE of typing (1.06 (0.15)), or reading at a desk (1.04 (0.20))
Shibata 2019 [41] Quality = good	Cross-sectional Japan	Random sampling from paper-based registry of residential addresses, stratified by age and gender	Older adults aged 65-84, n=287 (38%) Mean age 74.5 (5.2)	Average time spent watching TV on weekdays and weekend days in the past 7 days (completed when accelerometer was fitted) *Continuous (min/day)*	Educational status, gender, age, BMI, marital status, number of chronic conditions, accelerometer wear time	Time spent in accelerometer-measured* prolonged sedentary time (≥30 minute bouts; minutes per day) Number of breaks in sedentary time per hour (number per day) * Waist-worn accelerometer with waking wear protocol (valid data set: ≥10h on ≥4 days including a weekend day)	TV time was associated with more time spent in prolonged sedentary time (*b*=2.93 (95%CI 2.09, 3.76)) and with fewer hourly breaks in sedentary time per day (*b*=-0.06 (95%CI -0.08, -0.04))
van der Berg 2019 [46] Quality = fair	Experimental (within subjects) Netherlands	Recruited from the university (convenience)	Adults aged 18 to 48 without mobility limitations, n=18 (44%) Mean age 27.3 (6.5)	Lab-based exposure (7 minutes) seated in a chair without armrests while watching TV on a computer screen *Experimental aim: to identify whether 'dynamic sitting (e.g., fidgeting) can be identified from triaxial accelerometry counts*	N/A	Accelerometer vector magnitude* while watching TV compared to natural sitting (allowed to move freely), imposed sitting (motionless), imposed fidgeting (told to move the upper body side to side and back to front), while reading a newspaper, while writing a letter, while playing cards, while typing on a computer,	Accelerometer vector magnitude was significantly* higher while watching TV compared to natural and 'motionless' sitting, reading, writing, and typing; it was significantly lower while watching TV compared to imposed fidgeting. There was no difference in vector magnitude between TV and playing cards or between TV and preparing food while seated. **Based on confidence intervals crossing zero in our calculation of*

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						and while preparing food while seated *Paper reported mean acceleration in the vertical, anteroposterior, and mediolateral planes separately; we converted this to mean vector magnitude by taking the square root of the summed squares of all three axes	*the standardized mean difference between the two groups; the paper itself did not test this (see Additional file 1)*
Wagnild 2020 [27] Quality = fair	Cross-sectional UK	Pregnant women recruited from two NHS (public) antenatal clinics	Pregnant women at 20 weeks' gestation classed as at risk of gestational diabetes, n=167 (100%) Mean age 31 (5)	Usual TV time during the second trimester *Categorical (<2, ≥2h/day)*	Age, BMI, children at home, marital status, smoking status, waking time, recruitment site	activPAL-measured* prolonged sedentary time (time spent in bouts of sedentary time lasting ≥30minutes) dichotomized at the median: >137.1min/day for all waking hours, >53.9min/day for evening hours (6-11pm) at 20 weeks' gestation activPAL-measured breaks in sedentary time (number per day) for all waking hours and evening hours only * Thigh-worn accelerometer with 24-hour wear protocol (valid data set: ≥4 days of ≥24 hour wear)	Those with high TV time had significantly fewer breaks in sedentary time across all waking hours (exp(*b*) 0.92 (95%CI 0.86, 0.998)); there were no differences in likelihood of high prolonged sedentary time (OR 1.20 (95%CI 0.83, 1.73)) Those with high TV time had significantly higher sedentary time in the evening hours (*b*= 9.86 (95%CI 0.50, 19.20)). There were no differences in high prolonged sedentary time (OR 1.15 (95%CI 0.82, 1.63)) or breaks in sedentary time (exp(*b*)= 0.95 (95%CI 0.87, 1.04)) in the evening hours.
Physical inactivity							
Bennett 2006 [49] Quality = good	Cross-sectional USA *Open Doors to Health study*	Random and census sampling within low-income housing communities	Adults aged ≥18 living in low-income housing communities in Boston, n=486 (64%)	Watching TV on an average weekday and average weekend day (weighted average) *Continuous (h/day)*	Education, employment status, age, gender, BMI, race/ethnicity	Pedometer-measured steps per day (7-day wear period)	Associated with fewer steps per day (*b*= -144 (95%CI -276, -12)) and lower likelihood of achieving ≥10,000 steps per day (OR 0.84 (95%CI 0.71, 0.99))

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			No mean age or range provided				
Bertrais 2004 [47] Quality = good	Cross-sectional France *SUVIMAX*	Volunteer sample enrolled in RCT testing antioxidant supplementation in prevention of heart disease and cancer	French adults aged 35-60 (women) and 45-60 (men), n=7404 (54%) Mean age for men 55.4 (4.7), women 53.2 (5.3)	Watching TV 'in general' *Categorical (<1, 1-2, 2-3, ≥3h/day)*	Education, age, smoking status, urban/rural location	Meeting LTPA guidelines (≥150min or ≥60min/week of moderate or vigorous LTPA, respectively) based on Modifiable Activity Questionnaire (over past 12 months)	No association for men (OR 1.15 (95%CI 0.87, 1.51)) or women (OR 0.88 (95%CI 0.69, 1.12))
Burton 2012 [48] Quality = good	Cross-sectional Australia *2007 HABITAT study*	Multistage probability sampling	Middle-aged Australian adults, n=9121 (55%) Age range 40-64	Sitting on a usual weekday and weekend day watching TV (including DVDs, videos, video games) *Continuous (h/day)*	Education, age, current health status, BMI, smoking status	LTPA + walking for transport: no activity (<90MET.min/week), low (90 to <450), recommended (450 to <900), high (900 to <1800), very high (≥1800 MET.min/week) (previous week)	Weekend TV time was associated with lower likelihood of high PA level among men (OR 0.91 (95%CI 0.83,0.98)); no other associations significant
Kim 2019b [51] Quality = good	Prospective (median follow-up 5.7 years) UK *UK Biobank*	Adults aged 40-69 who were registered with the National Health Service and lived within <25 miles of one of 22 assessment centers across the UK were invited to take part	Adults in the UK aged 40-69, n=91,648 (56%) Mean ages 55.6 (7.7) and 56.8 (7.9) for women and men, respectively	TV viewing on a typical day *Continuous (h/day)*	Age, BMI, ethnicity, smoking status, employment status, duration between baseline and accelerometry, wear time, season, severe medical conditions, grip strength, time spent in other sedentary behaviours and physical activities	Accelerometer-measured* average acceleration (milli-*g*) * Wrist-worn accelerometer with 24-hour wear protocol (valid data set: ≥72 hours of wear)	TV time at baseline was associated with a significant decrease in log-transformed average acceleration at follow-up in both women (*b*= -0.019 (95%CI -0.021, -0.018)) and men (b=-0.021 (95%CI -0.023, -0.019))
Lakerveld 2011 [52] Quality = good	Prospective Australia *AusDiab (1999-2000, 2004-2005)*	Stratified cluster sampling	Australian adults ≥25, n=4841 (55%) Overall mean or age range not provided	Total time spent watching TV/videos in past week *Categorical (<2, 2-4, ≥4h/day)*	Education, household income, employment status, age,	LTPA + walking for transport: reductions from sufficient (≥150min/week) to insufficient/no PA, or	Women in the highest TV group at baseline (≥4h/day) had higher likelihood of reducing LTPA at follow-up (OR 1.46 (95%CI 1.01,

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					baseline PA level, baseline waist circumference	insufficient to no PA at 5-year follow-up ('insufficient' defined as >0 but <150min/week) (in the previous week)	2.11)); no other associations significant
Shields 2008 [35] Quality = good	Cross-sectional Canada *2007 Canadian Community Health Survey*	Multistage stratified cluster design and random digit dialing	Canadian adults aged 20-64, n=42,612 (54%) Overall mean age not provided	Number of hours in a typical week over the past three months spent watching TV including videos *Categorical (≤5, 6-10, 11-14, 15-20, ≥21 h/week)*	Education, household income, age, marital status, population of residence, immigrant status Weighted to Canadian population	LTPA + walking or cycling to work/school over past 3 months, classed as inactive (<1.5kcal/kg/day), moderately active, active (≥3kcal/kg/day) (previous three months)	Men with ≥15h/week TV time had higher likelihood of being inactive compared to ≤5h/week OR (95%CI) 6-10h/week 1.1 (1.0, 1.2) 11-14h/week 1.0 (0.8, 1.1) 15-20h/week 1.4 (1.1, 1.6) ≥21h/week 1.3 (1.1, 1.5) Women with ≥6h/week TV time had higher likelihood of being inactive compared to ≤5h/week 6-10h/week 1.2 (1.1, 1.4) 11-14h/week 1.3 (1.1, 1.5) 15-20h/week 1.6 (1.4, 1.9) ≥21h/week 1.9 (1.7, 2.2)
Sugiyama 2008 [34] Quality = fair	Cross-sectional Australia *PLACE study*	Randomly selected addresses within 32 neighborhoods	Australian adults aged 20-65, n=2046 (64%) Mean age 44 (12.3)	Watching TV or videos in the past 7 days *Categorical ≤1, 1-2, >2h/day)*	Education, employment status, age, BMI, marital status, living with children	Total LTPA (minutes per day) over the past 7 days based on the International Physical Activity Questionnaire	Women with high TV time had significantly lower LTPA time (24.1 (41.7) min/day) than those with low TV time (39.0 (59.2) min/day); age-adjusted correlation between TV (continuous) and LTPA was -0.11 (p<0.001) There was no difference in mean LTPA time between men in the high (39.5 (54.7) min/day) vs low (45.7 (59.3)) categories of TV time; age-adjusted correlation between TV (continuous) and LTPA was -0.06 (p=0.08)
Tucker 2015 [50] Quality = good	Cross-sectional USA	Recruited from local area (convenience sample)	Healthy women, n=302 (100%) Mean age 40.2 (3.0)	Watching TV and videos per day or week *Categorical (≤1, 2, ≥3h/day)*	Education, age, PA, body fat percentage, BMI, season	Cardiorespiratory fitness (VO_2 max assessed by maximal graded treadmill test)	No difference in VO_2 max between the high (34.2 (6.4)) and low (35.5 (6.5)) TV groups in fully adjusted model
Sleep							

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Andrianasolo 2016 [53] Quality = good	Cross-sectional France *SU.VI.MAX-2*	Participants who had voluntarily enrolled in RCT assessing the effect of antioxidant supplementation on CVD and cancer incidence	Middle-aged and older adults who had taken part in SUVIMAX RCT, n=4179 (56%) Mean age 61.9 (6.1), range 47-74	Time usually spent watching television/video per day in past 12 months *Categorical (<9.9, 9.9-16, 16-21.5, >21.5h/week)*	Education, employment status, sex, age, marital status, BMI, sleep satisfaction, frequent nighttime awakening, history of sleep disorders, anxiety, interval between TV and outcome variable measurement	Excessive daytime sleepiness (score >10 on Epworth Sleepiness Scale) (timing unclear)	High TV was associated with lower likelihood of excessive daytime sleepiness (OR 0.73 (0.57, 0.94 for highest quartile versus lowest quartile of TV time))
Asaoka 2007 [59] Quality = fair	Intervention Japan	Voluntary sample recruited from pool of previous research participants (university sample) and employment service center (elderly sample)	University students (n=8, 62.5%) and elderly adults (n=8, 62.5%) Mean age for university students 19.9 (0.8), range 19-21; for elderly adults 73.6 (2.7), range 70-78	Logged TV time in daily activity diary (used only to confirm the efficacy of the intervention for changing TV time, not used as a predictor variable)	None	Rising time, bedtime, total sleep time, nap total sleep time, and daytime total sleep time (between 8am and 8pm) all derived from self-reported sleep logs	Compared to the control week, university students went to bed significantly earlier ($p<0.05$), slept for significantly longer in total ($p<0.05$) and during the daytime ($p<0.05$) when TV was limited to 30 minutes per day; there were no significant changes in rising time, bedtime, or total sleep time among the elderly ($p>0.05$).
Buman 2015 [54] Quality = good	Cross-sectional USA National Sleep Foundation's 2013 *Sleep in America* poll	Half of the sample was recruited via random digit dialing; the other half was obtained via an online e-Rewards panel	Adults currently living in the US, n=843 (51%) Mean age 42.1 (SD 11.1)	Sitting while watching television in past week *Continuous (h/day)*	Education, work status, age, sex, ethnicity, race, geographic region, web vs phone administration, BMI, physical activity	Self-reported sleep metrics referring to the past 2 weeks: short sleep duration (<7h) long sleep duration (>8.5h), long sleep onset latency (≥30min), waking up during the night (≥a few nights a week), waking up too early in the morning (≥a few nights a week), poor sleep quality rating, high risk for obstructive sleep apnea, excessive daytime sleepiness (score >10)	High TV time was associated with higher likelihood of poor sleep quality (OR 1.12 (95%CI 1.02, 1.24)), long sleep onset latency (OR 1.15 (95%CI 1.04, 1.27)), and waking up too early in the morning (OR 1.12 (95%CI 1.03, 1.23)); no association was found with short (OR 1.04 (95%CI 0.96, 1.14)) or long (OR 1.10 (0.93, 1.30)) sleep duration, excessive daytime sleepiness (OR 1.04 (95%CI 0.93, 1.18)) or waking up in the night (OR 1.08 (0.98, 1.18))

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Custers 2012 [55] Quality = fair	Cross-sectional Belgium	20 randomly selected addresses from 73 randomly selected cities/towns in Flanders	Adults aged ≥18 living in Flanders, n=711 (57%) Mean age 44.9 (17.4)	Hours spent watching TV on weekdays and weekend days *Categorical (10 categories ranging from 0 to ≥6h/day)*	Education, gender, age, frequency of going out (e.g., to pubs), exercise, whether they had a TV in the bedroom, internet access in bedroom, time spent using the internet	Usual bedtime, rise time, sleep window (rise time minus bedtime), separately for weekdays and weekend days; overall tiredness score (ranging -25 to +25)	TV time was only associated with later bedtime on weekends (β=0.10, p<0.05); no associations with weekend rise time (β=0.07), weekend sleep window (β=-0.02), weekday bedtime (β=0.05), weekday rise time (β=0.05), weekday sleep window (β=0.03), or overall level of tiredness (β=0.07), were reported (all p>0.05)
De Souza 2017 [56] Quality = good	Cross-sectional Brazil *PRO-MESTRE* study	Census of teachers working at the 20 largest schools in Londrina, Brazil	Public schoolteachers working at schools in Londrina, n=959 (68%) Mean = 41.6 (9.9) year, range 23-68	Time spent watching television each day (weighted for weekdays and weekend days) *Categorical (≤60, >60 to 120, and ≥120min/day)*	Gender, age, working hours, alcohol and coffee consumption, current smoking status, LTPA, eating while watching TV, obesity, anxiety, depression, chronic pain	Poor sleep quality in the past month (>5 on Pittsburgh Sleep Quality Index)	No association between TV time and poor sleep quality rating (OR 1.30 (95%CI 0.91, 1.85))
Kakinami 2017 [57] Quality = good	Cross-sectional Canada *Nicotine Dependence in Teens study*	Original cohort recruited from 10 secondary schools in Montreal	24-year-old adults from the cohort, n=658 (54%) Mean age 24 (0.6)	Hours of television (including video movies) usually watched in a single day (weighted for weekday and weekend day) *Continuous (h/day)*	Mother's education, age, sex, MVPA	Poor sleep quality (score ≥5 based on Pittsburgh Sleep Quality Index), and short (<7h) or long (>9h) sleep duration calculated by subtracting self-reported usual bedtime from usual rise time after subtracting out time it usually took to fall asleep during the past month	High TV time was associated with higher likelihood of poor sleep quality (OR 1.15 (95%CI 1.01, 1.28); no association between TV time and short (OR 1.11 (95%CI 0.99, 1.26) or long (OR 1.13 (95%CI 0.96, 1.32) sleep duration
Komada 2015 [58] Quality = fair	Experimental (between and within subjects) Japan	Convenience sample	Young Japanese men, n=13 (0%) Mean age 22.7 (0.9), range 21-24	Lab-based exposure (107 minutes) on three night-time occasions. Participants were randomized to normal	N/A	Change in salivary melatonin and cortisol levels (taken both before and after the 107-minute exposure)	The increase in melatonin levels was larger following the half-blue light exposure compared to the normal blue light exposure (p<0.05); there was no effect of

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				luminance or high luminance display groups. Regardless of group, all participants were exposed to both normal blue light and half blue light as well as a baseline (control) condition (random order).			luminance or luminance*blue light interaction on melatonin levels. There were no associations between blue light level, luminance, or their interaction on cortisol levels.

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Additional file 3. Summary of characteristics, measurements, and findings of included studies examining association between TV and dietary characteristics.

First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model (SEP listed first where applicable)**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
Daily/weekly TV and dietary characteristics							
Bowman 2006 [32] Quality = good	Cross-sectional USA *USDA Continuing Survey of Food Intakes by Individuals 1994-1996*	Stratified, multistage area probability sample of US households	American adults aged ≥20, n=9157 (52%) Mean age or range not reported	Two-day averages were used for estimating daily hours of television or videotape viewed *Categorical (<1, 1-2, >2h/day)*	Annual household income as a percentage of federal poverty level (low: 0-130%, medium: 131-350%, high: >350%), age, sex, race/ethnicity, region, urbanization	Two-day averages of dietary intake collected via interview-administered 24-hour dietary recalls to estimate daily kcal consumption (total, breakfast, lunch, dinner, snacks), macronutrients (as g/day and %kcal: total fat, saturated fat, carbohydrates, added sugars, dietary fiber, protein), regular soft drinks (g), energy-rich snack-type foods (g, including cakes, cookies, pastries, pies, sweet rolls, crackers, popcorn, pretzels, corn chips, fried potatoes including chips), grain mixtures (g, including pizza, burritos, tacos, egg rolls, quiche, spaghetti, rice, pasta mixtures)	Those with high TV time had significantly higher daily total energy intake (2033 (1994) vs 1896 (1065) kcal), dinner intake (783 (836) vs 737 (404) kcal), daily kcal from snacks (390 (684) vs 329 (514) g), consumption of regular soft drinks (266 (722) vs 185 (367) g), energy-rich snack foods (100 (114) vs 89 (110) g), added sugars (15 (8) vs 14 (9) %total kcal), and lower fiber intake (8 (4) vs 9 (4) g) than those with low TV time. There were no differences in breakfast kcal or lunch kcal, or %kcal from total fat, saturated fat, carbohydrates, or protein.
Charreire 2011 [36] Quality = fair	Cross-sectional France *SU.VI.MAX*	Volunteer sample enrolled in RCT testing antioxidant supplementation in prevention of heart disease and cancer	French adults aged 35-60 (women) and 45-60 (men), n=1359 (48%) Mean ages 54.2 (4.2) and 51.8 (4.6) for men and women, respectively	Time spent daily watching TV over the past 12 months *Categorical (<1, 1-2, 2-3, ≥3h/day)*	Education, age, smoking status, PA	Dietary patterns, defined by applying principal components analysis to average consumption patterns (based on six 24-hour food records 2 months apart from each participant), separately by sex. The three main dietary patterns used in analyses for men were alcohol/meat,	Among men, high TV time was associated with high convenience food (p-trend=0.04) consumption; no association with alcohol/meat or healthy food Among women, high TV time was associated with high alcohol/appetizer consumption (p-trend=0.03); no association with healthy food or convenience food

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						healthy food, convenience food; alcohol/appetizers, healthy food, convenience food were used for women.	
Compernolle 2016 [61] Quality = good	Cross-sectional Multiple European countries *SPOTLIGHT study*	Stratified random sampling within four types of neighborhoods (high/low SES x high/low residential density)	Adults aged ≥18 in Belgium, France, Hungary, the Netherlands, and the UK, n=6001 (56%) Mean age 51.9 (SD=16.4)	Time spent sitting while watching television during the last seven days (weighted average for weekday and weekend days) *Continuous (h/day)*	Education, neighborhood type, age, urban region, gender, BMI, MVPA	Likelihood of high weekly intake of: fruit (≥7portions), vegetables (≥7 portions), sugar-sweetened beverages (≥2glasses), alcohol (>2glasses), sweets (≥3portions), fast food (>0.5portions) based on food frequency questionnaire	TV time was associated with lower likelihood of high fruit (OR 0.984 (95%CI 0.976, 0.991)) and vegetable (OR 0.981 (95%CI 0.975, 0.987)) consumption, and higher likelihood of high sugar-sweetened beverage (OR 1.017 (95%CI 1.010, 1.025) and fast food (OR 1.007 (95%CI 1.003, 1.011)) consumption No association was reported between TV time and likelihood of high consumption of alcohol (OR 1.004 (95%CI 0.996, 1.011)) or sweets (OR 1.005 (95%CI 0.997, 1.014)).
Rehm 2008 [62] Quality = good	Cross-sectional USA *2005 NYC Department of Health and Mental Hygiene Community Health Survey*	Stratified random digit-dialing of NYC residents	Adults aged ≥18 living in New York City, n=9916 (60%) Overall mean age or age range not provided	Television viewing *Categorical (≤1, 2-4, ≥5h/day)*	Education, income to poverty ratio, age, gender, race/ethnicity, PA	Self-reported consumption of ≥1 sugar-sweetened soda (12oz can, bottle, or glass) on an average day	Likelihood of ≥1 sugar-sweetened soda per day was higher among those with higher TV time (OR 1.8 (95%CI 1.4, 2.3).
Robinson 2004 [64] Quality = fair	Cross-sectional UK *Southampton Women's Survey*	All women registered with GPs in Southampton were sent an invitation letter for the study	Women living in Southampton aged 20-34, n=6125 (100%) Mean age 27.8 (SD=4.2)	Time spent watching television *Number of hours*	Education, age, smoking status, currently dieting to lose weight, any strenuous exercise in past three months, sharing home with children	Prudent diet score, calculated based on 100-item food frequency questionnaire over the preceding 3 months collapsed into 49 groups and entered into a principal components analysis (standardized with mean of 0 and standard deviation of 1)	TV time was inversely associated with prudent diet score (β=-0.11 (95%CI -0.13, -0.09), p<0.001)

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Scully 2009 [63] Quality = good	Cross-sectional Australia	Stratified random sampling	Adults aged ≥18 in Victoria, Australia, n=1495 (NR%) Mean age or age range not provided	Average hours per day spent watching commercial TV (i.e., TV that included advertisements) *Categorical (≤1, 2, ≥3h/day)*	Education, area-level disadvantage, age, gender, urban/rural location	Likelihood of self-reported fast food consumption at least once a week over the past three months, recorded separately for breakfast, lunch, dinner, and snacks	There was no association between TV category and likelihood of eating fast food for breakfast (OR 1.21 (95%CI 0.66, 2.23)) or lunch (OR 1.08 (95%CI 0.76, 1.55)); however, there was an association between TV time and higher likelihood of having fast food for dinner (OR 1.45 (95%CI 1.04, 2.03)) or snacks (OR 1.81 (95%CI 1.20, 2.72))
Shields 2008 [35] Quality = good	Cross-sectional Canada *2007 Canadian Community Health Survey*	Multistage stratified cluster design and random digit dialing	Canadian adults aged 20-64, n=42,612 (54%) Overall mean age not provided	Number of hours in a typical week over the past three months spent watching TV *Categorical (≤5, 6-10, 11-14, 15-20, ≥21 h/week)*	Education, household income, age, marital status, population of residence, immigrant status Weighted to Canadian population	Likelihood of low fruit and vegetable consumption (<3 times per day)	TV time was associated with higher likelihood of low fruit and vegetable consumption among men (OR 1.7 (95%CI 1.5, 2.0)) and women (OR 1.9 (95%CI 1.7, 2.3))
Thomson 2008 [65] Quality = fair	Cross-sectional Canada	Random sample of students at the University of Alberta	University students aged 18-25, n=613 (64%) Mean age 20.8 (1.9)	Hours viewed on average day (sum of average weekday and weekend day) *Continuous (h/day)*	None	Energy-dense (ED) snack consumption in a typical week (score ranging from 0 to 28 based on frequency (never=0, >5 times per week=4) of consumption of 7 ED snacks (including salty snacks, candy, baked goods, soft/fruit drinks, ice cream, salted meats, snack bars) Healthy snack consumption (score ranging from 0 to 20) based on frequency (0 to 4, as above) for 5 healthy snacks (including fruit, dairy, vegetables, nuts, and breads/rolls/biscuits) Frequency of snacking while watching TV (score	TV time was correlated with higher frequency of energy-dense snack consumption (r=0.20, p<0.01) There was no association between TV time and frequency of healthy snack consumption (r=-0.02, p>0.05). TV time was correlated with frequency of snacking while watching TV (r=0.56, p<0.01)

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						ranging from 1=never to 5=everyday)	
Diet while watching TV							
Gemming 2015 [29] Quality = good	Cross-sectional New Zealand	Recruited from the community and university (convenience)	Healthy adults in Auckland, n=40 (50%) Mean ages 35 (SD=17) and 28 (SD=7) for men and women, respectively	Instances of television time (only while eating) as identified during processing of SenseCam (wearable camera) recordings across 3 days	NA	Energy intake (kJ), energy density (kJ/g), and duration (min) of meals and snacks across contexts (including during TV) measured by 24-hour multiple pass dietary recall on days that wearable camera was worn	Snacks at home when viewing television were higher in energy intake (+514 kJ (47, 1077)) compared to snacks when no television was viewed. Meals at home while watching TV were not significantly longer (3.1 min (-0.6, 6.7)) than meals not viewing screens, and were not significantly higher in energy (543kJ (-32, 1120)).
Ghosh Roy 2018 [37] Quality = good	Cross-sectional USA *African American Women's Daily Life Study*	Recruited from community and university (convenience)	African American women aged 25-65 in Chicago, n=79 (100%) Mean age 44.6 (SD=10.6)	In each instance of reported snack or sweetened beverage consumption (see 'outcome measurement'), participants were asked what they were doing while eating or drinking; watching television was an option	Education, per capita income, age, auto ownership, BMI, weekday or weekend day measurement, hunger status, fast food restaurant and convenience store density in home neighborhood	Self-reported consumption of ≥1 snack food item (including French fries, salty snacks, cookies or sweetened baked goods, chocolate/candy, ice cream/frozen dessert) or ≥1 sweetened beverage in response to smartphone prompts which came 5 times per day across one week (ecological momentary assessment)	Consumption of snack foods was significantly more likely to occur while watching TV versus while doing nothing else (OR 1.8 (95%CI 1.2, 2.7)); no such association was seen for sugar-sweetened beverage consumption (OR 0.9 (95%CI 0.5, 1.6)).
Laska 2011 [38] Quality = fair	Cross-sectional USA	Recruited from community and university (convenience)	Young adults aged 18-23 from four groups (attending university and living on campus; attending university and living independently off-campus; attending university and living with parents; not attending university and living independently), n=48	All instances of eating and drinking were logged on a PDA over a 7-day period, and followed up with contextual questions, including whether they were watching TV during that instance of eating	N/A	Proportion of food consumption while watching TV compared to while not doing anything else, separately by food group: calorically sweetened beverages, water, coffee/tea drinks, milk, cookies and sweetened baked goods, candy/gummy fruit snacks, fruits and vegetables (fresh or 'other'), frozen desserts, non-milk dairy products,	A larger proportion of calorically sweetened beverages (24% vs 15%) and smaller proportions of water (12% vs 17%), fruits and vegetables (26% vs 36%), and cereals/grains (25% vs 31%) were consumed while watching TV compared to while not doing anything else, respectively ($p<0.05$ for all). There were no significant differences among the other food groups listed in the previous column.

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			(12 from each group), 56% female Mean age approximately 21			entrees, cereals/grains, and fried side dishes	
Stroebele 2004 [39] Quality = fair	Cross-sectional USA	Recruited from university for course credit	University students, n=78 (82%) Mean age 22 (0.9)	Television time was recorded on a diary in 15-minute intervals. Instances of eating while watching TV were recorded on food diaries (see 'outcome measurement' column)	N/A	Meal sizes, caloric intake, and meal frequency, ascertained from a 7-day food diary in which participants recorded the volume of all food and drinks consumed, with a specification concerning whether the TV was on in each instance	Meal sizes did not differ with and without TV (data not shown). There was no correlation between TV time (h/day) and total caloric intake or meal frequency (data not shown).
Experimental dietary studies							
Study, quality	**Study design, country**	**Target sample Sample size (% female) Mean (SD) age or age range**	**Study aim and experimental protocol**		**Main outcome measurement**	**Findings**	
Anschutz 2011 [79] Quality = fair	Between subjects Netherlands	Non-overweight university students, n=82 (50%) Mean age (SD) for men and women was 20.9 (2.5) and 20.4 (1.5), respectively	Aim: to examine whether food commercial exposure would increase concurrent snack food intake among young adults Protocol: participants were randomized to 'food commercial' condition (30-minute nature TV clip interrupted by 3 food and 5 non-food commercials) or 'neutral commercial' condition (30-minute nature TV clip with 8 non-food commercials)		Food intake (kcal) based on weight of crisps and M&Ms consumed during the session Amount of food provided not specified	There was no effect of commercial condition on food intake (b=-0.52, SE=43.12, $p>0.05$) and the addition of commercial condition to the model had a negligible effect (change in R^2 <0.001); the interaction between sex and commercial condition was significant (women ate more when exposed to food commercials compared to neutral, while men ate more when exposed to neutral compared to food commercials)	
Bellisle 2004 [66] Quality = fair	Within subjects France	Women aged 18-50 with BMI between 18.5 and 24.9, n=48 (100%) Mean age 29.9 (1.4), range 18-50	Aim: To assess the impact of two environmental stimuli (TV and audio story) on meal intake Protocol: Participants ate a meal in four conditions each lasting minimum of 30 minutes and spaced one week apart: control (alone), while watching TV (no references to food or eating), while listening to a recorded detective story, and last meal (alone); order of TV and auditory conditions was randomized		Food intake (kJ) of meal (shepherd's pie and fruit sherbet) ~4850kJ (1160kcal) available	Compared to the baseline control condition (1751 (SE=84) kJ), meal size was larger in the TV (2023 (SE=84) kJ) and audio recording (2044 (SE=84) kJ) conditions ($p<0.001$). Meal size did not differ between the TV and audio recording conditions ($p>0.05$).	
Bellisle 2009 [69] Quality = fair	Within subjects France	Healthy women of normal weight, half with high and half	Aim: To assess the effects of dietary restraint and environmental factors on meal intake		Food intake (g, kJ) of meal (shepherd's pie and fruit sherbet)	Significantly less food (g, kJ) was consumed in the group meal condition compared to the others ($p<0.02$) in the pooled sample.	

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		with low dietary restraint, total n=40 (100%) Overall mean age or range not reported	Protocol: Participants ate a meal in five conditions (randomized order) each lasting a minimum of 30 minutes and spaced one week apart: alone in a quiet room, in groups of 3, alone with TV (no food cues), alone with TV (including food advertisements), alone while listening to audio detective story	~6000kJ (1430kcal) available	Consumption was not different between the other conditions (data not shown).
Blass 2006 [75] Quality = fair	Within subjects USA	Undergraduate students, n=20 (75%) Mean age or range not reported	Aim: to test whether more was eaten when watching TV compared to not Protocol: participants were assigned to either pizza or mac and cheese (assignment protocol not stated) and either watched TV for 30 minutes (program of choice with original commercials included) or listened to *Rachmaninoff's Second Symphony* for 30 minutes; the experiment was repeated a week later for the other condition (counterbalanced order)	Food intake (kcal) of pizza or macaroni and cheese 12-inch pizza or family-size bowl (~900g) of macaroni and cheese available	More calories were consumed with the TV on (793.7kcal) than with TV off (538.2 kcal), $p<0.001$. There was no interaction effect between food type and condition.
Bodenlos 2013 [84] Quality = fair	Between subjects USA	Undergraduate students, n=80 (73%) Mean age 19.5 (1.0), range 18-22	Aim: to test whether exposure to a cooking show affected caloric intake Protocol: Participants were randomized to watch either a cooking show or Planet Earth for 10 minutes, followed by a 'taste test' of cheese curls, chocolate covered candies, and carrots	Intake (kcal) of cheese curls, chocolate covered candies, and carrots 800kcal in total available	No significant difference in overall calories consumed between the two conditions ($F(1,74)=3.32$, $p=0.07$), $d=0.38$ Those in the cooking show group consumed significantly more calories from chocolate covered candies than those in the nature group ($F(1,74)=3.90$, $p=0.05$), $d=0.51$
Boyland 2017 [80]	Within subjects UK	Females recruited from university (staff and students) and surrounding area, n=55 (100%) Mean age 32.4 (9.8), range 20-62	Aim: to examine consumption responses to televised food commercials in overweight and lean adult females Protocol: Participants attended two sessions in a counterbalanced order: TV show including commercials for high-calorie palatable foods and TV show including non-food commercials. Following the show, participants were given pizza to eat.	Intake (kcal) of pizza 2056kcal of pizza available	No main effect of condition or weight status on pizza intake or interaction between weight status and condition (p values >0.28).
Braude 2014 [67] Quality = fair	Within (and between) subjects Australia	Female undergraduates, n=62 (100%) Mean age 19.6 (2.2), range 18-29	Aim: To test whether TV affected sensory specific satiety Protocol: Participants were randomized to a single- or variety-foods group; both groups took part in a TV (*Friends,* no commercials) and no-TV condition (counterbalanced order) lasting 20 minutes and spaced one week apart. In the single-foods group, participants chose their preference between Maltesers, Skittles, almonds, or salted potato chips and were given 80g of it. Participants in the variety-foods group were given a 20g bowl of each food (totaling 80g).	Food intake (kJ, number of items) Variety condition totaled ~1600kcal available; single food condition depended on which food was picked	Participants consumed significantly more energy ($z=2.37$, $p<0.025$) and more items of food ($z=2.07$, $p<0.05$) in the TV condition compared to the no-TV condition.

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Chapman 2014 [82] Quality = fair	Within subjects Sweden	Healthy female women, n=18 (100%) Mean age 22 (1.3)	Aim: to explore how the content of TV programs differentially impacts simultaneous eating behavior Protocol: Participants took part in three conditions (counterbalanced order) each lasting 30 minutes and spaced one week apart: watching an engaging TV program (comedy), a boring program (art lecture) or reading a non-engaging text about insects. Bowls of M&Ms and grapes were available in each condition.	Food intake (grams, kcal) 250 grams (~1200kcal) of M&Ms and 300 grams (~200kcal) of grapes were available	Significantly more food (grams) was consumed during the boring program (125g) compared to the engaging program (82g, $p<0.01$), but no difference in kcal (data not shown). No differences in grams or kcal consumption were reported between the engaging program vs text condition (109.3g vs 81.9g, $p=0.05$) or the boring program vs text conditions (+15.5 grams, $p=0.26$) (kcal data not shown).
Hetherington 2006 [68] Quality = fair	Within subjects UK	University staff and students, n=37 (43%) Mean age 28.3 (1.8), range 18 to 54	Aim: To test whether the social facilitation of eating occurs as a function of distraction (by comparing consumption in social contexts with TV and control conditions) Protocol: Participants took part in four conditions in a counterbalanced order at least 3 days apart: eating alone, eating with TV (game show, duration not fixed), eating with strangers, and eating with friends. The same buffet meal was available in all conditions.	Food intake (kJ) A buffet-style meal (13,743kJ/3283kcal) with 9 different food items was available	Participants consumed significantly more when eating in front of the TV (4350 (SE=252) kJ) and when eating with friends (4565) SE=272) kJ) compared to the control condition (3861 (SE=200) kJ). No differences were found between eating with strangers and any other condition.
Higgs 2009 [77] Quality = fair	Within subjects UK	Female undergraduates of normal weight, n=16 (100%) Mean age 19 (1)	Aim: to examine whether watching TV during lunch would increase afternoon snack intake due to impaired memory Protocol: Participants had a standardized lunch either in quiet solitude (control) or with the TV on (comedy show with no references to food). Both conditions lasted 10 minutes and all participants ate the entire lunch in that time period. At least 2.5 hours later on the same day, participants were given three kinds of cookies broken into small pieces to taste test.	Amount (g) of cookies consumed Three kinds of cookies (40g of each) were available, totaling ~600kcal available	Participants consumed significantly more grams of cookies after the lunch-with-TV condition compared to after the lunch-without-TV condition ($F(1,15)=12.6$, $p<0.01$)
Lyons 2012 [76] Quality = fair	Between subjects USA	Adults aged 18-35, n=120 (50%) Mean age 24.1 (4.4)	Aim: to compare differences in energy intake among 2 sedentary screen behaviors (TV and video games) and one potentially active screen behavior (motion-controlled video game). Protocol: Participants were randomized to a TV, video game (PS3), or motion-controlled video game (Wii or Xbox 360) group. Each condition lasted 1 hour. During the condition, snacks and beverages were available for consumption.	Amount (kcal) of snacks and soda consumed Snack foods included M&Ms, baked Lay's, Doritos, and trail mix (nuts with dried fruit) (amounts provided not specified); soda included cans (3 of each) of Coca-Cola, Diet Coke, and Mountain Dew.	Energy consumption between the three groups was not significantly different ($p=0.07$). Mean intake during TV was 716 (407) kcal; during video games was 747 (540) kcal; and during motion-controlled video game was 553 (498) kcal.

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Martin 2009 [71] Quality = good	Within subjects USA	Healthy adults aged 18 to 54 (BMI 20-35), n=48 (54%) Mean age 31.9 (SE=1.5)	Aim: to test the effect of TV viewing with and without ads on energy intake compared to a control and a reading condition Protocol: participants completed four conditions (each lasting 32 minutes) in a random order on two test days (experimental lunch and dinner after consuming a standardized breakfast): control, reading, TV without ads, TV with ads (half food-related, half not). The same buffet meal was available for all conditions.	Food intake (kcal) Buffet-style meals were available including 19 different items (totaling 4921kcal)	Energy intake did not vary by experimental condition ($F(3, 131)=0.30$, $p=0.81$); partial $\eta^2 \leq 0.01$
Mathur 2015 [83] Quality = fair	Within subjects Australia	Female undergraduates, n=45 (100%) Mean age 19.5 (SD=2.2), range 18-29	Aim: to test whether variability in engagingness of TV affects food intake Protocol: Participants completed two conditions in a randomized order one week apart: watching the same episode of *Friends* twice back-to-back ('same'), or 'watching two different episodes back-to-back ('different'), with each episode lasting 20 minutes and without advertisements). The first episode was watched without food; for the second episode, regardless of condition, they were given 80g each of their preferred two snacks (given the choices of almonds, Pringles, and M&Ms).	Food intake (g, kJ) of their first and second preferred snacks	Significantly more snack food was consumed in the same condition (mean 76.2g, SD=36.2) than the different condition (mean 66.7g, SD 37.3, $z=1.990$ (*one-tailed), $p<0.05$).
Mittal 2010, experiment 1 [70] Quality = fair	Between subjects Australia	Females from the university with BMI between 18 and 25, n=32 (100%) Mean ages 20.8 (SD 3.8) and 20.3 (SD 3.9) in the two groups	Aim: to assess whether snacking with concurrent TV would result in eating more at a test meal later on Protocol: Participants were assigned to either a snack-with-TV or snack-without-TV condition, both lasting 20 minutes (TV program was choice between *Seinfeld* and *Friends*). In both conditions, they consumed as much as they wanted of their snack (potato chips, chocolate balls, and coke or orange juice). Then, after 45 minutes of psychological tests, they were given lunch (test meal).	Food intake (kJ) of the snack in the TV versus no-TV condition 2080kJ (~500kcal) offered (chocolate balls, Pringles, Coke/orange juice) Food intake (kJ) of the test meal following the snack conditions 2470kJ (~600kcal) offered (sandwiches, biscuits, crackers and dip)	Energy consumption of the snack did not differ between the snack-with-TV (1855.9 (264.2)kJ) and snack-without-TV (1667.7 (362.0) kJ) conditions (test statistics not shown). Energy consumption of the test meal was significantly higher among those who had watched TV with their snack (1584.6 (516.4) kJ vs 1354.9 (335.6) kJ), $p<0.05$ (*one-tailed)
Mittal 2010, experiment 2 [70]	Between subjects Australia	Females from the university, n=84 (100%)	Aim: to determine whether the content of TV shows watched while snacking affect recall	Food intake (kJ) between the snack-TV conditions	There was no difference in the energy consumption of the snack across the four groups (test statistics not shown). Mean snack consumptions (kJ) were 2308.3 (262.4) for boring, 2291.6

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Quality = fair		Overall mean age not reported but around 21 years	accuracy of food eaten during the show and intake on a later TV-free test meal Protocol: Participants were assigned to one of four conditions, each lasting 20 minutes: snack-without-TV, snack-with-boring-TV, snack-with-funny-TV, or snack-with-sad-TV. They consumed their snack (potato chips, chocolate balls, and coke or orange juice). Then, after 45 minutes of psychological tests, they were given lunch (test meal).	2411kJ (575kcal) offered (chocolate balls, Pringles, Coke/orange juice) Food intake (kJ) of the test meal following the snack conditions 4021kJ (960kcal) offered (sandwiches, crackers and dip, ice cream and chocolate, Coke/orange juice)	(223.5) for sad, 2250.2 (289.7) for funny, and 2194.9 (322.0) for control. Energy consumption of the test meal was significantly higher in all three TV conditions compared to the no-TV condition (post-hoc pairwise $p<0.02$ in all three cases), but consumption did not differ between the three TV groups (test statistics not shown). Mean test meal consumptions (kJ) were 2507.0 (438.2) for boring, 2842.0 (452.4) for sad, 2637.6 (540.3) for funny, and 2147.9 (527.2) for control.
Moray 2007 [74] Quality = fair	Within subjects USA	University students, n=20 (50%) Mean age 20.8 (range 18 to 23)	Aim: to determine whether TV promotes increased food consumption because it impairs one's ability to accurately estimate food intake Protocol: participants completed two conditions in counterbalanced orders, each lasting 25 minutes: eat with and without TV. After the meal, participants were asked to estimate how much food they had just consumed, using a visible tablespoon as a reference for 'one unit' (which was equivalent to 1.5oz of food).	Food intake (oz) between the TV and no-TV conditions Accuracy of food estimation, determined as estimated units minus actual units consumed 16oz macaroni and cheese provided	There was no difference in amount of food consumed between the TV and no-TV condition (2.5%, $p>0.05$). Unit estimation error was greater ($T_{0.05}$ 2.2, $p=0.022$) in the TV vs no-TV condition.
Ogden 2013 [73] Quality = fair	Between subjects UK	Females aged ≥18 who had a manual driving license, n=81 (100%) Mean age 22 (5.18), range 18-40	Aim: to compare the impact of different forms of distraction on eating behavior Protocol: participants were randomized to one of four conditions, each lasting 7 minutes: driving (simulation with a manual transmission), TV (*Friends*), social interaction (conversing with one of the researchers), or sitting alone. 100g of hula hoops (potato snacks) were available in each condition.	Grams of hula hoops consumed 100g were provided (~520kcal)	Those in the TV condition consumed more (28.61 (24.44) g) than those in the social (14.16 (12.33)g, $p<0.01$) and driving (14.02 (10.34)g, $p<0.05$) conditions, but no significant difference was seen compared to the alone condition (18.21 (20.9)g, $p=0.06$).
Ogden 2017 [78] Quality = fair	Between subjects UK	Females from the university, n=60 (100%) Mean age 24 (3.3)	Aim: to compare the impact of distraction and dietary restraint on food intake during a subsequent taste test Protocol: participants were randomized to one of three conditions, each lasting 5 minutes: watching TV, walking (along the corridor), and social interaction (talking with another research participant). During each condition, participants were told to consume a cereal bar. After the	Mass and energy (kcal) of food consumed during the taste test 150g of each of the foods were provided	There was no main effect of condition on total mass or total calories consumed during the taste test (values not shown).

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			condition, participants took part in a taste test lasting 7 minutes during which they could consume as much of the foods (M&Ms, carrot sticks, grapes, and hula hoops) as they wished.		
Rosenthal 2017 [72] Quality = good	Within subjects USA	Young adults aged 18 to 35 with BMI between 18.5 and 24.9kg/m^2 n=20 (85%) Mean age 22.3 (3.7)	Aim: to test the independent and interactive effects of TV and portion size on food intake Protocol: participants took part in four randomized conditions one week apart: small portion with and without TV and large portion with and without TV. In the 'no TV' condition, participants ate quietly with no other activities for 30 minutes; the TV condition was a 30-minute episode of *Scandal*.	Food intake (grams and kcals) Small portion condition provided 1083 kcal (650 grams) of macaroni and cheese plus salad with dressing; the large portion provided 2166 kcal (1300 grams) of the same foods	Only a main effect of portion size was found for grams and kcals consumed. No main effect for TV was found for food intake ($F(1,16)=0.39$, $p=0.54$, $d=0.11$) and $F(1,16)=1.10$, $p=0.31$, $d=0.17$)) measured as grams and kcals, respectively. No interactions for TV and portion size were found (>0.05).
Wonderlich-Tierney 2013 [81] Quality = fair	Between subjects USA	Undergraduates, n=83 (52%) Mean age 19.6 (3.5)	Aim: to examine the impact of TV advertisements on food intake Protocol: participants were randomized to one of three groups, each lasting 60 minutes: TV with food advertisements, TV with non-food advertisements, and TV with no advertisements. A jar of cookies was available throughout each condition.	Number of cookies eaten 26 cookies were available in each condition	There was no effect of condition on number of cookies eaten ($F(2,69)=1.50$, $p=0.23$), $R^2=0.04$

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N/A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5-6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6-7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7-8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7-8

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	27
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Supplementary files 2-4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	11-23, supplementary files 2-4
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Additional file 1; pp13, 18, 19, 22
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	23-28
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	27-28
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	28
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	28

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From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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How is television time linked to cardiometabolic health in adults? A critical systematic review of the evidence for an effect of watching television on eating, movement, affect and sleep

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ABSTRACT

Objective: To improve our understanding of how TV time is linked to cardiometabolic health among adults by systematically and critically evaluating the evidence that watching TV is associated with increased food consumption, lack of movement or negative affect, or affects subsequent sleep.

Design: Systematic review

Data sources: Web of Science and PubMed

Eligibility criteria for selecting studies: Studies that provided quantitative evidence of dietary intake, characteristics of sitting, or affect while watching TV, or on subsequent sleep on the same day among samples of healthy adults (≥18 years old).

Data extraction and synthesis: Study quality was assessed using the NHLBI Quality Assessment Tools. Due to heterogeneity of study designs and measurements, the findings were synthesised using narrative summary accompanied by custom plots.

Results: We identified 31 studies that met the inclusion criteria. Most of the associations reported by the studies included in this review were weak or inconsistent. There was no strong evidence to suggest that food consumption is higher while watching TV than in other contexts or that TV is a particularly 'sedentary' behavior. Affect was less likely to be positive while watching TV than in other contexts but was not more likely to be negative. Two small studies suggest that TV may impact sleep via suppressing melatonin and delaying bedtime.

Conclusion: There is currently no strong evidence to suggest that TV might impact cardiometabolic health via increasing food consumption, being linked with prolonged/inactive sitting, affect, or subsequent sleep. Additional research is required to understand how TV fits within everyday lives and relates to eating, sitting, affect and sleep to improve our understanding of how it might impact cardiometabolic health.

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STRENGTHS AND LIMITATIONS OF THE STUDY

- This review synthesizes for the first time a diverse body of evidence that describes what happens while watching TV (or immediately after) amongst adults, including food consumption, characteristics of sitting, mood/affect, and sleep, to shed light on mechanisms on which TV time might impact cardiometabolic health
- Custom plots were produced to visualize the findings of individual studies, including the magnitude, direction, and statistical significance of associations
- The outcomes and designs of included studies were too heterogeneous for estimating overall summary estimates using meta-analysis

Keywords: TV time, sedentary behavior, cardiometabolic health, diet, mental health, sleep

INTRODUCTION

Television time has been associated with poor cardiometabolic health outcomes among adults, including cardiovascular mortality,[1, 2] incidence of type 2 diabetes,[1, 2] and increases in adiposity.[3] Television time is generally assumed to be sedentary and sedentary time is now a well-established risk factor for poor cardiometabolic health.[1, 4] However, the association between TV time and poor health outcomes is stronger than the association between total sitting time and poor health outcomes[1, 5] or than time spent sitting in other contexts, such as sitting at work.[6-9] For studies based entirely on self-reported measures, lower measurement error in the assessment of TV time than in the assessment of other sedentary behaviors is likely to contribute to such a difference.[10-12] However, the relationship between self-reported TV time and cardiometabolic outcomes is also stronger than the relationship between objectively assessed sedentary time and those same outcomes,[13-15] a finding that is unlikely to be attributable to measurement error. It is

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possible that TV time has a particularly adverse impact on cardiometabolic health and this suggestion is repeatedly made in the literature.[1, 5, 7, 16]

Several possible explanations for the larger observed effects of TV time have been put forth. Most commonly, it has been suggested that dietary intake while watching TV (particularly snacking) may contribute to its effect.[1, 5, 7, 16] It has also been suggested that the way in which sitting occurs while watching TV may be important, for example if it is characterized by particularly low energy expenditure or low levels of muscular activation.[1, 5, 16] Detrimental impacts of TV on sleep and mental health (potentially partly via short-term affective responses) have also been noted as potential mechanisms by which TV might adversely impact cardiometabolic health.[16-18] However, the evidence lending support to these suggested possible explanations has not, to our knowledge, been exhaustively or critically evaluated.

The purpose of this review is to improve our understanding of how TV time is linked to cardiometabolic health among adults by systematically and critically evaluating the available literature that provides evidence regarding food consumption, characteristics of sitting, or affect while watching TV, or on the impact of watching TV on subsequent sleep.

METHODS

Eligibility criteria

Studies were eligible for inclusion in the review if they provided quantitative evidence of dietary intake, characteristics of sitting, or affect while watching TV or subsequently on the same day as watching TV, or on sleep subsequently on the same day. Studies could be experimental or observational as long as they described behaviors while watching TV or on

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the same day subsequent to watching TV. Participants in the studies had to be non-institutionalized adults (≥18 years old) who were not exclusively characterized by a particular health status (e.g., only overweight or obese adults). All studies had to be published in English in 2000 or later as TV viewing practices change over time.

Information sources and search strategy

The literature was searched in March 2021 using Web of Science Core Collections and PubMed. The search strategy included terms to identify studies that measured TV time across all disciplines (Supplementary file 1). The reference lists of all included papers were also checked. We had initially set out to address a broader research question, aiming to capture all studies that examined TV time in relation to any factors that might impact cardiometabolic health; the search strategy was therefore intentionally broad and we did not specify outcome variables of interest *a priori*. Given the diversity of studies identified, we subsequently chose to narrow the scope of our research question to focus only on studies that measured the variable of interest while watching TV (or immediately after) amongst adults.

Study selection

The titles and abstracts of the retrieved studies were screened for relevance. A subsample (10%) was independently screened by both authors to ensure agreement and consistency; any discrepancies were discussed until an agreed application of inclusion and exclusion criteria was reached, then one author (JW) screened the remainder of titles and abstracts. The full-texts of studies with relevant abstracts were consulted to determine eligibility. Both authors confirmed the eligibility of each full-text, and any cases of disagreement were resolved through discussion.

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Quality assessment

The quality of each eligible full-text was assessed based on the NHLBI Quality Assessment Tools for Observational Cohort and Cross-Sectional Studies (for observational studies) and for Controlled Intervention Studies (for experimental and intervention studies) (https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools). In order to draw on the highest standards of evidence, studies deemed to have low internal validity based on the quality appraisal were excluded from the review. The quality assessment of each study was confirmed by both authors and any disagreements were discussed until consensus was reached.

Data extraction

The data from each eligible study were extracted into a spreadsheet with pre-defined columns for the general study details (location, date), study design, methodology, sample characteristics, statistical analyses (including covariates), and main findings describing the behavior(s) of interest while watching TV.

Synthesis of results

The results of the associations between TV time and the outcome of interest were organized by categories of outcome variables that emerged from the search (diet, characteristics of sitting, affect, sleep). Because of heterogeneity of study outcomes and study designs, meta-analysis was not used. Therefore, the results were synthesized using narrative summary accompanied by visualization of study findings. We constructed custom plots designed to illustrate each study's findings to provide a visual summary of the strength of the evidence for each outcome. We plotted the reported effect size if the paper reported this; if studies did not report effect size but reported sufficient information to calculate it (e.g., mean and

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standard deviation), we calculated the standardized mean difference using the R package *metafor* (Supplementary file 2). If 95% confidence intervals or standard errors were reported, these were converted to standard deviations using the formula suggested in the Cochrane handbook (https://handbook-5-1.cochrane.org/chapter_7). When calculating the effect size within experimental studies that used within-subjects designs, we imputed correlations of 0.5 to account for repeated measurements; sensitivity analyses using correlations of 0.6 and 0.8 were also conducted to examine the extent to which the imputed correlation might affect the results (shown in Supplementary file 1). While we acknowledge that the interpretation of the magnitude of effect sizes is arbitrary and context-specific, we classified effect sizes as negligible, small, medium, or large based on standard conventions[19] for the purposes of plotting (standardized mean difference of <0.20, 0.20-0.49, 0.50-0.79, ≥0.80; odds ratio of <1.50 1.50-2.49, 2.50-4.29, ≥4.30, correlation of <0.10, 0.10-0.29, 0.30-0.49, ≥0.50, respectively).

Within the plots, the included papers are ordered from most robust to least robust evidence, primarily on the basis of study design (e.g., within-subjects prioritized over between-subjects) and sample size. Whether the finding of the study was statistically significant is denoted through a filled (black) symbol with the direction of the association denoted by a positive sign (+), negative sign (-), zero (0), or question mark (?) if the direction could not be ascertained; non-significant associations are unfilled. The strength of the effect size is denoted by number of boxes (one=negligible, two=small, three=medium, four=large) based on the reported or calculated effect size. Where the effect size was not reported and could not be calculated in a standardized way (e.g., regression coefficient, graphical presentation of means), we plotted the findings with a triangle, filled (or not) to denote statistical significance. Significance was determined based on 1) the calculated or reported confidence

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intervals not crossing zero, or 2) the study reported a p-value <0.05 where effect size with confidence intervals was not reported and could not be calculated.

Patient and public involvement

No patients involved

RESULTS

Search results

The flow of studies through the review process is shown in Figure 1. From the original 13,036 unique records retrieved by the database searches, 31 were eligible for inclusion. An additional 5 papers were retrieved through reference lists and 5 papers were excluded, leaving 31 studies included in the review (see Supplementary files 3-4 for study details, including information on statistical modelling with covariates listed).

Characteristics of included studies

Twenty-five studies were experimental, primarily focused on links between TV and food consumption (n=19); the remainder focused on energy expenditure of sitting or movement while watching TV (n=4) or the effects of TV on subsequent sleep (n=2). Most experimental studies used a within-subjects study design (n=17) and were conducted in Europe (n=11) or North America (n=9), Australia (n=3), and Japan (n=2). Sample sizes ranged from 13 to 120; ten studies included only female participants while one study included only men.

Six studies were observational, including studies examining food consumption (n=4) and affect (n=2) while watching TV in free-living contexts. Five of these studies were based in the USA and one in New Zealand with sample sizes ranging from 40 to 538.

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Food and drink consumption while watching TV

Experimental studies

TV versus doing nothing

There is some evidence to suggest that more food is consumed while watching TV compared with while doing nothing else in a laboratory setting (Figure 2; Supplementary file 3). Nine experimental studies compared food consumption while watching TV compared with a control condition spent alone with no distraction.[20-28] Seven studies found that participants had higher consumption in terms of energy intake[20-22, 24, 26] or mass[26-28] during the TV condition compared with the control condition (generally small effect sizes), although this did not reach significance in most cases. In sensitivity analyses, effect sizes increased from small to medium in two studies[20, 22] when a larger correlation coefficient was imputed (Supplementary file 2).

TV versus other activity

There is little evidence to suggest that more food is consumed while watching TV compared with while doing other activities in a laboratory setting (Figure 2; Supplementary file 3). Seven studies compared food consumption while watching TV with consumption while doing other tasks, including listening to audio stories,[20, 23] listening to classical music,[29] reading,[25] simulated driving,[27] playing video games,[30] eating with a researcher or strangers,[22, 27] and eating with friends.[22] Five of these studies reported no differences in the amount of food consumed between the TV condition and the other experimental condition they tested.[20, 22, 23, 25, 30] The remaining two studies reported higher consumption while watching TV compared with while listening to classical music,[29] driving,[27] or while

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conversing with a researcher.[27] Sensitivity analyses suggested larger imputed correlations did not materially change the findings (Supplementary file 2).

Effect of TV on later consumption on the same day

Three studies, all in laboratory settings, tested whether the consumption of a snack while watching TV (compared with while not watching TV) would result in higher food consumption at a meal or snack later on in the same day (consumed without TV)[24, 31, 32] (Supplementary file 3). Two of these studies found that when the earlier snack had been consumed while watching TV, significantly more was eaten in the subsequent TV-free meal/snack;[24, 31] the third did not find subsequent consumption was higher after the eating-with-TV exposure.[32]

The role of TV content

There is little evidence to suggest that the content of TV being watched influences food consumption in experimental settings (Supplementary file 3). Four studies examined food consumption when exposed to food advertisements compared with either neutral (non-food) advertisements[33-35] or no advertisements,[25, 35] all reporting no main effect of commercial condition (although Anschutz et al.[33] reported a significant interaction with gender such that women ate more when exposed to food commercials compared with neutral commercials but the opposite was true for men). Three studies evaluated the role of TV content on food consumption;[24, 36, 37] two of these reported that participants ate more when watching boring or repeated showings of TV compared with engaging or novel TV,[36, 37] while the third reported no difference in consumption when participants watched funny, boring, or sad TV shows.[24] One study[38] compared snack consumption following exposure to either a clip of a cooking show or a nature show, reporting no difference in

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overall consumption between the groups (although more chocolate balls were consumed following exposure to the cooking show than the nature show).

Observational studies

Four observational studies examined dietary patterns while watching television in free-living settings among voluntary/convenience samples of adults in New Zealand,[39] African-American women in Chicago,[40] and young adults in the USA[41, 42] (Figure 3; Supplementary file 3).

The most detailed evidence comes from the study by Gemming et al.[39] in New Zealand, who linked 24-hour dietary recalls to wearable camera recordings on three separate measurement days, allowing for an in-depth assessment of the contextual details (e.g., location, environment, social context, presence of screens) surrounding each eating episode noted in the recall. In their sample, 17% of eating episodes (22% of meals and 12% of snacks) took place while watching TV. Snacks consumed while watching television were higher in energy (by 514kJ (123kcal)) than snacks consumed when no TV was viewed. There was no difference in meal energy intake or duration of meals or snacks while watching TV compared with when no screen was viewed.

The remaining three studies examined the contexts of self-reported eating and drinking using ecological momentary assessment[40] and food diary entries annotated with contextual information.[41, 42] These studies found that 30%[40] and 26%[41] of snacking and eating occasions, respectively, took place while watching TV. Amongst young adults in the USA, Stroebele and de Castro[42] reported that an average of 1.03 meals per day took place with the TV on, but meal size did not differ with or without the TV on. Compared to food and

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drink consumed while 'not doing anything else,' a larger proportion of snacks[40] and sugar-sweetened beverages[41] and smaller proportions of water, fruit, vegetables, and cereals/grains[41] were consumed while watching TV compared to while doing nothing else.

Characteristics of sitting while watching TV

Four experimental studies examined associations between watching TV and characteristics of sitting, specifically energy expenditure[43-45] and level of movement.[46] Three experimental studies compared the energy expenditure (EE) of watching TV with the EE of other sedentary behaviors.[43-45] The most robust evidence indicated that the EE of sitting while watching TV was lower than the EEs of typing, playing a handheld video game (both small effect sizes), and playing a screen-based video game while sitting (large effect size; Figure 4).[45] The other two studies found no difference between the EE of watching TV and the EE of typing at a desk,[44] reading at a desk,[44] or working on a laptop computer[43] in their samples (Figure 4; Supplementary file 4), although their relatively small sample sizes and lack of *a priori* power calculations may mean their non-significant findings were due to underpowered samples.

A laboratory-based study compared the level of movement captured using hip-worn, triaxial accelerometry while watching TV (seated on a chair without armrests) compared with other sedentary behaviors among adults.[46] Total movement (which we calculated as vector magnitude of acceleration) was higher while watching TV compared with while reading, writing, typing, and sitting both 'naturally' (i.e., with free movement) and motionless, but did not significantly differ from movement while playing cards or preparing food while seated (Figure 4).

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In sensitivity analyses using larger imputed correlation coefficients, the effect sizes slightly increased for the difference in the EE of TV compared with typing and for the difference in vector magnitude of acceleration for TV versus reading, preparing food, and sitting naturally (Supplementary file 2).

Affect during TV viewing

In two studies, participants self-reported their affect while watching TV within their everyday lives (Figure 5; Supplementary file 4). Using a day reconstruction approach in which leisure activities and affect whilst engaged in those activities from the previous day were reported, Kuykendall et al.[47] found that respondents generally experienced poorer affect while watching TV compared with other leisure activities, in terms of lower positive activated and deactivated affect (energetic/alert/vigorous and peaceful/at ease/serene, respectively) and higher negative deactivated affect (bored/tired/dull). However, negative activated affect (anxious/jittery/nervous) was generally lower while watching TV compared with other leisure activities. In a separate sample using similar methods, levels of relaxation were higher while watching TV than during physical activity (but similar to other leisure activities), levels of happiness and satisfaction were lower when watching TV compared with other activities (except for physical activity, which was similar to TV), and feelings of sadness did not differ between TV and other leisure activities.

In a study using experience sampling methods over the course of a week, Goodwin et al.[48] found that mean positive affect scores were lower while watching TV compared with while engaging in other leisure activities (e.g., hobbies) or while doing productive activities (e.g., work or volunteering); negative affect score did not differ while watching TV compared with while doing any other activities. In sensitivity analysis, only the effect size of the difference

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in positive affect during TV versus productive activities increased (from small to medium) with larger imputed correlations (Supplementary file 2).

The impact of TV viewing on sleep

One small experimental study (n=13)[49] and one small intervention study (n=16)[50], both in Japan, were included here (Supplementary file 4). Komada et al.[49] tested the effect of TV displays on melatonin and cortisol secretion at night in a laboratory setting. Melatonin and cortisol are hormones involved in the body's sleep-wake cycles, and the production of melatonin in the body promotes sleep and is known to be suppressed by bright light.[51] Komada et al.[49] found that a TV display with higher blue light intensity was associated with a suppressive effect on melatonin levels but not on cortisol levels. Asaoka et al.[50] conducted an intervention in which university students and elderly adults limited their TV time to 30 minutes a day for one week, following a week-long 'control' in which participants went about normal activities. During the week of restricted TV time, the university students went to bed significantly earlier and slept significantly longer compared to the control week; the intervention had no effect on sleep duration among the elderly.

DISCUSSION

This review aimed to improve our understanding of how TV might impact cardiometabolic health by evaluating and synthesizing available evidence on the effects of watching TV on food consumption, characteristics of sitting, affect and subsequent sleep. Most of the associations found were weak or inconsistent. There was no strong evidence from laboratory or free-living settings to suggest that food consumption while watching TV is higher than food consumption in other contexts. There was also no clear evidence to suggest that sitting while watching TV (in laboratory contexts) might be a particularly 'sedentary' behavior. The

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association between TV and affect was complex. Positive affect and certain aspects of negative affect (e.g., anxiousness or nervousness) tended to be lower while watching TV compared with other leisure activities, while other aspects of negative affect (e.g., sadness) did not differ; feelings of tiredness or boredom were higher while watching TV compared with other leisure activities. There was a very small amount of evidence to suggest TV may impact sleep by suppressing melatonin and by impacting bedtime, at least in younger people.

The impact of watching TV on food consumption is weak. In laboratory contexts in which fixed (and often large) amounts of food were placed in front of participants, participants tended to consume more while watching TV compared with while doing nothing else but more often than not this difference was not statistically significant, and there was very little evidence that consumption was higher while watching TV than during other potentially distracting activities such as listening to music. There is a very small amount of evidence that consumption of food while watching television (compared with doing nothing) may be associated with greater subsequent intake than consumption while not watching television. Evidence from free-living contexts, in which television viewing conditions and food availability are quite different, suggested that energy intake while watching TV versus not did not significantly differ for meals but differed by a small amount for each snacking episode.[39, 42] Observational studies found 17-26% of eating events and 12-30% of snacking events occurred while watching TV;[39-41] however, no evidence was available to describe what proportion of TV-watching was accompanied by eating to determine whether TV 'triggers' food consumption. It is worth highlighting that the studies cited most often as evidence to suggest links between TV and snacking or higher food intake are not based on measurements of food consumption *while watching TV*; rather, this evidence shows that those with higher TV time have higher total energy consumption, higher snack intake, or less-

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healthy diets than those with lower TV time with the causality of this association entirely unclear.[52-56] Further research in free-living contexts is needed to improve our understanding of the possible co-occurrence of TV watching and food consumption to test the hypothesis that watching TV leads to greater energy intake or greater intake of foods associated with cardiometabolic risk. The use of devices such as wearable cameras[e.g., 39] may prove particularly useful for this to assess the contexts of eating episodes (with and without TV), the contexts of TV viewing episodes (with and without eating), and the co-occurrence of the two.

There was little evidence to suggest that TV might be a particularly 'sedentary' behavior. Compared to other seated activities, one of three experimental studies found that energy expenditure was lower while watching TV and another experimental study showed incidental movement was *higher* while watching TV. The ecological validity of these findings is unclear as the settings for TV watching in these studies (e.g., on chairs without armrests, for short and fixed amounts of time) do not reflect TV-watching conditions in free-living contexts. We did not find any studies that examined whether watching TV may induce prolonged uninterrupted bouts of sitting, which have been shown to interfere with glucose and lipid metabolism.[57, 58] We also did not find any studies that examined the characteristics of sitting while watching TV in free-living contexts. There is evidence that those with higher self-reported TV time have higher free-living accelerometer-measured prolonged sedentary time[59] and fewer breaks in sedentary time,[59, 60] but these studies do not provide evidence that this prolonged sedentary time or lack of breaks is directly linked with time spent watching TV. Further research in free-living contexts should make use of mixed methodologies, for example by combining accelerometry with wearable cameras, using electromyographic shorts

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and diaries, or applying direct observation methods [see 61, 62 for review] to improve our understanding of characteristics of sitting specifically during TV viewing.

There is little evidence to support the hypothesis that TV may impact cardiometabolic health via impacting affect. The two studies that examined affect while watching TV in free-living contexts found that positive affect and feelings of happiness tended to be lower while watching TV than during other leisure activities, while the relationship between negative affect and TV depended on what aspect of negative affect was measured; overall negative affect and feelings of sadness were not higher or lower while watching TV, boredom and tiredness was higher, and anxiousness and nervousness were lower while watching TV compared with during other leisure activities.[47, 48] These findings suggest that TV may be a form of leisure that is 'numbing' in that it may ease negative feelings but not necessarily increase positive feelings. The role of TV in everyday lives as a method of stress management and relaxation and as a way to escape negative or depressed feelings has been previously described.[63-66] In light of this, it is possible that measured mood or affect while watching TV may represent 'emotional spillover' from activities preceding TV time.[47] From a measurement perspective, it would be useful to assess how affect might change over the course of TV viewing instances to better understand how TV might alter emotional states throughout the course of viewing.

Mental health status may be an important confounder of the association between TV and cardiometabolic health. Depression, for example, has been prospectively associated with both high TV time[67] and poor cardiometabolic health outcomes.[68] There is mixed evidence suggesting prospective associations between TV time and depression or depressive

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symptoms.[69-72] Further research is needed to disentangle the relationships between TV, mood/affect and mental health, and cardiometabolic health outcomes.

There is a very small body of evidence to suggest that television time may impact sleep, which may in turn impact cardiometabolic health. One small study included in this review found that blue light from a TV display suppressed melatonin,[49] which may lead to a disruption in circadian rhythms, and another small study found that a reduction in TV time led to an earlier bedtime in young people.[50] Thus, TV time may be linked to sleep disruption and later bedtimes, but much more evidence is needed to demonstrate this effect. The hypothesis that watching TV reduces sleep quantity or quality should be tested in people during their everyday lives, and might involve using wearable cameras to assess TV time and accelerometry to assess sleep quality and quantity.

This review did not identify strong evidence for any mechanism through which TV time has been posited to adversely impact cardiometabolic health. It is possible, however, that these factors may interact or have a multiplicative effect. For example, the combination of low energy expenditure of sitting while watching TV (based on laboratory evidence) with the slightly higher energy intake from snacks while watching TV could potentially produce a larger effect than either of these aspects on their own.

Importantly, even if such associations are observed in everyday lives, causality requires careful consideration as they may be the result of confounding, whereby, for example, chronic stress may prompt consumption of calorically-dense foods[73] and may also prompt TV-watching as a way to relax, perhaps concurrently. More generally, while the association between socioeconomic position (SEP) and TV time is likely to vary globally across cultural

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and economic contexts, low SEP is consistently linked to high TV time in 'Western' countries such as the UK,[74] the US,[75] Australia,[76] and Canada.[77] Within these contexts, low SEP itself is associated with some of the outcomes considered here, as well as with poor cardiometabolic health outcomes[78-81] and may thus confound the relationship between TV viewing, diet, sitting characteristics, affect and sleep, and health outcomes. More empirical evidence, particularly using ethnographic or mixed-methods approaches, is needed to understand the place of TV in everyday lives and its potential role in the development of cardiometabolic disease, with particular attention to the question of whether watching television is causally implicated in changing health-related practices. Experimental studies are also useful in helping disentangle causality.

The findings of this review must be interpreted in light of the limitations in the evidence base. The vast majority of included studies reported statistically significant results but often with very small effect sizes, suggesting possible publication bias. The associations seen in this review should therefore be interpreted with caution as statistically significant findings may be over-represented in the literature base. The majority of included studies examining food consumption and characteristics of sitting were laboratory-based, thus the ecological validity of such findings is unclear. Methods that capture these kinds of data in free-living contexts, such as use of wearable cameras or electromyographic shorts, may be useful for providing insights into the ways in which TV occurs in real-life settings. These methods will also be relevant for examining the contexts and possible health-related consequences of screen-related practices more broadly beyond just TV time. A key strength of this review is that it is the first to comprehensively examine and evaluate evidence of the assertions commonly made in the literature explaining why TV is particularly detrimental to cardiometabolic health.

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CONCLUSIONS

There is currently no strong evidence to suggest that TV impacts cardiometabolic health via impacts on food consumption, prolonged/inactive sitting, affect, or sleep. Further research exploring the putative mechanism(s) by which TV viewing might impact cardiometabolic health is required. We suggest that studies make use of innovative and/or mixed methods, such as wearable cameras, participant observation, or diaries, to improve our understanding of how TV fits within everyday lives and how it may impact cardiometabolic health outcomes, or whether relationships between TV time and cardiometabolic health reflect confounding.

DECLARATIONS

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Competing interests

The authors declare that they have no competing interests

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Ethics approval and consent to participate

Not applicable

Consent for publication

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Not applicable

Availability of data and material

All data generated or analyzed during this study are included in this published article and its supplementary information files

Authors' contributions

JW led the study. Both authors (JW and TP) contributed to the design, database searches, data collection, and analysis. JW constructed the plots. Both authors (JW and TP) contributed to and approved the final manuscript.

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Figure 1. Flow of studies through the review process

Figure 2. Summary plot of effect sizes for experimental effects of TV on food consumption compared with doing nothing else or doing something else

Figure 3. Summary plot of eating patterns while watching TV in free-living contexts.

Figure 4. Summary plot of effect sizes for characteristics of sitting while watching TV

Figure 5. Summary plot of effect sizes for affect while watching TV

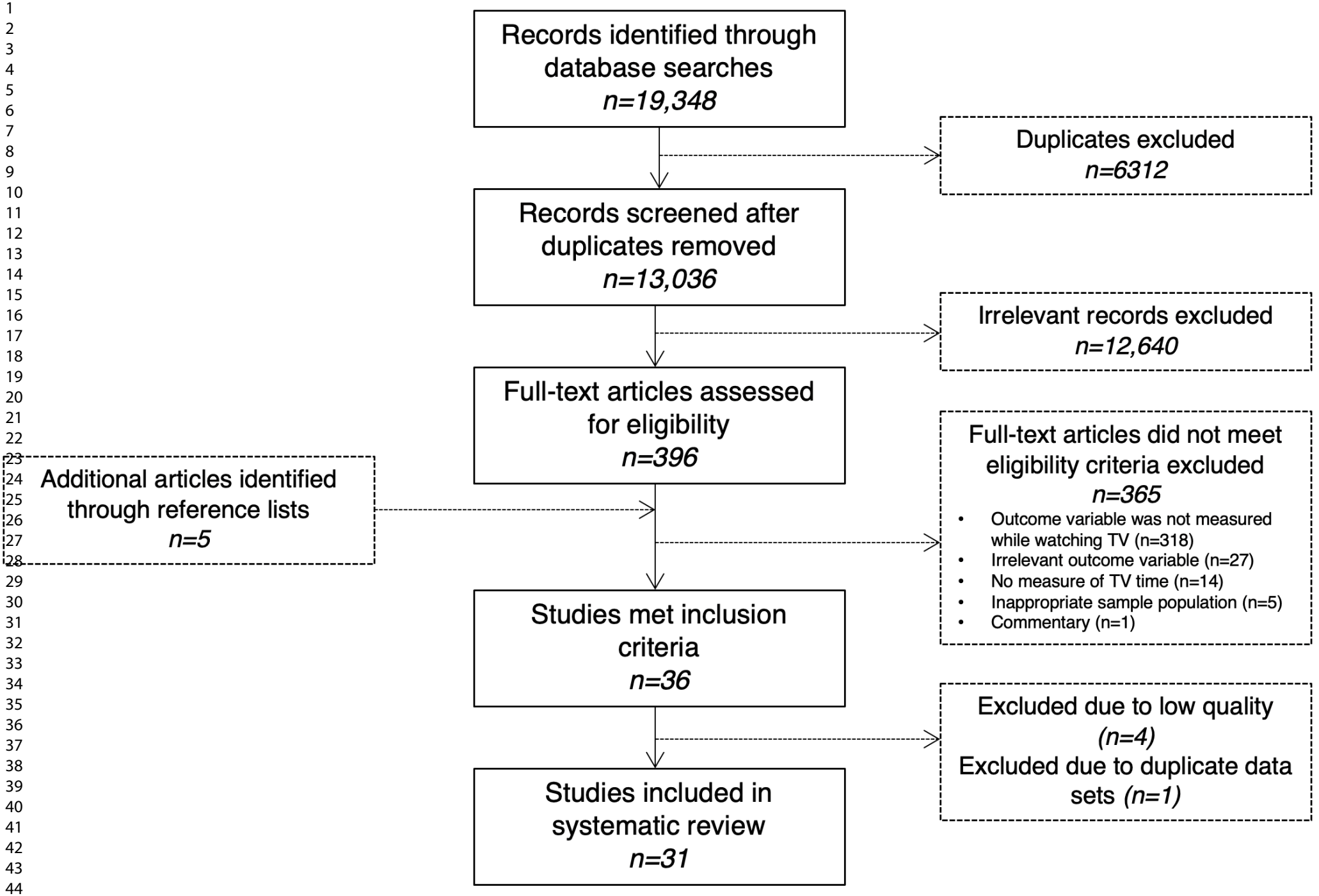
Supplementary file 1. Search strategy.

Supplementary file 2. Effect size calculations, with sensitivity analyses conducted for studies that used within-subjects or repeated-measures designs.

Supplementary file 3. Summary of characteristics, measurements, and findings of studies examining food and drink consumption while watching TV.

Supplementary file 4. Summary of characteristics, measurements, and findings of included studies examining characteristics of sitting or affect while watching TV and sleep after watching TV

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Records identified through database searches
n=19,348
Duplicates excluded
n=6312
Records screened after duplicates removed
n=13,036
Irrelevant records excluded
n=12,640
Full-text articles assessed for eligibility
n=396
Full-text articles did not meet eligibility criteria excluded
n=365
Outcome variable was not measured while watching TV (n=318)
Irrelevant outcome variable (n=27)
No measure of TV time (n=14)
Inappropriate sample population (n=5)
Commentary (n=1)
Additional articles identified through reference lists
n=5
Studies met inclusion criteria
n=36
Excluded due to low quality (n=4)
Excluded due to duplicate data sets (n=1)
Studies included in systematic review
n=31

Study	Study design (n)	Outcome measure	Association	
			Overall	Women
TV vs doing nothing				
Hetherington 2006	Within subjects (n=37)	Meal intake (kJ)	+ +	
Martin 2009	Within subjects (n=48)	Meal intake (kcal)	-	
Rosenthal 2017	Within subjects (n=20)	Meal intake (kcal, g)	+	
Moray 2007	Within subjects (n=20)	Meal intake (ounces)	+	
Braude 2014	Within subjects (n=62)	Snack intake (kJ)		+
Bellisle 2004	Within subjects (n=48)	Meal intake (kJ)		+ +
Bellisle 2009	Within subjects (n=40)	Meal intake (kJ, g)		?
Ogden 2013	Between subjects (n=81)	Snack intake (g)		+ +
Mittal 2010	Between subjects (n=32)	Snack intake (kJ)		+ + +
TV vs other condition				
Hetherington 2006	Within subjects (n=37)	With friends (kJ)	-	
Hetherington 2006	Within subjects (n=37)	With strangers (kJ)	+	
Martin 2009	Within subjects (n=48)	While reading (kcal)	+	
Bellisle 2004	Within subjects (n=48)	Listening to audio story (kJ)		-
Bellisle 2009	Within subjects (n=40)	Listening to audio story (kJ, g)		?
Bellisle 2009	Within subjects (n=40)	With strangers (kJ, g)		+
Blass 2006	Within subjects (n=20)	Listening to music (kcal)	+	
Lyons 2012	Between subjects (n=120)	With handheld video games (kcal)	-	
		With motion-controlled video games (kcal)	- -	
Ogden 2013	Between subjects (n=81)	While driving (g)		+ + +
Ogden 2013	Between subjects (n=81)	While talking to researcher (g)		+ + +

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Study	Method	Variable of interest	While watching TV compared to	Finding
Gemming 2015	**Wearable cameras** worn for three days linked with 24-hour diet recalls on those days (n=40)	Snack energy intake (kJ)	At home with no TV	▲+
		Meal energy intake (kJ)	At home with no TV	△+
Ghosh Roy 2018	**Ecological momentary assessment** (5 prompts per day) to ascertain the context of any snack or beverage consumption that occurred since the previous prompt (n=79)	Snacks consumed (%)	While doing nothing else	■+ ■+
		Sugar-sweetened beverages consumed (%)	While doing nothing else	□-
Laska 2011	**7-day food diary** with contextual information provided for each instance of eating or drinking (n=48)	Sugar-sweetened beverages consumed (%)	While doing nothing else	▲+
		Water consumed (%)	While doing nothing else	▲-
		Fruit and vegetables consumed (%)	While doing nothing else	▲-
		Cereals and grains consumed (%)	While doing nothing else	▲-
		Cookies and sweetened baked goods consumed (%)	While doing nothing else	△-
		Candy/gummy fruit snacks consumed (%)	While doing nothing else	△0
		Frozen desserts consumed (%)	While doing nothing else	△0
		Salty snacks consumed (%)	While doing nothing else	△+
		Fried side dishes consumed (%)	While doing nothing else	△-
Stroebele 2004	**7-day food diary** in which participants specified whether the TV was on in each instance (n=78)	Meal size (kJ)	Without the TV on	△?

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Study	Study design (n)	Outcome measure	Association		
			Overall	Women	Men
Experimental studies					
Energy expenditure (EE)					
Mansoubi 2015	Within subjects (n=51)	EE TV vs typing	- -		
		EE TV vs handheld game (PSP)	- -		
		EE TV vs seated video game (Wii)	- - - -		
Newton 2013	Within subjects (n=25)	EE TV vs typing	-		
		EE TV vs reading	-		
Creasy 2016	Between subjects (n=36)	EE TV vs computer	- -		
Level of movement					
van der Berg 2019	Within subjects (n=18)	Acceleration TV vs reading	+ + +		
		Acceleration TV vs writing	+ + + +		
		Acceleration TV vs typing	+ + + +		
		Acceleration TV vs playing cards	0		
		Acceleration TV vs preparing food while seated	+		
		Acceleration TV vs 'natural' sitting	+ + +		
		Acceleration TV vs sitting motionless	+ + + +		

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Study	Study design (n)	Outcome measure	Finding
Kuykendall 2020	Cross-sectional (n=538)	*Positive activated affect*	
		TV vs physically active leisure	- - - -
		TV vs social leisure	- - - -
		TV vs cognitively stimulating leisure	- -
		TV vs creative leisure	- - -
		TV vs cultural activities	- - - -
		Positive deactivated affect	
		TV vs physically active leisure	△-
		TV vs social leisure	△+
		TV vs cognitively stimulating leisure	-
		TV vs creative leisure	- -
		TV vs cultural activities	△+
		Negative activated affect	
		TV vs physically active leisure	-
		TV vs social leisure	-
		TV vs cognitively stimulating leisure	-
		TV vs creative leisure	△-
		TV vs cultural activities	△-
		Negative deactivated affect	
		TV vs physically active leisure	+ +
		TV vs social leisure	+ +
		TV vs cognitively stimulating leisure	+ +
		TV vs creative leisure	+ +
		TV vs cultural activities	+ +
Goodwin 2005	Cross-sectional (n=69)	*Positive affect*	
		TV vs other leisure	- -
		TV vs productive activities	- -
		TV vs maintenance activities	□-
		Negative affect	
		TV vs other leisure	□0
		TV vs productive activities	□-
		TV vs maintenance activities	□0

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Search strategy for PubMed

All fields: “television time” OR “television viewing” OR “television watching” OR sedentary behavio?r OR “sedentary time” OR “screen time” OR “watching television” OR “viewing television”

Limits: Date range 2000 to present

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Supplementary file 2. Effect size calculations, with sensitivity analyses conducted for studies that used within-subjects or repeated-measures designs.

Study	Condition comparison or outcome variable	Standardized mean difference (95% CI)	Notes
Characteristics of sitting			
Creasy 2016	Energy expenditure (kcal/15min) TV vs computer	-0.23 (-0.88, 0.43)	
Mansoubi 2015	Energy expenditure (METs) TV vs typing	-0.41 (-0.70, -0.12)	Imputed correlation 0.5, total sample
		-0.45 (-0.74, -0.17)	Imputed correlation 0.6, total sample
		-0.61 (-0.91, -0.31)	Imputed correlation 0.8, total sample
	Energy expenditure (METs) TV vs PSP	-0.30 (-0.58, -0.02)	Imputed correlation 0.5, total sample
		-0.33 (-0.62, -0.05)	Imputed correlation 0.6, total sample
		-0.47 (-0.76, -0.18)	Imputed correlation 0.8, total sample
	Energy expenditure (METs) TV vs seated Wii	-1.66 (-2.08, -1.24)	Imputed correlation 0.5, total sample
		-1.78 (-2.22, -1.33)	Imputed correlation 0.6, total sample
		-2.11 (-2.61, -1.62)	Imputed correlation 0.8, total sample
Newton 2013	Energy expenditure (METs) TV vs typing	-0.19 (-0.59, 0.20)	Imputed correlation 0.5
		-0.22 (-0.61, 0.18)	Imputed correlation 0.6
		-0.31 (-0.71, 0.09)	Imputed correlation 0.8
	Energy expenditure (METs) TV vs reading	-0.05 (-0.45, 0.34)	Imputed correlation 0.5
		-0.06 (-0.45, 0.33)	Imputed correlation 0.6
		-0.08 (-0.47, 0.31)	Imputed correlation 0.8
van der Berg 2019	Vector magnitude (cpm) TV vs reading	0.79 (0.26, 1.32)	Imputed correlation 0.5
		0.84 (0.30, 1.37)	Imputed correlation 0.6
		0.96 (0.40, 1.52)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs writing	0.97 (0.41, 1.53)	Imputed correlation 0.5
		0.99 (0.43, 1.55)	Imputed correlation 0.6
		1.03 (0.46, 1.60)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs typing	0.92 (0.37, 1.47)	Imputed correlation 0.5
		0.95 (0.40, 1.51)	Imputed correlation 0.6
		1.04 (0.47, 1.62)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs playing cards	0.00 (-0.46, 0.46)	Imputed correlation 0.5
		0.00 (-0.46, 0.46)	Imputed correlation 0.6
		0.01 (-0.46, 0.47)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs preparing food while seated	0.14 (-0.32, 0.61)	Imputed correlation 0.5
		0.16 (-0.31, 0.62)	Imputed correlation 0.6
		0.22 (-0.24, 0.69)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs natural sitting	0.71 (0.19, 1.22)	Imputed correlation 0.5
		0.78 (0.25, 1.31)	Imputed correlation 0.6

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		1.02 (0.45, 1.59)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs imposed sitting	0.94 (0.39,1.50)	Imputed correlation 0.5
		0.97 (0.41, 1.52)	Imputed correlation 0.6
		1.01 (0.44, 1.58)	Imputed correlation 0.8
Diet			
Bellisle 2004	Food intake (kJ), TV vs control	0.46 (0.16, 0.76)	Imputed correlation 0.5
		0.51 (0.21, 0.82)	Imputed correlation 0.6
		0.73 (0.41, 1.05)	Imputed correlation 0.8
Hetherington 2006	Food intake (kJ), TV vs control	0.34 (0.01, 0.67)	Imputed correlation 0.5
		0.38 (0.05, 0.71)	Imputed correlation 0.6
		0.52 (0.18, 0.86	Imputed correlation 0.8
Mittal 2011, experiment 1	Food intake (kJ), TV vs control	0.59 (-0.13, 1.29)	Snack consumed while watching TV
	Food intake (kJ), TV vs control	0.51 (-0.19, 1.22)	Meal consumed after TV/no-TV condition
Ogden 2013	Food intake (g), TV vs control	0.45 (-0.17, 1.07)	
Bellisle 2004	Food intake (kJ), TV vs audio story	-0.04 (-0.32, 0.25)	Imputed correlation 0.5
		-0.04 (-0.32, 0.24)	Imputed correlation 0.6
		-0.06 (-0.34, 0.23)	Imputed correlation 0.8
Hetherington 2006	Food intake (kJ), TV vs eating with friends	-0.13 (-0.46, 0.19)	Imputed correlation 0.5
		-0.15 (-0.47, 0.18)	Imputed correlation 0.6
		-0.21 (-0.53, 0.12)	Imputed correlation 0.8
Lyons 2012	Food intake (kcal), TV vs handheld video game	-0.06 (-0.50, 0.37)	Total intake
	Food intake (kcal), TV vs motion-controlled video game	0.36 (-0.09, 0.80)	Total intake
Ogden 2013	Food intake (g), TV vs driving	0.77 (0.13, 1.40)	
	Food intake (g), TV vs talking to researcher	0.73 (0.08, 1.37)	
Mood			
Goodwin 2005	Positive affect, TV vs other leisure	-0.25 (-0.49, -0.01)	Imputed correlation 0.5
		-0.28 (-0.52, -0.04)	Imputed correlation 0.6
		-0.39 (-0.63, -0.14)	Imputed correlation 0.8
	Positive affect, TV vs productive activities	-0.39 (-0.64, -0.15)	Imputed correlation 0.5
		-0.44 (-0.69, -0.19)	Imputed correlation 0.6
		-0.62 (-0.88, -0.36)	Imputed correlation 0.8
	Positive affect, TV vs maintenance activities	-0.01 (-0.25, 0.22)	Imputed correlation 0.5
		-0.01 (-0.25, 0.22)	Imputed correlation 0.6
		-0.02 (-0.25, 0.22)	Imputed correlation 0.8
	Negative affect, TV vs other leisure	0.00 (-0.24, 0.24)	Imputed correlation 0.5
		0.00 (-0.24, 0.24)	Imputed correlation 0.6
		0.00 (-0.24, 0.24)	Imputed correlation 0.8
	Negative affect, TV vs productive activities	-0.08 (-0.31, 0.16)	Imputed correlation 0.5
		-0.09 (-0.32, 0.15)	Imputed correlation 0.6

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		-0.12 (-0.36, 0.12)	Imputed correlation 0.8
	Negative affect, TV vs maintenance activities	0.00 (-0.23, 0.24)	Imputed correlation 0.5
		0.00 (-0.23, 0.24)	Imputed correlation 0.6
		0.01 (-0.23, 0.24)	Imputed correlation 0.8

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Supplementary File 3. Summary of characteristics, measurements, and findings of studies examining food and drink consumption while watching TV.

First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
Observational/free-living studies							
Gemming 2015 [39] Quality = good	Cross-sectional New Zealand	Recruited from the community and university (convenience)	Healthy adults in Auckland, n=40 (50%) Mean ages 35 (SD=17) and 28 (SD=7) for men and women, respectively	Instances of television time (only while eating) as identified during processing of SenseCam (wearable camera) recordings across 3 days	NA	Energy intake (kJ), energy density (kJ/g), and duration (min) of meals and snacks across contexts (including during TV) measured by 24-hour multiple pass dietary recall on days that wearable camera was worn	Snacks at home when viewing television were higher in energy intake (+514 kJ (47, 1077)) compared to snacks when no television was viewed. Meals at home while watching TV were not significantly longer (3.1 min (-0.6, 6.7)) than meals not viewing screens, and were not significantly higher in energy (543kJ (-32, 1120)).
Ghosh Roy 2018 [40] Quality = good	Cross-sectional USA *African American Women's Daily Life Study*	Recruited from community and university (convenience)	African American women aged 25-65 in Chicago, n=79 (100%) Mean age 44.6 (SD=10.6)	In each instance of reported snack or sweetened beverage consumption (see 'outcome measurement'), participants were asked what they were doing while eating or drinking; watching television was an option	Education, per capita income, age, auto ownership, BMI, weekday or weekend day measurement, hunger status, fast food restaurant and convenience store density in home neighborhood	Self-reported consumption of ≥1 snack food item (including French fries, salty snacks, cookies or sweetened baked goods, chocolate/candy, ice cream/frozen dessert) or ≥1 sweetened beverage in response to smartphone prompts which came 5 times per day across one week (ecological momentary assessment)	Consumption of snack foods was significantly more likely to occur while watching TV versus while doing nothing else (OR 1.8 (95%CI 1.2, 2.7)); no such association was seen for sugar-sweetened beverage consumption (OR 0.9 (95%CI 0.5, 1.6)).
Laska 2011 [41] Quality = fair	Cross-sectional USA	Recruited from community and university (convenience)	Young adults aged 18-23 from four groups (attending university and living	All instances of eating and drinking were logged on a PDA over a 7-day	N/A	Proportion of food consumption while watching TV compared to while not doing	A larger proportion of calorically sweetened beverages (24% vs 15%) and smaller proportions of water

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			on campus; attending university and living independently off-campus; attending university and living with parents; not attending university and living independently), n=48 (12 from each group), 56% female Mean age approximately 21	period, and followed up with contextual questions, including whether they were watching TV during that instance of eating		anything else, separately by food group: calorically sweetened beverages, water, coffee/tea drinks, milk, cookies and sweetened baked goods, candy/gummy fruit snacks, fruits and vegetables (fresh or 'other'), frozen desserts, non-milk dairy products, entrees, cereals/grains, and fried side dishes	(12% vs 17%), fruits and vegetables (26% vs 36%), and cereals/grains (25% vs 31%) were consumed while watching TV compared to while not doing anything else, respectively ($p<0.05$ for all). There were no significant differences among the other food groups listed in the previous column.
Stroebele 2004 [42] Quality = fair	Cross-sectional USA	Recruited from university for course credit	University students, n=78 (82%) Mean age 22 (0.9)	Television time was recorded on a diary in 15-minute intervals. Instances of eating while watching TV were recorded on food diaries (see 'outcome measurement' column)	N/A	Meal sizes, caloric intake, and meal frequency, ascertained from a 7-day food diary in which participants recorded the volume of all food and drinks consumed, with a specification concerning whether the TV was on in each instance	Meal sizes did not differ with and without TV (data not shown). There was no correlation between TV time (h/day) and total caloric intake or meal frequency (data not shown).
Experimental/laboratory studies							
Study, quality	**Study design, country**	**Target sample Sample size (% female) Mean (SD) age or age range**	**Study aim and experimental protocol**		**Main outcome measurement**	**Findings**	
Anschutz 2011 [33] Quality = fair	Between subjects Netherlands	Non-overweight university students, n=82 (50%) Mean age (SD) for men and women was 20.9 (2.5) and 20.4 (1.5), respectively	Aim: to examine whether food commercial exposure would increase concurrent snack food intake among young adults Protocol: participants were randomized to 'food commercial' condition (30-minute nature TV clip interrupted by 3 food and 5 non-food commercials) or 'neutral commercial' condition (30-minute nature TV clip with 8 non-food commercials)		Food intake (kcal) based on weight of crisps and M&Ms consumed during the session Amount of food provided not specified	There was no effect of commercial condition on food intake (b=-0.52, SE=43.12, $p>0.05$) and the addition of commercial condition to the model had a negligible effect (change in R^2 <0.001); the interaction between sex and commercial condition was significant (women ate more when exposed to food commercials compared to neutral, while men ate more when exposed to neutral compared to food commercials)	

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Bellisle 2004 [20] Quality = fair	Within subjects France	Women aged 18-50 with BMI between 18.5 and 24.9, n=48 (100%) Mean age 29.9 (1.4), range 18-50	Aim: To assess the impact of two environmental stimuli (TV and audio story) on meal intake Protocol: Participants ate a meal in four conditions each lasting minimum of 30 minutes and spaced one week apart: control (alone), while watching TV (no references to food or eating), while listening to a recorded detective story, and last meal (alone); order of TV and auditory conditions was randomized	Food intake (kJ) of meal (shepherd's pie and fruit sherbet) ~4850kJ (1160kcal) available	Compared to the baseline control condition (1751 (SE=84) kJ), meal size was larger in the TV (2023 (SE=84) kJ) and audio recording (2044 (SE=84) kJ) conditions ($p<0.001$). Meal size did not differ between the TV and audio recording conditions ($p>0.05$).
Bellisle 2009 [23] Quality = fair	Within subjects France	Healthy women of normal weight, half with high and half with low dietary restraint, total n=40 (100%) Overall mean age or range not reported	Aim: To assess the effects of dietary restraint and environmental factors on meal intake Protocol: Participants ate a meal in five conditions (randomized order) each lasting a minimum of 30 minutes and spaced one week apart: alone in a quiet room, in groups of 3, alone with TV (no food cues), alone with TV (including food advertisements), alone while listening to audio detective story	Food intake (g, kJ) of meal (shepherd's pie and fruit sherbet) ~6000kJ (1430kcal) available	Significantly less food (g, kJ) was consumed in the group meal condition compared to the others ($p<0.02$) in the pooled sample. Consumption was not different between the other conditions (data not shown).
Blass 2006 [29] Quality = fair	Within subjects USA	Undergraduate students, n=20 (75%) Mean age or range not reported	Aim: to test whether more was eaten when watching TV compared to not Protocol: participants were assigned to either pizza or mac and cheese (assignment protocol not stated) and either watched TV for 30 minutes (program of choice with original commercials included) or listened to *Rachmaninoff's Second Symphony* for 30 minutes; the experiment was repeated a week later for the other condition (counterbalanced order)	Food intake (kcal) of pizza or macaroni and cheese 12-inch pizza or family-size bowl (~900g) of macaroni and cheese available	More calories were consumed with the TV on (793.7kcal) than with TV off (538.2 kcal), $p<0.001$. There was no interaction effect between food type and condition.
Bodenlos 2013 [38] Quality = fair	Between subjects USA	Undergraduate students, n=80 (73%) Mean age 19.5 (1.0), range 18-22	Aim: to test whether exposure to a cooking show affected caloric intake Protocol: Participants were randomized to watch either a cooking show or Planet Earth for 10 minutes, followed by a 'taste test' of cheese curls, chocolate covered candies, and carrots	Intake (kcal) of cheese curls, chocolate covered candies, and carrots 800kcal in total available	No significant difference in overall calories consumed between the two conditions ($F(1,74)=3.32$, $p=0.07$), $d=0.38$ Those in the cooking show group consumed significantly more calories from chocolate covered candies than those in the nature group ($F(1,74)=3.90$, $p=0.05$), $d=0.51$
Boyland 2017 [34]	Within subjects UK	Females recruited from university (staff and	Aim: to examine consumption responses to televised food commercials in overweight and lean adult females	Intake (kcal) of pizza	No main effect of condition or weight status on pizza intake or interaction between weight status and condition (p values >0.28).

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		students) and surrounding area, n=55 (100%) Mean age 32.4 (9.8), range 20-62	Protocol: Participants attended two sessions in a counterbalanced order: TV show including commercials for high-calorie palatable foods and TV show including non-food commercials. Following the show, participants were given pizza to eat.	2056kcal of pizza available	
Braude 2014 [21] Quality = fair	Within (and between) subjects Australia	Female undergraduates, n=62 (100%) Mean age 19.6 (2.2), range 18-29	Aim: To test whether TV affected sensory specific satiety Protocol: Participants were randomized to a single- or variety-foods group; both groups took part in a TV (*Friends,* no commercials) and no-TV condition (counterbalanced order) lasting 20 minutes and spaced one week apart. In the single-foods group, participants chose their preference between Maltesers, Skittles, almonds, or salted potato chips and were given 80g of it. Participants in the variety-foods group were given a 20g bowl of each food (totaling 80g).	Food intake (kJ, number of items) Variety condition totaled ~1600kcal available; single food condition depended on which food was picked	Participants consumed significantly more energy (z=2.37, p<0.025) and more items of food (z=2.07, p<0.05) in the TV condition compared to the no-TV condition.
Chapman 2014 [36] Quality = fair	Within subjects Sweden	Healthy female women, n=18 (100%) Mean age 22 (1.3)	Aim: to explore how the content of TV programs differentially impacts simultaneous eating behavior Protocol: Participants took part in three conditions (counterbalanced order) each lasting 30 minutes and spaced one week apart: watching an engaging TV program (comedy), a boring program (art lecture) or reading a non-engaging text about insects. Bowls of M&Ms and grapes were available in each condition.	Food intake (grams, kcal) 250 grams (~1200kcal) of M&Ms and 300 grams (~200kcal) of grapes were available	Significantly more food (grams) was consumed during the boring program (125g) compared to the engaging program (82g, p<0.01), but no difference in kcal (data not shown). No differences in grams or kcal consumption were reported between the engaging program vs text condition (109.3g vs 81.9g, p=0.05) or the boring program vs text conditions (+15.5 grams, p=0.26) (kcal data not shown).
Hetherington 2006 [22] Quality = fair	Within subjects UK	University staff and students, n=37 (43%) Mean age 28.3 (1.8), range 18 to 54	Aim: To test whether the social facilitation of eating occurs as a function of distraction (by comparing consumption in social contexts with TV and control conditions) Protocol: Participants took part in four conditions in a counterbalanced order at least 3 days apart: eating alone, eating with TV (game show, duration not fixed), eating with strangers, and eating with friends. The same buffet meal was available in all conditions.	Food intake (kJ) A buffet-style meal (13,743kJ/3283kcal) with 9 different food items was available	Participants consumed significantly more when eating in front of the TV (4350 (SE=252) kJ) and when eating with friends (4565) SE=272) kJ) compared to the control condition (3861 (SE=200) kJ). No differences were found between eating with strangers and any other condition.

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Higgs 2009 [31] Quality = fair	Within subjects UK	Female undergraduates of normal weight, n=16 (100%) Mean age 19 (1)	Aim: to examine whether watching TV during lunch would increase afternoon snack intake due to impaired memory Protocol: Participants had a standardized lunch either in quiet solitude (control) or with the TV on (comedy show with no references to food). Both conditions lasted 10 minutes and all participants ate the entire lunch in that time period. At least 2.5 hours later on the same day, participants were given three kinds of cookies broken into small pieces to taste test.	Amount (g) of cookies consumed Three kinds of cookies (40g of each) were available, totaling ~600kcal available	Participants consumed significantly more grams of cookies after the lunch-with-TV condition compared to after the lunch-without-TV condition ($F(1,15)=12.6$, $p<0.01$)
Lyons 2012 [30] Quality = fair	Between subjects USA	Adults aged 18-35, n=120 (50%) Mean age 24.1 (4.4)	Aim: to compare differences in energy intake among 2 sedentary screen behaviors (TV and video games) and one potentially active screen behavior (motion-controlled video game). Protocol: Participants were randomized to a TV, video game (PS3), or motion-controlled video game (Wii or Xbox 360) group. Each condition lasted 1 hour. During the condition, snacks and beverages were available for consumption.	Amount (kcal) of snacks and soda consumed Snack foods included M&Ms, baked Lay's, Doritos, and trail mix (nuts with dried fruit) (amounts provided not specified); soda included cans (3 of each) of Coca-Cola, Diet Coke, and Mountain Dew.	Energy consumption between the three groups was not significantly different ($p=0.07$). Mean intake during TV was 716 (407) kcal; during video games was 747 (540) kcal; and during motion-controlled video game was 553 (498) kcal.
Martin 2009 [25] Quality = good	Within subjects USA	Healthy adults aged 18 to 54 (BMI 20-35), n=48 (54%) Mean age 31.9 (SE=1.5)	Aim: to test the effect of TV viewing with and without ads on energy intake compared to a control and a reading condition Protocol: participants completed four conditions (each lasting 32 minutes) in a random order on two test days (experimental lunch and dinner after consuming a standardized breakfast): control, reading, TV without ads, TV with ads (half food-related, half not). The same buffet meal was available for all conditions.	Food intake (kcal) Buffet-style meals were available including 19 different items (totaling 4921kcal)	Energy intake did not vary by experimental condition ($F(3, 131)=0.30$, $p=0.81$); partial $\eta^2 \leq 0.01$
Mathur 2015 [37] Quality = fair	Within subjects Australia	Female undergraduates, n=45 (100%)	Aim: to test whether variability in engagingness of TV affects food intake Protocol: Participants completed two conditions in a randomized order one week	Food intake (g, kJ) of their first and second preferred snacks	Significantly more snack food was consumed in the same condition (mean 76.2g, SD=36.2) than the different condition (mean 66.7g, SD 37.3, $z=1.990$ (*one-tailed), $p<0.05$).

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		Mean age 19.5 (SD=2.2), range 18-29	apart: watching the same episode of *Friends* twice back-to-back ('same'), or 'watching two different episodes back-to-back ('different'), with each episode lasting 20 minutes and without advertisements). The first episode was watched without food; for the second episode, regardless of condition, they were given 80g each of their preferred two snacks (given the choices of almonds, Pringles, and M&Ms).		
Mittal 2010, experiment 1 [24] Quality = fair	Between subjects Australia	Females from the university with BMI between 18 and 25, n=32 (100%) Mean ages 20.8 (SD 3.8) and 20.3 (SD 3.9) in the two groups	Aim: to assess whether snacking with concurrent TV would result in eating more at a test meal later on Protocol: Participants were assigned to either a snack-with-TV or snack-without-TV condition, both lasting 20 minutes (TV program was choice between *Seinfeld* and *Friends*). In both conditions, they consumed as much as they wanted of their snack (potato chips, chocolate balls, and coke or orange juice). Then, after 45 minutes of psychological tests, they were given lunch (test meal).	Food intake (kJ) of the snack in the TV versus no-TV condition 2080kJ (~500kcal) offered (chocolate balls, Pringles, Coke/orange juice) Food intake (kJ) of the test meal following the snack conditions 2470kJ (~600kcal) offered (sandwiches, biscuits, crackers and dip)	Energy consumption of the snack did not differ between the snack-with-TV (1855.9 (264.2)kJ) and snack-without-TV (1667.7 (362.0) kJ) conditions (test statistics not shown). Energy consumption of the test meal was significantly higher among those who had watched TV with their snack (1584.6 (516.4) kJ vs 1354.9 (335.6) kJ), $p<0.05$ (*one-tailed)
Mittal 2010, experiment 2 [24] Quality = fair	Between subjects Australia	Females from the university, n=84 (100%) Overall mean age not reported but around 21 years	Aim: to determine whether the content of TV shows watched while snacking affect recall accuracy of food eaten during the show and intake on a later TV-free test meal Protocol: Participants were assigned to one of four conditions, each lasting 20 minutes: snack-without-TV, snack-with-boring-TV, snack-with-funny-TV, or snack-with-sad-TV. They consumed their snack (potato chips, chocolate balls, and coke or orange juice). Then, after 45 minutes of psychological tests, they were given lunch (test meal).	Food intake (kJ) between the snack-TV conditions 2411kJ (575kcal) offered (chocolate balls, Pringles, Coke/orange juice) Food intake (kJ) of the test meal following the snack conditions	There was no difference in the energy consumption of the snack across the four groups (test statistics not shown). Mean snack consumptions (kJ) were 2308.3 (262.4) for boring, 2291.6 (223.5) for sad, 2250.2 (289.7) for funny, and 2194.9 (322.0) for control. Energy consumption of the test meal was significantly higher in all three TV conditions compared to the no-TV condition (post-hoc pairwise $p<0.02$ in all three cases), but consumption did not differ between the three TV groups (test statistics not shown). Mean test meal consumptions (kJ) were 2507.0 (438.2) for boring, 2842.0 (452.4) for sad, 2637.6 (540.3) for funny, and 2147.9 (527.2) for control.

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				4021kJ (960kcal) offered (sandwiches, crackers and dip, ice cream and chocolate, Coke/orange juice)	
Moray 2007 [28] Quality = fair	Within subjects USA	University students, n=20 (50%) Mean age 20.8 (range 18 to 23)	Aim: to determine whether TV promotes increased food consumption because it impairs one's ability to accurately estimate food intake Protocol: participants completed two conditions in counterbalanced orders, each lasting 25 minutes: eat with and without TV. After the meal, participants were asked to estimate how much food they had just consumed, using a visible tablespoon as a reference for 'one unit' (which was equivalent to 1.5oz of food).	Food intake (oz) between the TV and no-TV conditions Accuracy of food estimation, determined as estimated units minus actual units consumed 16oz macaroni and cheese provided	There was no difference in amount of food consumed between the TV and no-TV condition (2.5%, p>0.05). Unit estimation error was greater ($T_{0.05}$ 2.2, p=0.022) in the TV vs no-TV condition.
Ogden 2013 [27] Quality = fair	Between subjects UK	Females aged ≥18 who had a manual driving license, n=81 (100%) Mean age 22 (5.18), range 18-40	Aim: to compare the impact of different forms of distraction on eating behavior Protocol: participants were randomized to one of four conditions, each lasting 7 minutes: driving (simulation with a manual transmission), TV (*Friends*), social interaction (conversing with one of the researchers), or sitting alone. 100g of hula hoops (potato snacks) were available in each condition.	Grams of hula hoops consumed 100g were provided (~520kcal)	Those in the TV condition consumed more (28.61 (24.44) g) than those in the social (14.16 (12.33)g, p<0.01) and driving (14.02 (10.34)g, p<0.05) conditions, but no significant difference was seen compared to the alone condition (18.21 (20.9)g, p=0.06).
Ogden 2017 [32] Quality = fair	Between subjects UK	Females from the university, n=60 (100%) Mean age 24 (3.3)	Aim: to compare the impact of distraction and dietary restraint on food intake during a subsequent taste test Protocol: participants were randomized to one of three conditions, each lasting 5 minutes: watching TV, walking (along the corridor), and social interaction (talking with another research participant). During each condition, participants were told to consume a cereal bar. After the condition, participants took part in a taste test lasting 7 minutes during which they could consume as much of the foods (M&Ms, carrot sticks, grapes, and hula hoops) as they wished.	Mass and energy (kcal) of food consumed during the taste test 150g of each of the foods were provided	There was no main effect of condition on total mass or total calories consumed during the taste test (values not shown).

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Rosenthal 2017 [26] Quality = good	Within subjects USA	Young adults aged 18 to 35 with BMI between 18.5 and 24.9kg/m^2 n=20 (85%) Mean age 22.3 (3.7)	Aim: to test the independent and interactive effects of TV and portion size on food intake Protocol: participants took part in four randomized conditions one week apart: small portion with and without TV and large portion with and without TV. In the 'no TV' condition, participants ate quietly with no other activities for 30 minutes; the TV condition was a 30-minute episode of *Scandal*.	Food intake (grams and kcals) Small portion condition provided 1083 kcal (650 grams) of macaroni and cheese plus salad with dressing; the large portion provided 2166 kcal (1300 grams) of the same foods	Only a main effect of portion size was found for grams and kcals consumed. No main effect for TV was found for food intake ($F(1,16)=0.39$, $p=0.54$, $d=0.11$) and ($F(1,16)=1.10$, $p=0.31$, $d=0.17$)) measured as grams and kcals, respectively. No interactions for TV and portion size were found (>0.05).
Wonderlich-Tierney 2013 [35] Quality = fair	Between subjects USA	Undergraduates, n=83 (52%) Mean age 19.6 (3.5)	Aim: to examine the impact of TV advertisements on food intake Protocol: participants were randomized to one of three groups, each lasting 60 minutes: TV with food advertisements, TV with non-food advertisements, and TV with no advertisements. A jar of cookies was available throughout each condition.	Number of cookies eaten 26 cookies were available in each condition	There was no effect of condition on number of cookies eaten ($F(2,69)=1.50$, $p=0.23$), $R^2=0.04$

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Supplementary file 4. Summary of characteristics, measurements, and findings of included studies examining characteristics of sitting or mood while watching TV and sleep after watching TV

First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
Characteristics of sitting							
Creasy 2016 [43] Quality = good	Experimental (between subjects) USA	Recruited at university (convenience)	Adults aged 18 to 40 with BMI between 20-35, n=74 (58%)* Mean age 24.2 (3.6) *In full study. Only those whose baseline conditions were sitting while watching TV or sitting at the computer (n=18 in each condition) are described here	Lab-based exposure (15 minutes) *Experimental aim: to examine the cumulative energy expenditure of various combinations of sitting, standing, and walking*	N/A	Energy expenditure (kcal per 15min, indirect calorimetry)	TV EE (18.66 (4.01) kcal/15min) not significantly* different from computer EE (19.63 (4.37) kcal/15min) **Based on confidence intervals crossing zero in our calculation of the standardized mean difference between the two groups; the paper itself did not test this (see Additional file 1)*
Mansoubi 2015 [45] Quality = good	Experimental (within subjects) UK	Recruited at university and in community in 2x2 format to obtain equal numbers of men/women and healthy weight/obese participants (convenience)	Adults aged ≥18, 47% of whom were obese, n=51 (51%) Overall mean age not reported	Lab-based exposure (10 minutes) *Experimental aim: to measure the energy expenditure of sedentary behaviours in normal weight and obese participants*	N/A	Energy expenditure (METs, indirect calorimetry)	TV EE (METs) (1.33 (0.24)) was significantly* lower than typing (1.45 (0.32)), playing a handheld computer game (1.41 (0.28)), and playing a screen-based computer game (2.06 (0.50)) * Based on confidence intervals crossing zero in our calculation of the standardized mean change between the conditions; the paper itself did not test this (see Additional file 1)
Newton 2013 [44] Quality = fair	Experimental (within subjects) USA	Recruited in community (convenience)	African American adults aged ≥18, n=25 (60%)	Lab-based exposure (30 minutes) *Experimental aim: to examine the energy*	N/A	Energy expenditure (METs, whole-room calorimetry)	TV EE (METs) (1.03 (0.15)) was not significantly different from EE of typing (1.06 (0.15)), or reading at a desk (1.04 (0.20))

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First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
			Mean age 38.2 (11.4), range 20-56	*expenditure of common sedentary behaviours*			
van der Berg 2019 [46] Quality = fair	Experimental (within subjects) Netherlands	Recruited from the university (convenience)	Adults aged 18 to 48 without mobility limitations, n=18 (44%) Mean age 27.3 (6.5)	Lab-based exposure (7 minutes) seated in a chair without armrests while watching TV on a computer screen *Experimental aim: to identify whether 'dynamic sitting (e.g., fidgeting) can be identified from triaxial accelerometry counts*	N/A	Accelerometer vector magnitude* while watching TV compared to natural sitting (allowed to move freely), imposed sitting (motionless), imposed fidgeting (told to move the upper body side to side and back to front), while reading a newspaper, while writing a letter, while playing cards, while typing on a computer, and while preparing food while seated *Paper reported mean acceleration in the vertical, anteroposterior, and mediolateral planes separately; we converted this to mean vector magnitude by taking the square root of the summed squares of all three axes	Accelerometer vector magnitude was significantly* higher while watching TV compared to natural and 'motionless' sitting, reading, writing, and typing; it was significantly lower while watching TV compared to imposed fidgeting. There was no difference in vector magnitude between TV and playing cards or between TV and preparing food while seated. **Based on confidence intervals crossing zero in our calculation of the standardized mean difference between the two groups; the paper itself did not test this (see Additional file 1)*
Mood							

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First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
Goodwin 2005 [48] Quality = fair	Cross-sectional USA	Convenience sampling	Older adults, n=69 (72%) Mean age 72.0 (6.4)	TV was reported on a paper diary when it was the current activity in response to pager prompts which came 6 times a day for 7 days between 8am and 8pm	N/A NB: affect while watching TV was compared to affect while engaged in other leisure activities (e.g., hobbies, reading), maintenance activities (e.g., cleaning, meal preparation), and productive activities (e.g., work or volunteering) which were reported in response to the pager prompts	Positive and negative affect 'in the moment' when prompted by the pager (i.e., while watching TV if that was the current activity). Mood descriptors were rated on a 5-point Likert scale; positive affect descriptors included things like 'enthusiastic', and negative affect descriptors included things like 'upset'	Positive affect was significantly lower while watching TV (mean 26.07 (SD 9.27)) compared to while doing other leisure activities (28.39 (9.35)) and productive activities (29.78 (9.36)), but was not significantly different compared to while doing maintenance activities (26.19 (9.65)) (see Figure 7). Negative affect while watching TV (10.871 (2.93)) did not significantly differ compared to while doing other leisure activities (10.873 (2.86)), during productive activities (11.09 (2.70)), or during maintenance activities (10.86 (2.74)).
Kuykendall 2020 [47] Quality = good	Cross-sectional USA	Recruited adults who had completed at least 500 assignments on Amazon Mechanical Turk with ≥96% approval rating	Full-time working adults *Study 1a:* n=264 (50%) Mean age 38.7 *Study 1b:* n=538 (50%) Mean age 37.2	*Study 1a & 1b:* Day reconstruction in which all activity episodes from the prior day were reported	N/A	*Study 1a:* For each leisure activity reported in the day reconstruction, the extent to which participants felt the following on a scale of 1 (not at all) to 11 (extremely) was reported: detached, relaxed, stressed,	*Study 1a:* Levels of relaxation were higher when watching TV than when engaging in physical activities; relaxation was similar while watching TV compared to other leisure activities. Levels of satisfaction and happiness were higher during all other leisure activities compared to while watching TV (except for physical activity, for which happiness was

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First author, year Quality assessment	Study design, country, and name (if applicable)	Recruitment strategy	Target sample Sample size (% female) Mean (SD) age or age range	TV measurement details with operationalization used in analyses	Covariates included in model	Outcome measurement(s)	Finding for the association between TV and outcome measurement(s)
						autonomy, meaning, mastery, affiliation, happy, sad, satisfied. *Study 1b:* For each leisure activity reported in the day reconstruction, the extent to which participants felt the following: positive activated (energetic, alert, vigorous); negative activated (anxious, jittery, nervous); negative deactivated (bored, tired, dull); or positive deactivated (peaceful, at ease, serene)	similar to TV). Feelings of sadness did not significantly differ while watching TV versus during other leisure activities. *Study 1b:* Positive activated affect was lowest while watching TV compared to all other activities; positive deactivated affect was lower while watching TV than some (but not all) activities. Negative activated affect was lower while watching TV compared to during active, social, and cognitively stimulating activities. Negative deactivated affect was higher while watching TV than during all other activities.
Sleep							
Asaoka 2007 [50] Quality = fair	Intervention Japan	Voluntary sample recruited from pool of previous research participants (university sample) and employment service center (elderly sample)	University students (n=8, 62.5%) and elderly adults (n=8, 62.5%) Mean age for university students 19.9 (0.8), range 19-21; for elderly adults 73.6 (2.7), range 70-78	Logged TV time in daily activity diary (used only to confirm the efficacy of the intervention for changing TV time, not used as a predictor variable)	None	Rising time, bedtime, total sleep time, nap total sleep time, and daytime total sleep time (between 8am and 8pm) all derived from self-reported sleep logs	Compared to the control week, university students went to bed significantly earlier ($p<0.05$), slept for significantly longer in total ($p<0.05$) and during the daytime ($p<0.05$) when TV was limited to 30 minutes per day; there were no significant changes in rising time, bedtime, or total sleep time among the elderly ($p>0.05$).
Komada 2015 [49] Quality = fair	Experimental (between and within subjects)	Convenience sample	Young Japanese men, n=13 (0%)	Lab-based exposure (107 minutes) on three night-time occasions.	N/A	Change in salivary melatonin and cortisol levels (taken both	The increase in melatonin levels was larger following the half-blue light exposure compared to

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First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
	Japan		Mean age 22.7 (0.9), range 21-24	Participants were randomized to normal luminance or high luminance display groups. Regardless of group, all participants were exposed to both normal blue light and half blue light as well as a baseline (control) condition (random order).		before and after the 107-minute exposure)	the normal blue light exposure ($p<0.05$); there was no effect of luminance or luminance*blue light interaction on melatonin levels. There were no associations between blue light level, luminance, or their interaction on cortisol levels.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N/A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4-5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supp. File 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6-8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6-8

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	19
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8, Fig 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Supp files 3-4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8-14, supplementary files 3-4
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Supp fil 2; pp9, 12, 13
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-19
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	19
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	19-20
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	20

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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How is television time linked to cardiometabolic health in adults? A critical systematic review of the evidence for an effect of watching television on eating, movement, affect and sleep

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How is television time linked to cardiometabolic health in adults? A critical systematic review of the evidence for an effect of watching television on eating, movement, affect and sleep

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ABSTRACT

Objective: To improve our understanding of how TV time is linked to cardiometabolic health among adults by systematically and critically evaluating the evidence that watching TV is associated with increased food consumption, lack of movement or negative affect, or affects subsequent sleep.

Design: Systematic review

Data sources: Web of Science and PubMed

Eligibility criteria: Studies that provided quantitative evidence of dietary intake, characteristics of sitting, or affect while watching TV, or on subsequent sleep on the same day among samples of healthy adults (≥18 years old).

Data extraction and synthesis: Study quality was assessed using the NHLBI Quality Assessment Tools; studies deemed to be of low quality were excluded from the review. Due to heterogeneity of study designs and measurements, the findings were synthesised using narrative summary accompanied by custom plots.

Results: We identified 31 studies that met the inclusion criteria. Most of the associations reported by the studies included in this review were weak or inconsistent. There was no strong evidence to suggest that food consumption is higher while watching TV than in other contexts or that TV is a particularly 'sedentary' behavior. Affect was less likely to be positive while watching TV than in other contexts but was not more likely to be negative. Two small studies suggest that TV may impact sleep via suppressing melatonin and delaying bedtime.

Conclusion: There is currently no strong evidence to suggest that TV might impact cardiometabolic health via increasing food consumption, being linked with prolonged/inactive sitting, affect, or subsequent sleep. Additional research is required to understand how TV fits within everyday lives and relates to eating, sitting, affect and sleep to improve our understanding of how it might impact cardiometabolic health.

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STRENGTHS AND LIMITATIONS OF THE STUDY

- This review synthesizes for the first time a diverse body of evidence that describes what happens while watching TV (or immediately after) amongst adults, including food consumption, characteristics of sitting, mood/affect, and sleep, to shed light on mechanisms on which TV time might impact cardiometabolic health
- A systematic approach was taken in gathering and appraising the evidence
- Custom plots were produced to visualize the findings of individual studies, including the magnitude, direction, and statistical significance of associations
- The outcomes and designs of included studies were too heterogeneous for estimating overall summary estimates using meta-analysis

Keywords: TV time, sedentary behavior, cardiometabolic health, diet, mental health, sleep

INTRODUCTION

Television time has been associated with poor cardiometabolic health outcomes among adults, including cardiovascular mortality,[1, 2] incidence of type 2 diabetes,[1, 2] and increases in adiposity.[3] Television time is generally assumed to be sedentary and sedentary time is now a well-established risk factor for poor cardiometabolic health.[1, 4] However, the association between TV time and poor health outcomes is stronger than the association between total sitting time and poor health outcomes[1, 5] or than time spent sitting in other contexts, such as sitting at work.[6-9] For studies based entirely on self-reported measures, lower measurement error in the assessment of TV time than in the assessment of other sedentary behaviors is likely to contribute to such a difference.[10-12] However, the relationship between self-reported TV time and cardiometabolic outcomes is also stronger

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than the relationship between objectively assessed sedentary time and those same outcomes,[13-15] a finding that is unlikely to be attributable to measurement error. It is possible that TV time has a particularly adverse impact on cardiometabolic health and this suggestion is repeatedly made in the literature.[1, 5, 7, 16]

Several possible explanations for the larger observed effects of TV time have been put forth. Most commonly, it has been suggested that dietary intake while watching TV (particularly snacking) may contribute to its effect.[1, 5, 7, 16] It has also been suggested that the way in which sitting occurs while watching TV may be important, for example if it is characterized by particularly low energy expenditure or low levels of muscular activation.[1, 5, 16] Detrimental impacts of TV on sleep and mental health (potentially partly via short-term affective responses) have also been noted as potential mechanisms by which TV might adversely impact cardiometabolic health.[16-18] However, the evidence lending support to these suggested possible explanations has not, to our knowledge, been exhaustively or critically evaluated.

The purpose of this critical systematic review is to improve our understanding of how TV time is linked to cardiometabolic health among adults. To this end, we have critically evaluated the available literature that provides evidence regarding food consumption, characteristics of sitting, or affect while watching TV, or on the impact of watching TV on subsequent sleep.

METHODS

Eligibility criteria

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Studies were eligible for inclusion in the review if they provided quantitative evidence of dietary intake, characteristics of sitting, or affect while watching TV or subsequently on the same day as watching TV, or on sleep subsequently on the same day. Studies could be experimental or observational as long as they described behaviors while watching TV or on the same day subsequent to watching TV. Participants in the studies had to be non-institutionalized adults (≥18 years old) who were not exclusively characterized by a particular health status (e.g., only overweight or obese adults). All studies had to be published in English in 2000 or later as TV viewing practices change over time.

Information sources and search strategy

The literature was searched in March 2021 using Web of Science Core Collections and PubMed. The search strategy included terms to identify studies that measured TV time across all disciplines (Supplementary file 1). The reference lists of all included papers were also checked. We had initially set out to address a broader research question, aiming to capture all studies that examined TV time in relation to any factors that might impact cardiometabolic health; the search strategy was therefore intentionally broad and we did not specify outcome variables of interest *a priori*. Given the diversity of studies identified, we subsequently chose to narrow the scope of our research question to focus only on studies that measured the variable of interest while watching TV (or immediately after) amongst adults.

Study selection

The titles and abstracts of the retrieved studies were screened for relevance. A subsample (10%) was independently screened by both authors to ensure agreement and consistency; any discrepancies were discussed until an agreed application of inclusion and exclusion criteria was reached, then one author (JW) screened the remainder of titles and abstracts. The full-

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texts of studies with relevant abstracts were consulted to determine eligibility. Both authors confirmed the eligibility of each full-text, and any cases of disagreement were resolved through discussion.

Quality assessment

The quality of each eligible full-text was assessed based on the NHLBI Quality Assessment Tools for Observational Cohort and Cross-Sectional Studies (for observational studies) and for Controlled Intervention Studies (for experimental and intervention studies) (https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools). In order to draw on the highest standards of evidence, studies deemed to have low internal validity based on the quality appraisal were excluded from the review. The quality assessment of each study was confirmed by both authors and any disagreements were discussed until consensus was reached.

Data extraction

The data from each eligible study were extracted into a spreadsheet with pre-defined columns for the general study details (location, date), study design, methodology, sample characteristics, statistical analyses (including covariates), and main findings describing the behavior(s) of interest while watching TV.

Synthesis of results

The results of the associations between TV time and the outcome of interest were organized by categories of outcome variables that emerged from the search (diet, characteristics of sitting, affect, sleep). Because of heterogeneity of study outcomes and study designs, meta-analysis was not used. Therefore, the results were synthesized using narrative summary

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accompanied by visualization of study findings. We constructed custom plots designed to illustrate each study's findings to provide a visual summary of the strength of the evidence for each outcome. We plotted the reported effect size if the paper reported this; if studies did not report effect size but reported sufficient information to calculate it (e.g., mean and standard deviation), we calculated the standardized mean difference using the R package *metafor* (Supplementary file 2). If 95% confidence intervals or standard errors were reported, these were converted to standard deviations using the formula suggested in the Cochrane handbook (https://handbook-5-1.cochrane.org/chapter_7). When calculating the effect size within experimental studies that used within-subjects designs, we imputed correlations of 0.5 to account for repeated measurements; sensitivity analyses using correlations of 0.6 and 0.8 were also conducted to examine the extent to which the imputed correlation might affect the results (shown in Supplementary file 1). While we acknowledge that the interpretation of the magnitude of effect sizes is arbitrary and context-specific, for the purposes of plotting we classified effect sizes as negligible (standardized mean difference (SMD) <0.20, odds ratio (OR) <1.50, or correlation (*r*) <0.10), small (SMD 0.20-0.49, OR 1.50-2.49, *r* 0.10-0.29), medium (SMD 0.50-0.79, OR 2.50-4.29, *r* 0.30-0.49), or large (SMD ≥0.80, OR ≥4.30, *r* ≥0.50) based on standard conventions.[19]

Within the plots, the included papers are ordered from most robust to least robust evidence, primarily on the basis of study design (e.g., within-subjects prioritized over between-subjects) and sample size. Whether the finding of the study was statistically significant is denoted through a filled (black) symbol with the direction of the association denoted by a positive sign (+), negative sign (-), zero (0), or question mark (?) if the direction could not be ascertained; non-significant associations are unfilled. The strength of the effect size is denoted by number of boxes (one=negligible, two=small, three=medium, four=large) based

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on the reported or calculated effect size. Where the effect size was not reported and could not be calculated in a standardized way (e.g., regression coefficient, graphical presentation of means), we plotted the findings with a triangle, filled (or not) to denote statistical significance. Significance was determined based on 1) the calculated or reported confidence intervals not crossing zero, or 2) the study reported a p-value <0.05 where effect size with confidence intervals was not reported and could not be calculated.

Patient and public involvement

No patients involved

RESULTS

Search results

The flow of studies through the review process is shown in Figure 1. From the original 13,036 unique records retrieved by the database searches, 31 were eligible for inclusion. An additional 5 papers were retrieved through reference lists and 5 papers were excluded (4 due to poor study quality), leaving 31 studies included in the review (see Supplementary files 3-4 for study details, including information on statistical modelling with covariates listed).

Characteristics of included studies

Twenty-five studies were experimental, primarily focused on links between TV and food consumption (n=19); the remainder focused on energy expenditure of sitting or movement while watching TV (n=4) or the effects of TV on subsequent sleep (n=2). Most experimental studies used a within-subjects study design (n=17) and were conducted in Europe (n=11) or North America (n=9), Australia (n=3), and Japan (n=2). Sample sizes ranged from 13 to 120; ten studies included only female participants while one study included only men.

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Six studies were observational, including studies examining food consumption (n=4) and affect (n=2) while watching TV in free-living contexts. Five of these studies were based in the USA and one in New Zealand with sample sizes ranging from 40 to 538.

Food and drink consumption while watching TV

Experimental studies

TV versus doing nothing

There is some evidence to suggest that more food is consumed while watching TV compared with while doing nothing else in a laboratory setting (Figure 2; Supplementary file 3). Nine experimental studies compared food consumption while watching TV compared with a control condition spent alone with no distraction.[20-28] Seven studies found that participants had higher consumption in terms of energy intake[20-22, 24, 26] or mass[26-28] during the TV condition compared with the control condition (generally small effect sizes), although this did not reach significance in most cases. In sensitivity analyses, effect sizes increased from small to medium in two studies[20, 22] when a larger correlation coefficient was imputed (Supplementary file 2).

TV versus other activity

There is little evidence to suggest that more food is consumed while watching TV compared with while doing other activities in a laboratory setting (Figure 2; Supplementary file 3). Seven studies compared food consumption while watching TV with consumption while doing other tasks, including listening to audio stories,[20, 23] listening to classical music,[29] reading,[25] simulated driving,[27] playing video games,[30] eating with a researcher or strangers,[22, 27] and eating with friends.[22] Five of these studies reported no differences in

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the amount of food consumed between the TV condition and the other experimental condition they tested.[20, 22, 23, 25, 30] The remaining two studies reported higher consumption while watching TV compared with while listening to classical music,[29] driving,[27] or while conversing with a researcher.[27] Sensitivity analyses suggested larger imputed correlations did not materially change the findings (Supplementary file 2).

Effect of TV on later consumption on the same day

Three studies, all in laboratory settings, tested whether the consumption of a snack while watching TV (compared with while not watching TV) would result in higher food consumption at a meal or snack later on in the same day (consumed without TV)[24, 31, 32] (Supplementary file 3). Two of these studies found that when the earlier snack had been consumed while watching TV, significantly more was eaten in the subsequent TV-free meal/snack;[24, 31] the third did not find subsequent consumption was higher after the eating-with-TV exposure.[32]

The role of TV content

There is little evidence to suggest that the content of TV being watched influences food consumption in experimental settings (Supplementary file 3). Four studies examined food consumption when exposed to food advertisements compared with either neutral (non-food) advertisements[33-35] or no advertisements,[25, 35] all reporting no main effect of commercial condition (although Anschutz et al.[33] reported a significant interaction with gender such that women ate more when exposed to food commercials compared with neutral commercials but the opposite was true for men). Three studies evaluated the role of TV content on food consumption;[24, 36, 37] two of these reported that participants ate more when watching boring or repeated showings of TV compared with engaging or novel TV,[36,

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37] while the third reported no difference in consumption when participants watched funny, boring, or sad TV shows.[24] One study[38] compared snack consumption following exposure to either a clip of a cooking show or a nature show, reporting no difference in overall consumption between the groups (although more chocolate balls were consumed following exposure to the cooking show than the nature show).

Observational studies

Four observational studies examined dietary patterns while watching television in free-living settings among voluntary/convenience samples of adults in New Zealand,[39] African-American women in Chicago,[40] and young adults in the USA[41, 42] (Figure 3; Supplementary file 3).

The most detailed evidence comes from the study by Gemming et al.[39] in New Zealand, who linked 24-hour dietary recalls to wearable camera recordings on three separate measurement days, allowing for an in-depth assessment of the contextual details (e.g., location, environment, social context, presence of screens) surrounding each eating episode noted in the recall. In their sample, 17% of eating episodes (22% of meals and 12% of snacks) took place while watching TV. Snacks consumed while watching television were higher in energy (by 514kJ (123kcal)) than snacks consumed when no TV was viewed. There was no difference in meal energy intake or duration of meals or snacks while watching TV compared with when no screen was viewed.

The remaining three studies examined the contexts of self-reported eating and drinking using ecological momentary assessment[40] and food diary entries annotated with contextual information.[41, 42] These studies found that 30%[40] and 26%[41] of snacking and eating

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occasions, respectively, took place while watching TV. Amongst young adults in the USA, Stroebele and de Castro[42] reported that an average of 1.03 meals per day took place with the TV on, but meal size did not differ with or without the TV on. Compared to food and drink consumed while 'not doing anything else,' a larger proportion of snacks[40] and sugar-sweetened beverages[41] and smaller proportions of water, fruit, vegetables, and cereals/grains[41] were consumed while watching TV compared to while doing nothing else.

Characteristics of sitting while watching TV

Four experimental studies examined associations between watching TV and characteristics of sitting, specifically energy expenditure[43-45] and level of movement.[46] Three experimental studies compared the energy expenditure (EE) of watching TV with the EE of other sedentary behaviors.[43-45] The most robust evidence indicated that the EE of sitting while watching TV was lower than the EEs of typing, playing a handheld video game (both small effect sizes), and playing a screen-based video game while sitting (large effect size; Figure 4).[45] The other two studies found no difference between the EE of watching TV and the EE of typing at a desk,[44] reading at a desk,[44] or working on a laptop computer[43] in their samples (Figure 4; Supplementary file 4), although their relatively small sample sizes and lack of *a priori* power calculations may mean their non-significant findings were due to underpowered samples.

A laboratory-based study compared the level of movement captured using hip-worn, triaxial accelerometry while watching TV (seated on a chair without armrests) compared with other sedentary behaviors among adults.[46] Total movement (which we calculated as vector magnitude of acceleration) was higher while watching TV compared with while reading, writing, typing, and sitting both 'naturally' (i.e., with free movement) and motionless, but did

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not significantly differ from movement while playing cards or preparing food while seated (Figure 4).

In sensitivity analyses using larger imputed correlation coefficients, the effect sizes slightly increased for the difference in the EE of TV compared with typing and for the difference in vector magnitude of acceleration for TV versus reading, preparing food, and sitting naturally (Supplementary file 2).

Affect during TV viewing

In two studies, participants self-reported their affect while watching TV within their everyday lives (Figure 5; Supplementary file 4). Using a day reconstruction approach in which leisure activities and affect whilst engaged in those activities from the previous day were reported, Kuykendall et al.[47] found that respondents generally experienced poorer affect while watching TV compared with other leisure activities, in terms of lower positive activated and deactivated affect (energetic/alert/vigorous and peaceful/at ease/serene, respectively) and higher negative deactivated affect (bored/tired/dull). However, negative activated affect (anxious/jittery/nervous) was generally lower while watching TV compared with other leisure activities. In a separate sample using similar methods, levels of relaxation were higher while watching TV than during physical activity (but similar to other leisure activities), levels of happiness and satisfaction were lower when watching TV compared with other activities (except for physical activity, which was similar to TV), and feelings of sadness did not differ between TV and other leisure activities.

In a study using experience sampling methods over the course of a week, Goodwin et al.[48] found that mean positive affect scores were lower while watching TV compared with while

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engaging in other leisure activities (e.g., hobbies) or while doing productive activities (e.g., work or volunteering); negative affect score did not differ while watching TV compared with while doing any other activities. In sensitivity analysis, only the effect size of the difference in positive affect during TV versus productive activities increased (from small to medium) with larger imputed correlations (Supplementary file 2).

The impact of TV viewing on sleep

One small experimental study (n=13)[49] and one small intervention study (n=16)[50], both in Japan, were included here (Supplementary file 4). Komada et al.[49] tested the effect of TV displays on melatonin and cortisol secretion at night in a laboratory setting. Melatonin and cortisol are hormones involved in the body's sleep-wake cycles, and the production of melatonin in the body promotes sleep and is known to be suppressed by bright light.[51] Komada et al.[49] found that a TV display with higher blue light intensity was associated with a suppressive effect on melatonin levels but not on cortisol levels. Asaoka et al.[50] conducted an intervention in which university students and elderly adults limited their TV time to 30 minutes a day for one week, following a week-long 'control' in which participants went about normal activities. During the week of restricted TV time, the university students went to bed significantly earlier and slept significantly longer compared to the control week; the intervention had no effect on sleep duration among the elderly.

DISCUSSION

This critical systematic review aimed to improve our understanding of how TV might impact cardiometabolic health by evaluating and synthesizing available fair- and high-quality evidence on the effects of watching TV on food consumption, characteristics of sitting, affect and subsequent sleep. Most of the associations found were weak or inconsistent. There was

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no strong evidence from laboratory or free-living settings to suggest that food consumption while watching TV is higher than food consumption in other contexts. There was also no clear evidence to suggest that sitting while watching TV (in laboratory contexts) might be a particularly 'sedentary' behavior. The association between TV and affect was complex. Positive affect and certain aspects of negative affect (e.g., anxiousness or nervousness) tended to be lower while watching TV compared with other leisure activities, while other aspects of negative affect (e.g., sadness) did not differ; feelings of tiredness or boredom were higher while watching TV compared with other leisure activities. There was a very small amount of evidence to suggest TV may impact sleep by suppressing melatonin and by impacting bedtime, at least in younger people.

The impact of watching TV on food consumption is weak. In laboratory contexts in which fixed (and often large) amounts of food were placed in front of participants, participants tended to consume more while watching TV compared with while doing nothing else but more often than not this difference was not statistically significant, and there was very little evidence that consumption was higher while watching TV than during other potentially distracting activities such as listening to music. There is a very small amount of evidence that consumption of food while watching television (compared with doing nothing) may be associated with greater subsequent intake than consumption while not watching television. Evidence from free-living contexts, in which television viewing conditions and food availability are quite different, suggested that energy intake while watching TV versus not did not significantly differ for meals but differed by a small amount for each snacking episode.[39, 42] Observational studies found 17-26% of eating events and 12-30% of snacking events occurred while watching TV;[39-41] however, no evidence was available to describe what proportion of TV-watching was accompanied by eating to determine whether

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TV 'triggers' food consumption. It is worth highlighting that the studies cited most often as evidence to suggest links between TV and snacking or higher food intake are not based on measurements of food consumption *while watching TV*; rather, this evidence shows that those with higher TV time have higher total energy consumption, higher snack intake, or less-healthy diets than those with lower TV time with the causality of this association entirely unclear.[52-56] Further research in free-living contexts is needed to improve our understanding of the possible co-occurrence of TV watching and food consumption to test the hypothesis that watching TV leads to greater energy intake or greater intake of foods associated with cardiometabolic risk. The use of devices such as wearable cameras[e.g., 39] may prove particularly useful for this to assess the contexts of eating episodes (with and without TV), the contexts of TV viewing episodes (with and without eating), and the co-occurrence of the two.

There was little evidence to suggest that TV might be a particularly 'sedentary' behavior. Compared to other seated activities, one of three experimental studies found that energy expenditure was lower while watching TV and another experimental study showed incidental movement was *higher* while watching TV. The ecological validity of these findings is unclear as the settings for TV watching in these studies (e.g., on chairs without armrests, for short and fixed amounts of time) do not reflect TV-watching conditions in free-living contexts. We did not find any studies that examined whether watching TV may induce prolonged uninterrupted bouts of sitting, which have been shown to interfere with glucose and lipid metabolism.[57, 58] We also did not find any studies that examined the characteristics of sitting while watching TV in free-living contexts. There is evidence that those with higher self-reported TV time have higher free-living accelerometer-measured prolonged sedentary time[59] and fewer breaks in sedentary time,[59, 60] but these studies do not provide evidence that this

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prolonged sedentary time or lack of breaks is directly linked with time spent watching TV. Further research in free-living contexts should make use of mixed methodologies, for example by combining accelerometry with wearable cameras, using electromyographic shorts and diaries, or applying direct observation methods [see 61, 62 for review] to improve our understanding of characteristics of sitting specifically during TV viewing.

There is little evidence to support the hypothesis that TV may impact cardiometabolic health via impacting affect. The two studies that examined affect while watching TV in free-living contexts found that positive affect and feelings of happiness tended to be lower while watching TV than during other leisure activities, while the relationship between negative affect and TV depended on what aspect of negative affect was measured; overall negative affect and feelings of sadness were not higher or lower while watching TV, boredom and tiredness was higher, and anxiousness and nervousness were lower while watching TV compared with during other leisure activities.[47, 48] These findings suggest that TV may be a form of leisure that is 'numbing' in that it may ease negative feelings but not necessarily increase positive feelings. The role of TV in everyday lives as a method of stress management and relaxation and as a way to escape negative or depressed feelings has been previously described.[63-66] In light of this, it is possible that measured mood or affect while watching TV may represent 'emotional spillover' from activities preceding TV time.[47] From a measurement perspective, it would be useful to assess how affect might change over the course of TV viewing instances to better understand how TV might alter emotional states throughout the course of viewing.

Mental health status may be an important confounder of the association between TV and cardiometabolic health. Depression, for example, has been prospectively associated with both

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high TV time[67] and poor cardiometabolic health outcomes.[68] There is mixed evidence suggesting prospective associations between TV time and depression or depressive symptoms.[69-72] Further research is needed to disentangle the relationships between TV, mood/affect and mental health, and cardiometabolic health outcomes.

There is a very small body of evidence to suggest that television time may impact sleep, which may in turn impact cardiometabolic health. One small study included in this review found that blue light from a TV display suppressed melatonin,[49] which may lead to a disruption in circadian rhythms, and another small study found that a reduction in TV time led to an earlier bedtime in young people.[50] Thus, TV time may be linked to sleep disruption and later bedtimes, but much more evidence is needed to demonstrate this effect. The hypothesis that watching TV reduces sleep quantity or quality should be tested in people during their everyday lives, and might involve using wearable cameras to assess TV time and accelerometry to assess sleep quality and quantity.

This review did not identify strong evidence for any mechanism through which TV time has been posited to adversely impact cardiometabolic health. It is possible, however, that these factors may interact or have a multiplicative effect. For example, the combination of low energy expenditure of sitting while watching TV (based on laboratory evidence) with the slightly higher energy intake from snacks while watching TV could potentially produce a larger effect than either of these aspects on their own.

Importantly, even if such associations are observed in everyday lives, causality requires careful consideration as they may be the result of confounding, whereby, for example, chronic stress may prompt consumption of calorically-dense foods[73] and may also prompt

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TV-watching as a way to relax, perhaps concurrently. More generally, while the association between socioeconomic position (SEP) and TV time is likely to vary globally across cultural and economic contexts, low SEP is consistently linked to high TV time in 'Western' countries such as the UK,[74] the US,[75] Australia,[76] and Canada.[77] Within these contexts, low SEP itself is associated with some of the outcomes considered here, as well as with poor cardiometabolic health outcomes[78-81] and may thus confound the relationship between TV viewing, diet, sitting characteristics, affect and sleep, and health outcomes. More empirical evidence, particularly using ethnographic or mixed-methods approaches, is needed to understand the place of TV in everyday lives and its potential role in the development of cardiometabolic disease, with particular attention to the question of whether watching television is causally implicated in changing health-related practices. Experimental studies are also useful in helping disentangle causality.

The findings of this review must be interpreted in light of the limitations in the evidence base. The vast majority of included studies reported statistically significant results but often with very small effect sizes, suggesting possible publication bias. The associations seen in this review should therefore be interpreted with caution as statistically significant findings may be over-represented in the literature base. The majority of included studies examining food consumption and characteristics of sitting were laboratory-based, thus the ecological validity of such findings is unclear. Methods that capture these kinds of data in free-living contexts, such as use of wearable cameras or electromyographic shorts, may be useful for providing insights into the ways in which TV occurs in real-life settings. These methods will also be relevant for examining the contexts and possible health-related consequences of screen-related practices more broadly beyond just TV time. Key strengths of this review include its systematic approach and that it is the first to comprehensively examine and evaluate evidence

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of the assertions commonly made in the literature explaining why TV is particularly detrimental to cardiometabolic health.

CONCLUSIONS

There is currently no strong evidence to suggest that TV impacts cardiometabolic health via impacts on food consumption, prolonged/inactive sitting, affect, or sleep. Further research exploring the putative mechanism(s) by which TV viewing might impact cardiometabolic health is required. We suggest that studies make use of innovative and/or mixed methods, such as wearable cameras, participant observation, or diaries, to improve our understanding of how TV fits within everyday lives and how it may impact cardiometabolic health outcomes, or whether relationships between TV time and cardiometabolic health reflect confounding.

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Competing interests

The authors declare that they have no competing interests

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Ethics approval and consent to participate

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Not applicable

Consent for publication

Not applicable

Availability of data and material

All data generated or analyzed during this study are included in this published article and its supplementary information files

Authors' contributions

JW led the study. Both authors (JW and TP) contributed to the design, database searches, data collection, and analysis. JW constructed the plots. Both authors (JW and TP) contributed to and approved the final manuscript.

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Figure 1. Flow of studies through the review process

Figure 2. Summary plot of effect sizes for experimental effects of TV on food consumption compared with doing nothing else or doing something else. Strength of effect size is denoted by number of boxes (one=negligible to four=large); triangles denote cases where effect size could not be calculated. Filled (black) symbols denote significant associations; non-significant associations are unfilled.

Figure 3. Summary plot of eating patterns while watching TV in free-living contexts. Strength of effect size is denoted by number of boxes (one=negligible to four=large); triangles denote cases where effect size could not be calculated. Filled (black) symbols denote significant associations; non-significant associations are unfilled.

Figure 4. Summary plot of effect sizes for characteristics of sitting while watching TV. Strength of effect size is denoted by number of boxes (one=negligible to four=large); triangles denote cases where effect size could not be calculated. Filled (black) symbols denote significant associations; non-significant associations are unfilled.

Figure 5. Summary plot of effect sizes for affect while watching TV. Strength of effect size is denoted by number of boxes (one=negligible to four=large); triangles denote cases where effect size could not be calculated. Filled (black) symbols denote significant associations; non-significant associations are unfilled.

Supplementary file 1. Search strategy.

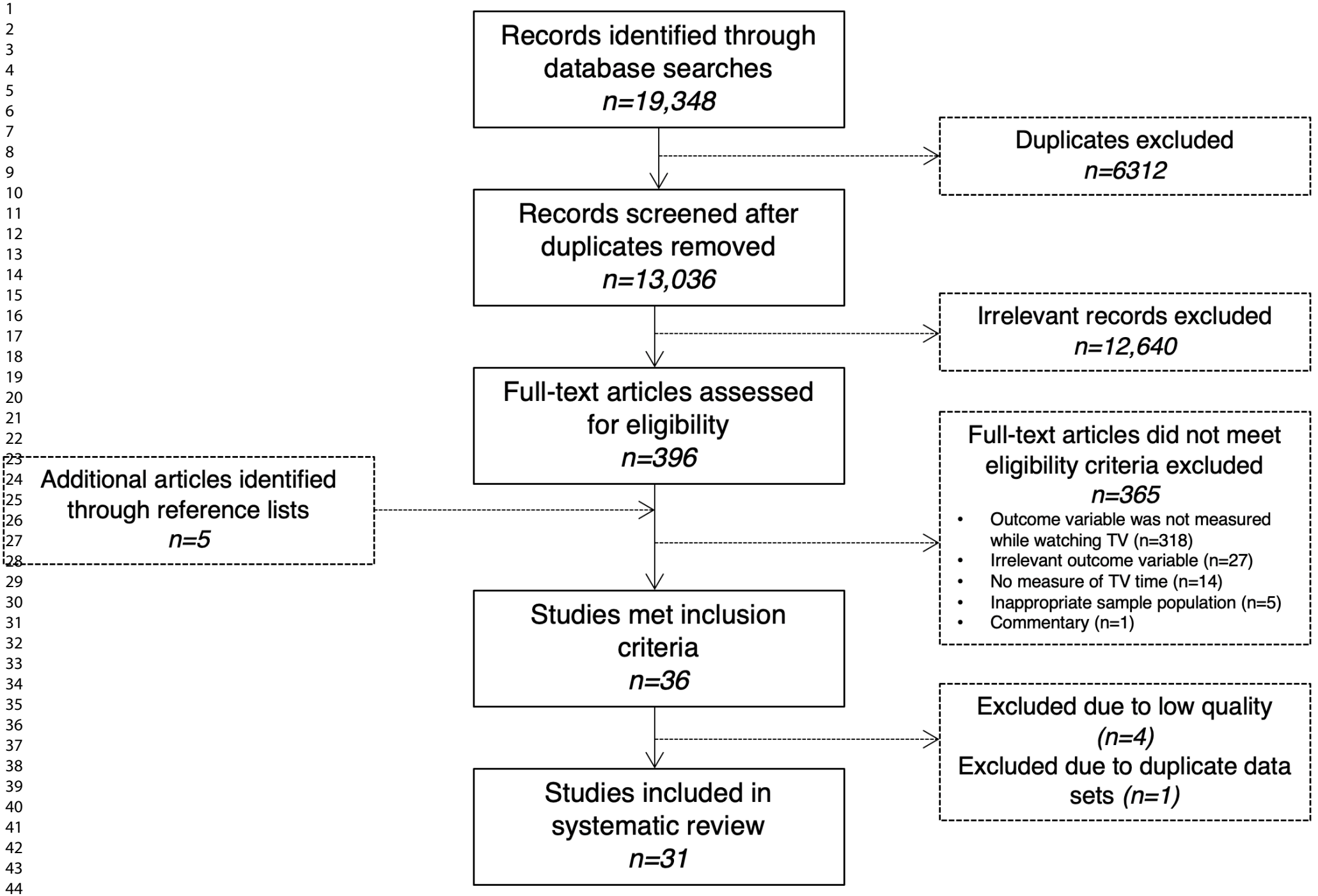
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Supplementary file 2. Effect size calculations, with sensitivity analyses conducted for studies that used within-subjects or repeated-measures designs.

Supplementary file 3. Summary of characteristics, measurements, and findings of studies examining food and drink consumption while watching TV.

Supplementary file 4. Summary of characteristics, measurements, and findings of included studies examining characteristics of sitting or affect while watching TV and sleep after watching TV

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Records identified through database searches
n=19,348
Duplicates excluded
n=6312
Records screened after duplicates removed
n=13,036
Irrelevant records excluded
n=12,640
Full-text articles assessed for eligibility
n=396
Full-text articles did not meet eligibility criteria excluded
n=365
Outcome variable was not measured while watching TV (n=318)
Irrelevant outcome variable (n=27)
No measure of TV time (n=14)
Inappropriate sample population (n=5)
Commentary (n=1)
Additional articles identified through reference lists
n=5
Studies met inclusion criteria
n=36
Excluded due to low quality (n=4)
Excluded due to duplicate data sets (n=1)
Studies included in systematic review
n=31

Study	Study design (n)	Outcome measure	Association	
			Overall	Women
TV vs doing nothing				
Hetherington 2006	Within subjects (n=37)	Meal intake (kJ)	+ +	
Martin 2009	Within subjects (n=48)	Meal intake (kcal)	-	
Rosenthal 2017	Within subjects (n=20)	Meal intake (kcal, g)	+	
Moray 2007	Within subjects (n=20)	Meal intake (ounces)	+	
Braude 2014	Within subjects (n=62)	Snack intake (kJ)		+
Bellisle 2004	Within subjects (n=48)	Meal intake (kJ)		+ +
Bellisle 2009	Within subjects (n=40)	Meal intake (kJ, g)		?
Ogden 2013	Between subjects (n=81)	Snack intake (g)		+ +
Mittal 2010	Between subjects (n=32)	Snack intake (kJ)		+ + +
TV vs other condition				
Hetherington 2006	Within subjects (n=37)	With friends (kJ)	-	
Hetherington 2006	Within subjects (n=37)	With strangers (kJ)	+	
Martin 2009	Within subjects (n=48)	While reading (kcal)	+	
Bellisle 2004	Within subjects (n=48)	Listening to audio story (kJ)		-
Bellisle 2009	Within subjects (n=40)	Listening to audio story (kJ, g)		?
Bellisle 2009	Within subjects (n=40)	With strangers (kJ, g)		+
Blass 2006	Within subjects (n=20)	Listening to music (kcal)	+	
Lyons 2012	Between subjects (n=120)	With handheld video games (kcal)	-	
		With motion-controlled video games (kcal)	- -	
Ogden 2013	Between subjects (n=81)	While driving (g)		+ + +
Ogden 2013	Between subjects (n=81)	While talking to researcher (g)		+ + +

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Study	Method	Variable of interest	While watching TV compared to	Finding
Gemming 2015	**Wearable cameras** worn for three days linked with 24-hour diet recalls on those days (n=40)	Snack energy intake (kJ)	At home with no TV	▲+
		Meal energy intake (kJ)	At home with no TV	△+
Ghosh Roy 2018	**Ecological momentary assessment** (5 prompts per day) to ascertain the context of any snack or beverage consumption that occurred since the previous prompt (n=79)	Snacks consumed (%)	While doing nothing else	■+ ■+
		Sugar-sweetened beverages consumed (%)	While doing nothing else	□-
Laska 2011	**7-day food diary** with contextual information provided for each instance of eating or drinking (n=48)	Sugar-sweetened beverages consumed (%)	While doing nothing else	▲+
		Water consumed (%)	While doing nothing else	▲-
		Fruit and vegetables consumed (%)	While doing nothing else	▲-
		Cereals and grains consumed (%)	While doing nothing else	▲-
		Cookies and sweetened baked goods consumed (%)	While doing nothing else	△-
		Candy/gummy fruit snacks consumed (%)	While doing nothing else	△0
		Frozen desserts consumed (%)	While doing nothing else	△0
		Salty snacks consumed (%)	While doing nothing else	△+
		Fried side dishes consumed (%)	While doing nothing else	△-
Stroebele 2004	**7-day food diary** in which participants specified whether the TV was on in each instance (n=78)	Meal size (kJ)	Without the TV on	△?

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Study	Study design (n)	Outcome measure	Association		
			Overall	Women	Men
Experimental studies					
Energy expenditure (EE)					
Mansoubi 2015	Within subjects (n=51)	EE TV vs typing	- -		
		EE TV vs handheld game (PSP)	- -		
		EE TV vs seated video game (Wii)	- - - -		
Newton 2013	Within subjects (n=25)	EE TV vs typing	-		
		EE TV vs reading	-		
Creasy 2016	Between subjects (n=36)	EE TV vs computer	- -		
Level of movement					
van der Berg 2019	Within subjects (n=18)	Acceleration TV vs reading	+ + +		
		Acceleration TV vs writing	+ + + +		
		Acceleration TV vs typing	+ + + +		
		Acceleration TV vs playing cards	0		
		Acceleration TV vs preparing food while seated	+		
		Acceleration TV vs 'natural' sitting	+ + +		
		Acceleration TV vs sitting motionless	+ + + +		

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Study	Study design (n)	Outcome measure	Finding
Kuykendall 2020	Cross-sectional (n=538)	*Positive activated affect*	
		TV vs physically active leisure	- - - -
		TV vs social leisure	- - - -
		TV vs cognitively stimulating leisure	- -
		TV vs creative leisure	- - -
		TV vs cultural activities	- - - -
		Positive deactivated affect	
		TV vs physically active leisure	△-
		TV vs social leisure	△+
		TV vs cognitively stimulating leisure	-
		TV vs creative leisure	- -
		TV vs cultural activities	△+
		Negative activated affect	
		TV vs physically active leisure	-
		TV vs social leisure	-
		TV vs cognitively stimulating leisure	-
		TV vs creative leisure	△-
		TV vs cultural activities	△-
		Negative deactivated affect	
		TV vs physically active leisure	+ +
		TV vs social leisure	+ +
		TV vs cognitively stimulating leisure	+ +
		TV vs creative leisure	+ +
		TV vs cultural activities	+ +
Goodwin 2005	Cross-sectional (n=69)	*Positive affect*	
		TV vs other leisure	- -
		TV vs productive activities	- -
		TV vs maintenance activities	[-]
		Negative affect	
		TV vs other leisure	[0]
		TV vs productive activities	[-]
		TV vs maintenance activities	[0]

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Search strategy for PubMed

All fields: “television time” OR “television viewing” OR “television watching” OR sedentary behavio?r OR “sedentary time” OR “screen time” OR “watching television” OR “viewing television”

Limits: Date range 2000 to present

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Supplementary file 2. Effect size calculations, with sensitivity analyses conducted for studies that used within-subjects or repeated-measures designs.

Study	Condition comparison or outcome variable	Standardized mean difference (95% CI)	Notes
Characteristics of sitting			
Creasy 2016	Energy expenditure (kcal/15min) TV vs computer	-0.23 (-0.88, 0.43)	
Mansoubi 2015	Energy expenditure (METs) TV vs typing	-0.41 (-0.70, -0.12)	Imputed correlation 0.5, total sample
		-0.45 (-0.74, -0.17)	Imputed correlation 0.6, total sample
		-0.61 (-0.91, -0.31)	Imputed correlation 0.8, total sample
	Energy expenditure (METs) TV vs PSP	-0.30 (-0.58, -0.02)	Imputed correlation 0.5, total sample
		-0.33 (-0.62, -0.05)	Imputed correlation 0.6, total sample
		-0.47 (-0.76, -0.18)	Imputed correlation 0.8, total sample
	Energy expenditure (METs) TV vs seated Wii	-1.66 (-2.08, -1.24)	Imputed correlation 0.5, total sample
		-1.78 (-2.22, -1.33)	Imputed correlation 0.6, total sample
		-2.11 (-2.61, -1.62)	Imputed correlation 0.8, total sample
Newton 2013	Energy expenditure (METs) TV vs typing	-0.19 (-0.59, 0.20)	Imputed correlation 0.5
		-0.22 (-0.61, 0.18)	Imputed correlation 0.6
		-0.31 (-0.71, 0.09)	Imputed correlation 0.8
	Energy expenditure (METs) TV vs reading	-0.05 (-0.45, 0.34)	Imputed correlation 0.5
		-0.06 (-0.45, 0.33)	Imputed correlation 0.6
		-0.08 (-0.47, 0.31)	Imputed correlation 0.8
van der Berg 2019	Vector magnitude (cpm) TV vs reading	0.79 (0.26, 1.32)	Imputed correlation 0.5
		0.84 (0.30, 1.37)	Imputed correlation 0.6
		0.96 (0.40, 1.52)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs writing	0.97 (0.41, 1.53)	Imputed correlation 0.5
		0.99 (0.43, 1.55)	Imputed correlation 0.6
		1.03 (0.46, 1.60)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs typing	0.92 (0.37, 1.47)	Imputed correlation 0.5
		0.95 (0.40, 1.51)	Imputed correlation 0.6
		1.04 (0.47, 1.62)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs playing cards	0.00 (-0.46, 0.46)	Imputed correlation 0.5
		0.00 (-0.46, 0.46)	Imputed correlation 0.6
		0.01 (-0.46, 0.47)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs preparing food while seated	0.14 (-0.32, 0.61)	Imputed correlation 0.5
		0.16 (-0.31, 0.62)	Imputed correlation 0.6
		0.22 (-0.24, 0.69)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs natural sitting	0.71 (0.19, 1.22)	Imputed correlation 0.5
		0.78 (0.25, 1.31)	Imputed correlation 0.6

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		1.02 (0.45, 1.59)	Imputed correlation 0.8
	Vector magnitude (cpm) TV vs imposed sitting	0.94 (0.39,1.50)	Imputed correlation 0.5
		0.97 (0.41, 1.52)	Imputed correlation 0.6
		1.01 (0.44, 1.58)	Imputed correlation 0.8
Diet			
Bellisle 2004	Food intake (kJ), TV vs control	0.46 (0.16, 0.76)	Imputed correlation 0.5
		0.51 (0.21, 0.82)	Imputed correlation 0.6
		0.73 (0.41, 1.05)	Imputed correlation 0.8
Hetherington 2006	Food intake (kJ), TV vs control	0.34 (0.01, 0.67)	Imputed correlation 0.5
		0.38 (0.05, 0.71)	Imputed correlation 0.6
		0.52 (0.18, 0.86	Imputed correlation 0.8
Mittal 2011, experiment 1	Food intake (kJ), TV vs control	0.59 (-0.13, 1.29)	Snack consumed while watching TV
	Food intake (kJ), TV vs control	0.51 (-0.19, 1.22)	Meal consumed after TV/no-TV condition
Ogden 2013	Food intake (g), TV vs control	0.45 (-0.17, 1.07)	
Bellisle 2004	Food intake (kJ), TV vs audio story	-0.04 (-0.32, 0.25)	Imputed correlation 0.5
		-0.04 (-0.32, 0.24)	Imputed correlation 0.6
		-0.06 (-0.34, 0.23)	Imputed correlation 0.8
Hetherington 2006	Food intake (kJ), TV vs eating with friends	-0.13 (-0.46, 0.19)	Imputed correlation 0.5
		-0.15 (-0.47, 0.18)	Imputed correlation 0.6
		-0.21 (-0.53, 0.12)	Imputed correlation 0.8
Lyons 2012	Food intake (kcal), TV vs handheld video game	-0.06 (-0.50, 0.37)	Total intake
	Food intake (kcal), TV vs motion-controlled video game	0.36 (-0.09, 0.80)	Total intake
Ogden 2013	Food intake (g), TV vs driving	0.77 (0.13, 1.40)	
	Food intake (g), TV vs talking to researcher	0.73 (0.08, 1.37)	
Mood			
Goodwin 2005	Positive affect, TV vs other leisure	-0.25 (-0.49, -0.01)	Imputed correlation 0.5
		-0.28 (-0.52, -0.04)	Imputed correlation 0.6
		-0.39 (-0.63, -0.14)	Imputed correlation 0.8
	Positive affect, TV vs productive activities	-0.39 (-0.64, -0.15)	Imputed correlation 0.5
		-0.44 (-0.69, -0.19)	Imputed correlation 0.6
		-0.62 (-0.88, -0.36)	Imputed correlation 0.8
	Positive affect, TV vs maintenance activities	-0.01 (-0.25, 0.22)	Imputed correlation 0.5
		-0.01 (-0.25, 0.22)	Imputed correlation 0.6
		-0.02 (-0.25, 0.22)	Imputed correlation 0.8
	Negative affect, TV vs other leisure	0.00 (-0.24, 0.24)	Imputed correlation 0.5
		0.00 (-0.24, 0.24)	Imputed correlation 0.6
		0.00 (-0.24, 0.24)	Imputed correlation 0.8
	Negative affect, TV vs productive activities	-0.08 (-0.31, 0.16)	Imputed correlation 0.5
		-0.09 (-0.32, 0.15)	Imputed correlation 0.6

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		-0.12 (-0.36, 0.12)	Imputed correlation 0.8
	Negative affect, TV vs maintenance activities	0.00 (-0.23, 0.24)	Imputed correlation 0.5
		0.00 (-0.23, 0.24)	Imputed correlation 0.6
		0.01 (-0.23, 0.24)	Imputed correlation 0.8

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Supplementary File 3. Summary of characteristics, measurements, and findings of studies examining food and drink consumption while watching TV.

First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
Observational/free-living studies							
Gemming 2015 [39] Quality = good	Cross-sectional New Zealand	Recruited from the community and university (convenience)	Healthy adults in Auckland, n=40 (50%) Mean ages 35 (SD=17) and 28 (SD=7) for men and women, respectively	Instances of television time (only while eating) as identified during processing of SenseCam (wearable camera) recordings across 3 days	NA	Energy intake (kJ), energy density (kJ/g), and duration (min) of meals and snacks across contexts (including during TV) measured by 24-hour multiple pass dietary recall on days that wearable camera was worn	Snacks at home when viewing television were higher in energy intake (+514 kJ (47, 1077)) compared to snacks when no television was viewed. Meals at home while watching TV were not significantly longer (3.1 min (-0.6, 6.7)) than meals not viewing screens, and were not significantly higher in energy (543kJ (-32, 1120)).
Ghosh Roy 2018 [40] Quality = good	Cross-sectional USA *African American Women's Daily Life Study*	Recruited from community and university (convenience)	African American women aged 25-65 in Chicago, n=79 (100%) Mean age 44.6 (SD=10.6)	In each instance of reported snack or sweetened beverage consumption (see 'outcome measurement'), participants were asked what they were doing while eating or drinking; watching television was an option	Education, per capita income, age, auto ownership, BMI, weekday or weekend day measurement, hunger status, fast food restaurant and convenience store density in home neighborhood	Self-reported consumption of ≥1 snack food item (including French fries, salty snacks, cookies or sweetened baked goods, chocolate/candy, ice cream/frozen dessert) or ≥1 sweetened beverage in response to smartphone prompts which came 5 times per day across one week (ecological momentary assessment)	Consumption of snack foods was significantly more likely to occur while watching TV versus while doing nothing else (OR 1.8 (95%CI 1.2, 2.7)); no such association was seen for sugar-sweetened beverage consumption (OR 0.9 (95%CI 0.5, 1.6)).
Laska 2011 [41] Quality = fair	Cross-sectional USA	Recruited from community and university (convenience)	Young adults aged 18-23 from four groups (attending university and living	All instances of eating and drinking were logged on a PDA over a 7-day	N/A	Proportion of food consumption while watching TV compared to while not doing	A larger proportion of calorically sweetened beverages (24% vs 15%) and smaller proportions of water

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			on campus; attending university and living independently off-campus; attending university and living with parents; not attending university and living independently), n=48 (12 from each group), 56% female Mean age approximately 21	period, and followed up with contextual questions, including whether they were watching TV during that instance of eating		anything else, separately by food group: calorically sweetened beverages, water, coffee/tea drinks, milk, cookies and sweetened baked goods, candy/gummy fruit snacks, fruits and vegetables (fresh or 'other'), frozen desserts, non-milk dairy products, entrees, cereals/grains, and fried side dishes	(12% vs 17%), fruits and vegetables (26% vs 36%), and cereals/grains (25% vs 31%) were consumed while watching TV compared to while not doing anything else, respectively ($p<0.05$ for all). There were no significant differences among the other food groups listed in the previous column.
Stroebele 2004 [42] Quality = fair	Cross-sectional USA	Recruited from university for course credit	University students, n=78 (82%) Mean age 22 (0.9)	Television time was recorded on a diary in 15-minute intervals. Instances of eating while watching TV were recorded on food diaries (see 'outcome measurement' column)	N/A	Meal sizes, caloric intake, and meal frequency, ascertained from a 7-day food diary in which participants recorded the volume of all food and drinks consumed, with a specification concerning whether the TV was on in each instance	Meal sizes did not differ with and without TV (data not shown). There was no correlation between TV time (h/day) and total caloric intake or meal frequency (data not shown).
Experimental/laboratory studies							
Study, quality	**Study design, country**	**Target sample Sample size (% female) Mean (SD) age or age range**	**Study aim and experimental protocol**		**Main outcome measurement**	**Findings**	
Anschutz 2011 [33] Quality = fair	Between subjects Netherlands	Non-overweight university students, n=82 (50%) Mean age (SD) for men and women was 20.9 (2.5) and 20.4 (1.5), respectively	Aim: to examine whether food commercial exposure would increase concurrent snack food intake among young adults Protocol: participants were randomized to 'food commercial' condition (30-minute nature TV clip interrupted by 3 food and 5 non-food commercials) or 'neutral commercial' condition (30-minute nature TV clip with 8 non-food commercials)		Food intake (kcal) based on weight of crisps and M&Ms consumed during the session Amount of food provided not specified	There was no effect of commercial condition on food intake (b=-0.52, SE=43.12, $p>0.05$) and the addition of commercial condition to the model had a negligible effect (change in R^2 <0.001); the interaction between sex and commercial condition was significant (women ate more when exposed to food commercials compared to neutral, while men ate more when exposed to neutral compared to food commercials)	

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Bellisle 2004 [20] Quality = fair	Within subjects France	Women aged 18-50 with BMI between 18.5 and 24.9, n=48 (100%) Mean age 29.9 (1.4), range 18-50	Aim: To assess the impact of two environmental stimuli (TV and audio story) on meal intake Protocol: Participants ate a meal in four conditions each lasting minimum of 30 minutes and spaced one week apart: control (alone), while watching TV (no references to food or eating), while listening to a recorded detective story, and last meal (alone); order of TV and auditory conditions was randomized	Food intake (kJ) of meal (shepherd's pie and fruit sherbet) ~4850kJ (1160kcal) available	Compared to the baseline control condition (1751 (SE=84) kJ), meal size was larger in the TV (2023 (SE=84) kJ) and audio recording (2044 (SE=84) kJ) conditions ($p<0.001$). Meal size did not differ between the TV and audio recording conditions ($p>0.05$).
Bellisle 2009 [23] Quality = fair	Within subjects France	Healthy women of normal weight, half with high and half with low dietary restraint, total n=40 (100%) Overall mean age or range not reported	Aim: To assess the effects of dietary restraint and environmental factors on meal intake Protocol: Participants ate a meal in five conditions (randomized order) each lasting a minimum of 30 minutes and spaced one week apart: alone in a quiet room, in groups of 3, alone with TV (no food cues), alone with TV (including food advertisements), alone while listening to audio detective story	Food intake (g, kJ) of meal (shepherd's pie and fruit sherbet) ~6000kJ (1430kcal) available	Significantly less food (g, kJ) was consumed in the group meal condition compared to the others ($p<0.02$) in the pooled sample. Consumption was not different between the other conditions (data not shown).
Blass 2006 [29] Quality = fair	Within subjects USA	Undergraduate students, n=20 (75%) Mean age or range not reported	Aim: to test whether more was eaten when watching TV compared to not Protocol: participants were assigned to either pizza or mac and cheese (assignment protocol not stated) and either watched TV for 30 minutes (program of choice with original commercials included) or listened to *Rachmaninoff's Second Symphony* for 30 minutes; the experiment was repeated a week later for the other condition (counterbalanced order)	Food intake (kcal) of pizza or macaroni and cheese 12-inch pizza or family-size bowl (~900g) of macaroni and cheese available	More calories were consumed with the TV on (793.7kcal) than with TV off (538.2 kcal), $p<0.001$. There was no interaction effect between food type and condition.
Bodenlos 2013 [38] Quality = fair	Between subjects USA	Undergraduate students, n=80 (73%) Mean age 19.5 (1.0), range 18-22	Aim: to test whether exposure to a cooking show affected caloric intake Protocol: Participants were randomized to watch either a cooking show or Planet Earth for 10 minutes, followed by a 'taste test' of cheese curls, chocolate covered candies, and carrots	Intake (kcal) of cheese curls, chocolate covered candies, and carrots 800kcal in total available	No significant difference in overall calories consumed between the two conditions ($F(1,74)=3.32$, $p=0.07$), $d=0.38$ Those in the cooking show group consumed significantly more calories from chocolate covered candies than those in the nature group ($F(1,74)=3.90$, $p=0.05$), $d=0.51$
Boyland 2017 [34]	Within subjects UK	Females recruited from university (staff and	Aim: to examine consumption responses to televised food commercials in overweight and lean adult females	Intake (kcal) of pizza	No main effect of condition or weight status on pizza intake or interaction between weight status and condition (p values >0.28).

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		students) and surrounding area, n=55 (100%) Mean age 32.4 (9.8), range 20-62	Protocol: Participants attended two sessions in a counterbalanced order: TV show including commercials for high-calorie palatable foods and TV show including non-food commercials. Following the show, participants were given pizza to eat.	2056kcal of pizza available	
Braude 2014 [21] Quality = fair	Within (and between) subjects Australia	Female undergraduates, n=62 (100%) Mean age 19.6 (2.2), range 18-29	Aim: To test whether TV affected sensory specific satiety Protocol: Participants were randomized to a single- or variety-foods group; both groups took part in a TV (*Friends,* no commercials) and no-TV condition (counterbalanced order) lasting 20 minutes and spaced one week apart. In the single-foods group, participants chose their preference between Maltesers, Skittles, almonds, or salted potato chips and were given 80g of it. Participants in the variety-foods group were given a 20g bowl of each food (totaling 80g).	Food intake (kJ, number of items) Variety condition totaled ~1600kcal available; single food condition depended on which food was picked	Participants consumed significantly more energy (z=2.37, p<0.025) and more items of food (z=2.07, p<0.05) in the TV condition compared to the no-TV condition.
Chapman 2014 [36] Quality = fair	Within subjects Sweden	Healthy female women, n=18 (100%) Mean age 22 (1.3)	Aim: to explore how the content of TV programs differentially impacts simultaneous eating behavior Protocol: Participants took part in three conditions (counterbalanced order) each lasting 30 minutes and spaced one week apart: watching an engaging TV program (comedy), a boring program (art lecture) or reading a non-engaging text about insects. Bowls of M&Ms and grapes were available in each condition.	Food intake (grams, kcal) 250 grams (~1200kcal) of M&Ms and 300 grams (~200kcal) of grapes were available	Significantly more food (grams) was consumed during the boring program (125g) compared to the engaging program (82g, p<0.01), but no difference in kcal (data not shown). No differences in grams or kcal consumption were reported between the engaging program vs text condition (109.3g vs 81.9g, p=0.05) or the boring program vs text conditions (+15.5 grams, p=0.26) (kcal data not shown).
Hetherington 2006 [22] Quality = fair	Within subjects UK	University staff and students, n=37 (43%) Mean age 28.3 (1.8), range 18 to 54	Aim: To test whether the social facilitation of eating occurs as a function of distraction (by comparing consumption in social contexts with TV and control conditions) Protocol: Participants took part in four conditions in a counterbalanced order at least 3 days apart: eating alone, eating with TV (game show, duration not fixed), eating with strangers, and eating with friends. The same buffet meal was available in all conditions.	Food intake (kJ) A buffet-style meal (13,743kJ/3283kcal) with 9 different food items was available	Participants consumed significantly more when eating in front of the TV (4350 (SE=252) kJ) and when eating with friends (4565) SE=272) kJ) compared to the control condition (3861 (SE=200) kJ). No differences were found between eating with strangers and any other condition.

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Higgs 2009 [31] Quality = fair	Within subjects UK	Female undergraduates of normal weight, n=16 (100%) Mean age 19 (1)	Aim: to examine whether watching TV during lunch would increase afternoon snack intake due to impaired memory Protocol: Participants had a standardized lunch either in quiet solitude (control) or with the TV on (comedy show with no references to food). Both conditions lasted 10 minutes and all participants ate the entire lunch in that time period. At least 2.5 hours later on the same day, participants were given three kinds of cookies broken into small pieces to taste test.	Amount (g) of cookies consumed Three kinds of cookies (40g of each) were available, totaling ~600kcal available	Participants consumed significantly more grams of cookies after the lunch-with-TV condition compared to after the lunch-without-TV condition ($F(1,15)=12.6$, $p<0.01$)
Lyons 2012 [30] Quality = fair	Between subjects USA	Adults aged 18-35, n=120 (50%) Mean age 24.1 (4.4)	Aim: to compare differences in energy intake among 2 sedentary screen behaviors (TV and video games) and one potentially active screen behavior (motion-controlled video game). Protocol: Participants were randomized to a TV, video game (PS3), or motion-controlled video game (Wii or Xbox 360) group. Each condition lasted 1 hour. During the condition, snacks and beverages were available for consumption.	Amount (kcal) of snacks and soda consumed Snack foods included M&Ms, baked Lay's, Doritos, and trail mix (nuts with dried fruit) (amounts provided not specified); soda included cans (3 of each) of Coca-Cola, Diet Coke, and Mountain Dew.	Energy consumption between the three groups was not significantly different ($p=0.07$). Mean intake during TV was 716 (407) kcal; during video games was 747 (540) kcal; and during motion-controlled video game was 553 (498) kcal.
Martin 2009 [25] Quality = good	Within subjects USA	Healthy adults aged 18 to 54 (BMI 20-35), n=48 (54%) Mean age 31.9 (SE=1.5)	Aim: to test the effect of TV viewing with and without ads on energy intake compared to a control and a reading condition Protocol: participants completed four conditions (each lasting 32 minutes) in a random order on two test days (experimental lunch and dinner after consuming a standardized breakfast): control, reading, TV without ads, TV with ads (half food-related, half not). The same buffet meal was available for all conditions.	Food intake (kcal) Buffet-style meals were available including 19 different items (totaling 4921kcal)	Energy intake did not vary by experimental condition ($F(3, 131)=0.30$, $p=0.81$); partial $\eta^2 \leq 0.01$
Mathur 2015 [37] Quality = fair	Within subjects Australia	Female undergraduates, n=45 (100%)	Aim: to test whether variability in engagingness of TV affects food intake Protocol: Participants completed two conditions in a randomized order one week	Food intake (g, kJ) of their first and second preferred snacks	Significantly more snack food was consumed in the same condition (mean 76.2g, SD=36.2) than the different condition (mean 66.7g, SD 37.3, $z=1.990$ (*one-tailed), $p<0.05$).

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		Mean age 19.5 (SD=2.2), range 18-29	apart: watching the same episode of *Friends* twice back-to-back ('same'), or 'watching two different episodes back-to-back ('different'), with each episode lasting 20 minutes and without advertisements). The first episode was watched without food; for the second episode, regardless of condition, they were given 80g each of their preferred two snacks (given the choices of almonds, Pringles, and M&Ms).		
Mittal 2010, experiment 1 [24] Quality = fair	Between subjects Australia	Females from the university with BMI between 18 and 25, n=32 (100%) Mean ages 20.8 (SD 3.8) and 20.3 (SD 3.9) in the two groups	Aim: to assess whether snacking with concurrent TV would result in eating more at a test meal later on Protocol: Participants were assigned to either a snack-with-TV or snack-without-TV condition, both lasting 20 minutes (TV program was choice between *Seinfeld* and *Friends*). In both conditions, they consumed as much as they wanted of their snack (potato chips, chocolate balls, and coke or orange juice). Then, after 45 minutes of psychological tests, they were given lunch (test meal).	Food intake (kJ) of the snack in the TV versus no-TV condition 2080kJ (~500kcal) offered (chocolate balls, Pringles, Coke/orange juice) Food intake (kJ) of the test meal following the snack conditions 2470kJ (~600kcal) offered (sandwiches, biscuits, crackers and dip)	Energy consumption of the snack did not differ between the snack-with-TV (1855.9 (264.2)kJ) and snack-without-TV (1667.7 (362.0) kJ) conditions (test statistics not shown). Energy consumption of the test meal was significantly higher among those who had watched TV with their snack (1584.6 (516.4) kJ vs 1354.9 (335.6) kJ), $p<0.05$ (*one-tailed)
Mittal 2010, experiment 2 [24] Quality = fair	Between subjects Australia	Females from the university, n=84 (100%) Overall mean age not reported but around 21 years	Aim: to determine whether the content of TV shows watched while snacking affect recall accuracy of food eaten during the show and intake on a later TV-free test meal Protocol: Participants were assigned to one of four conditions, each lasting 20 minutes: snack-without-TV, snack-with-boring-TV, snack-with-funny-TV, or snack-with-sad-TV. They consumed their snack (potato chips, chocolate balls, and coke or orange juice). Then, after 45 minutes of psychological tests, they were given lunch (test meal).	Food intake (kJ) between the snack-TV conditions 2411kJ (575kcal) offered (chocolate balls, Pringles, Coke/orange juice) Food intake (kJ) of the test meal following the snack conditions	There was no difference in the energy consumption of the snack across the four groups (test statistics not shown). Mean snack consumptions (kJ) were 2308.3 (262.4) for boring, 2291.6 (223.5) for sad, 2250.2 (289.7) for funny, and 2194.9 (322.0) for control. Energy consumption of the test meal was significantly higher in all three TV conditions compared to the no-TV condition (post-hoc pairwise $p<0.02$ in all three cases), but consumption did not differ between the three TV groups (test statistics not shown). Mean test meal consumptions (kJ) were 2507.0 (438.2) for boring, 2842.0 (452.4) for sad, 2637.6 (540.3) for funny, and 2147.9 (527.2) for control.

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				4021kJ (960kcal) offered (sandwiches, crackers and dip, ice cream and chocolate, Coke/orange juice)	
Moray 2007 [28] Quality = fair	Within subjects USA	University students, n=20 (50%) Mean age 20.8 (range 18 to 23)	Aim: to determine whether TV promotes increased food consumption because it impairs one's ability to accurately estimate food intake Protocol: participants completed two conditions in counterbalanced orders, each lasting 25 minutes: eat with and without TV. After the meal, participants were asked to estimate how much food they had just consumed, using a visible tablespoon as a reference for 'one unit' (which was equivalent to 1.5oz of food).	Food intake (oz) between the TV and no-TV conditions Accuracy of food estimation, determined as estimated units minus actual units consumed 16oz macaroni and cheese provided	There was no difference in amount of food consumed between the TV and no-TV condition (2.5%, p>0.05). Unit estimation error was greater ($T_{0.05}$ 2.2, p=0.022) in the TV vs no-TV condition.
Ogden 2013 [27] Quality = fair	Between subjects UK	Females aged ≥18 who had a manual driving license, n=81 (100%) Mean age 22 (5.18), range 18-40	Aim: to compare the impact of different forms of distraction on eating behavior Protocol: participants were randomized to one of four conditions, each lasting 7 minutes: driving (simulation with a manual transmission), TV (*Friends*), social interaction (conversing with one of the researchers), or sitting alone. 100g of hula hoops (potato snacks) were available in each condition.	Grams of hula hoops consumed 100g were provided (~520kcal)	Those in the TV condition consumed more (28.61 (24.44) g) than those in the social (14.16 (12.33)g, p<0.01) and driving (14.02 (10.34)g, p<0.05) conditions, but no significant difference was seen compared to the alone condition (18.21 (20.9)g, p=0.06).
Ogden 2017 [32] Quality = fair	Between subjects UK	Females from the university, n=60 (100%) Mean age 24 (3.3)	Aim: to compare the impact of distraction and dietary restraint on food intake during a subsequent taste test Protocol: participants were randomized to one of three conditions, each lasting 5 minutes: watching TV, walking (along the corridor), and social interaction (talking with another research participant). During each condition, participants were told to consume a cereal bar. After the condition, participants took part in a taste test lasting 7 minutes during which they could consume as much of the foods (M&Ms, carrot sticks, grapes, and hula hoops) as they wished.	Mass and energy (kcal) of food consumed during the taste test 150g of each of the foods were provided	There was no main effect of condition on total mass or total calories consumed during the taste test (values not shown).

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Rosenthal 2017 [26] Quality = good	Within subjects USA	Young adults aged 18 to 35 with BMI between 18.5 and 24.9kg/m^2 n=20 (85%) Mean age 22.3 (3.7)	Aim: to test the independent and interactive effects of TV and portion size on food intake Protocol: participants took part in four randomized conditions one week apart: small portion with and without TV and large portion with and without TV. In the 'no TV' condition, participants ate quietly with no other activities for 30 minutes; the TV condition was a 30-minute episode of *Scandal*.	Food intake (grams and kcals) Small portion condition provided 1083 kcal (650 grams) of macaroni and cheese plus salad with dressing; the large portion provided 2166 kcal (1300 grams) of the same foods	Only a main effect of portion size was found for grams and kcals consumed. No main effect for TV was found for food intake ($F(1,16)=0.39$, $p=0.54$, $d=0.11$) and ($F(1,16)=1.10$, $p=0.31$, $d=0.17$)) measured as grams and kcals, respectively. No interactions for TV and portion size were found (>0.05).
Wonderlich-Tierney 2013 [35] Quality = fair	Between subjects USA	Undergraduates, n=83 (52%) Mean age 19.6 (3.5)	Aim: to examine the impact of TV advertisements on food intake Protocol: participants were randomized to one of three groups, each lasting 60 minutes: TV with food advertisements, TV with non-food advertisements, and TV with no advertisements. A jar of cookies was available throughout each condition.	Number of cookies eaten 26 cookies were available in each condition	There was no effect of condition on number of cookies eaten ($F(2,69)=1.50$, $p=0.23$), $R^2=0.04$

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Supplementary file 4. Summary of characteristics, measurements, and findings of included studies examining characteristics of sitting or mood while watching TV and sleep after watching TV

First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
Characteristics of sitting							
Creasy 2016 [43] Quality = good	Experimental (between subjects) USA	Recruited at university (convenience)	Adults aged 18 to 40 with BMI between 20-35, n=74 (58%)* Mean age 24.2 (3.6) *In full study. Only those whose baseline conditions were sitting while watching TV or sitting at the computer (n=18 in each condition) are described here	Lab-based exposure (15 minutes) *Experimental aim: to examine the cumulative energy expenditure of various combinations of sitting, standing, and walking*	N/A	Energy expenditure (kcal per 15min, indirect calorimetry)	TV EE (18.66 (4.01) kcal/15min) not significantly* different from computer EE (19.63 (4.37) kcal/15min) **Based on confidence intervals crossing zero in our calculation of the standardized mean difference between the two groups; the paper itself did not test this (see Additional file 1)*
Mansoubi 2015 [45] Quality = good	Experimental (within subjects) UK	Recruited at university and in community in 2x2 format to obtain equal numbers of men/women and healthy weight/obese participants (convenience)	Adults aged ≥18, 47% of whom were obese, n=51 (51%) Overall mean age not reported	Lab-based exposure (10 minutes) *Experimental aim: to measure the energy expenditure of sedentary behaviours in normal weight and obese participants*	N/A	Energy expenditure (METs, indirect calorimetry)	TV EE (METs) (1.33 (0.24)) was significantly* lower than typing (1.45 (0.32)), playing a handheld computer game (1.41 (0.28)), and playing a screen-based computer game (2.06 (0.50)) * Based on confidence intervals crossing zero in our calculation of the standardized mean change between the conditions; the paper itself did not test this (see Additional file 1)
Newton 2013 [44] Quality = fair	Experimental (within subjects) USA	Recruited in community (convenience)	African American adults aged ≥18, n=25 (60%)	Lab-based exposure (30 minutes) *Experimental aim: to examine the energy*	N/A	Energy expenditure (METs, whole-room calorimetry)	TV EE (METs) (1.03 (0.15)) was not significantly different from EE of typing (1.06 (0.15)), or reading at a desk (1.04 (0.20))

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First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
			Mean age 38.2 (11.4), range 20-56	*expenditure of common sedentary behaviours*			
van der Berg 2019 [46] Quality = fair	Experimental (within subjects) Netherlands	Recruited from the university (convenience)	Adults aged 18 to 48 without mobility limitations, n=18 (44%) Mean age 27.3 (6.5)	Lab-based exposure (7 minutes) seated in a chair without armrests while watching TV on a computer screen *Experimental aim: to identify whether 'dynamic sitting (e.g., fidgeting) can be identified from triaxial accelerometry counts*	N/A	Accelerometer vector magnitude* while watching TV compared to natural sitting (allowed to move freely), imposed sitting (motionless), imposed fidgeting (told to move the upper body side to side and back to front), while reading a newspaper, while writing a letter, while playing cards, while typing on a computer, and while preparing food while seated *Paper reported mean acceleration in the vertical, anteroposterior, and mediolateral planes separately; we converted this to mean vector magnitude by taking the square root of the summed squares of all three axes	Accelerometer vector magnitude was significantly* higher while watching TV compared to natural and 'motionless' sitting, reading, writing, and typing; it was significantly lower while watching TV compared to imposed fidgeting. There was no difference in vector magnitude between TV and playing cards or between TV and preparing food while seated. **Based on confidence intervals crossing zero in our calculation of the standardized mean difference between the two groups; the paper itself did not test this (see Additional file 1)*
Mood							

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First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
Goodwin 2005 [48] Quality = fair	Cross-sectional USA	Convenience sampling	Older adults, n=69 (72%) Mean age 72.0 (6.4)	TV was reported on a paper diary when it was the current activity in response to pager prompts which came 6 times a day for 7 days between 8am and 8pm	N/A NB: affect while watching TV was compared to affect while engaged in other leisure activities (e.g., hobbies, reading), maintenance activities (e.g., cleaning, meal preparation), and productive activities (e.g., work or volunteering) which were reported in response to the pager prompts	Positive and negative affect 'in the moment' when prompted by the pager (i.e., while watching TV if that was the current activity). Mood descriptors were rated on a 5-point Likert scale; positive affect descriptors included things like 'enthusiastic', and negative affect descriptors included things like 'upset'	Positive affect was significantly lower while watching TV (mean 26.07 (SD 9.27)) compared to while doing other leisure activities (28.39 (9.35)) and productive activities (29.78 (9.36)), but was not significantly different compared to while doing maintenance activities (26.19 (9.65)) (see Figure 7). Negative affect while watching TV (10.871 (2.93)) did not significantly differ compared to while doing other leisure activities (10.873 (2.86)), during productive activities (11.09 (2.70)), or during maintenance activities (10.86 (2.74)).
Kuykendall 2020 [47] Quality = good	Cross-sectional USA	Recruited adults who had completed at least 500 assignments on Amazon Mechanical Turk with ≥96% approval rating	Full-time working adults *Study 1a:* n=264 (50%) Mean age 38.7 *Study 1b:* n=538 (50%) Mean age 37.2	*Study 1a & 1b:* Day reconstruction in which all activity episodes from the prior day were reported	N/A	*Study 1a:* For each leisure activity reported in the day reconstruction, the extent to which participants felt the following on a scale of 1 (not at all) to 11 (extremely) was reported: detached, relaxed, stressed,	*Study 1a:* Levels of relaxation were higher when watching TV than when engaging in physical activities; relaxation was similar while watching TV compared to other leisure activities. Levels of satisfaction and happiness were higher during all other leisure activities compared to while watching TV (except for physical activity, for which happiness was

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First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
						autonomy, meaning, mastery, affiliation, happy, sad, satisfied. *Study 1b:* For each leisure activity reported in the day reconstruction, the extent to which participants felt the following: positive activated (energetic, alert, vigorous); negative activated (anxious, jittery, nervous); negative deactivated (bored, tired, dull); or positive deactivated (peaceful, at ease, serene)	similar to TV). Feelings of sadness did not significantly differ while watching TV versus during other leisure activities. *Study 1b:* Positive activated affect was lowest while watching TV compared to all other activities; positive deactivated affect was lower while watching TV than some (but not all) activities. Negative activated affect was lower while watching TV compared to during active, social, and cognitively stimulating activities. Negative deactivated affect was higher while watching TV than during all other activities.
Sleep							
Asaoka 2007 [50] Quality = fair	Intervention Japan	Voluntary sample recruited from pool of previous research participants (university sample) and employment service center (elderly sample)	University students (n=8, 62.5%) and elderly adults (n=8, 62.5%) Mean age for university students 19.9 (0.8), range 19-21; for elderly adults 73.6 (2.7), range 70-78	Logged TV time in daily activity diary (used only to confirm the efficacy of the intervention for changing TV time, not used as a predictor variable)	None	Rising time, bedtime, total sleep time, nap total sleep time, and daytime total sleep time (between 8am and 8pm) all derived from self-reported sleep logs	Compared to the control week, university students went to bed significantly earlier ($p<0.05$), slept for significantly longer in total ($p<0.05$) and during the daytime ($p<0.05$) when TV was limited to 30 minutes per day; there were no significant changes in rising time, bedtime, or total sleep time among the elderly ($p>0.05$).
Komada 2015 [49] Quality = fair	Experimental (between and within subjects)	Convenience sample	Young Japanese men, n=13 (0%)	Lab-based exposure (107 minutes) on three night-time occasions.	N/A	Change in salivary melatonin and cortisol levels (taken both	The increase in melatonin levels was larger following the half-blue light exposure compared to

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First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
	Japan		Mean age 22.7 (0.9), range 21-24	Participants were randomized to normal luminance or high luminance display groups. Regardless of group, all participants were exposed to both normal blue light and half blue light as well as a baseline (control) condition (random order).		before and after the 107-minute exposure)	the normal blue light exposure ($p<0.05$); there was no effect of luminance or luminance*blue light interaction on melatonin levels. There were no associations between blue light level, luminance, or their interaction on cortisol levels.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N/A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4-5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supp. File 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6-8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6-8

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	19
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8, Fig 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Supp files 3-4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8-14, supplementary files 3-4
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Supp fil 2; pp9, 12, 13
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-19
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	19
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	19-20
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	20

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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